

Inequalities

Version 5.0

16-7-2022

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Introduction

Inequality is one major topic in algebra in the Olympiad. Proving inequalities is quite different from solving inequalities. Normally one cannot get anything by just verifying the inequalities for several particular examples, or by finding the equality cases. Therefore, one should learn some general ways which can help us prove the inequalities for all variables under consideration. In this note, we are going to introduce some standard results and methods in this area.

In section 1, we first introduce the basic operations of inequalities and basic techniques. In particular, we would learn the standard tricks in proving inequalities in a single variable and quadratic inequalities. These are fundamental but important, as we would always reduce inequalities into these forms. Those basic ideas will be used throughout the whole note.

It is always said that most classical Olympiad inequalities can be solved by the AM-GM inequality and the Cauchy-Schwarz inequality. However, the way of using these inequalities to solve a particular problem may not be too obvious. Sometimes it seems to be quite nonsense to treat the inequality in the way that the solution suggests so as to apply them. Nevertheless, we try to give some of the more natural way of applications of these classical inequalities in section 2. As generalizations of these inequalities, we would also introduce the power mean inequality and Hölder's inequality. In addition, we will talk about Chebyshev's inequality, which is another classical inequality.

Section 3 focuses on homogeneous inequalities. Many classical inequality problems have this feature, and there are some more standard results which are useful in solving such problems, including Muirhead's theorem and Schur's inequality. Such results are more useful if one intends to prove the inequality by expanding everything. More examples of this kind are contained in another note **Expansion Methods in Inequalities 1.0**.

Next, in section 4, we introduce the important trick called the local refinement method. This helps us solve an inequality problem step by step, by breaking up a difficult question into several simpler parts. In particular, the method of mixing variables will be introduced.

Then we move on to some more advanced technique in section 5, which is about calculus. Both differentiation and integration have some applications in Olympiad inequalities, and we will introduce some standard applications of these. In addition, we will introduce Jensen's inequality, which can be used to prove a lot of famous inequality theorems. Although calculus is considered as more advanced knowledge, it is much better if one can learn this and know some simple applications. More advanced knowledge and examples are contained in another note **Optimization of Multivariable Functions 1.0**.

Lastly, section 6 is about modern inequalities. This means inequality problems which are more popular nowadays. Among these, we see that induction is an important and basic trick in solving many problems of this kind. Also, to solve some inequality problems, sometimes it is helpful to use some combinatorial ideas. We will see such problems in this section.

1 Basic Operations and Techniques

In this section, we first go through the basic operations on inequalities. Although these are extremely elementary, it is not surprising that one would always forget these simple things when handling inequalities and deduce incorrect results. Being familiar with the most fundamental things is crucial before solving any problem. Afterwards, we shall introduce some basic techniques in proving inequalities, which will be used everywhere throughout this note.

1.1 Basic Operations on Inequalities

The real number system is far too complicated to talk about here. What we need to know is that the real numbers are composed of the zero element 0, positive elements and negative elements. In particular, the positive elements and the negative elements are in one-to-one correspondence. For any positive real number a , it corresponds to a negative real number $-a$, which is the unique number such that $a + (-a) = 0$.

There is an order relation $\geq$ in the real number system. For any two real numbers a and b , we have $a \geq b$ or $b \geq a$. In particular, $a \geq b$ and $b \geq a$ if and only if (iff) $a = b$. If $a \geq b$, then we also write $b \leq a$ to mean the same thing. In case of $a \geq b$ while $a \neq b$, we may also use the notation $a > b$, and also $b < a$. With these notations, a positive real number a is characterized by the fact that $a > 0$, while b is negative iff $b < 0$.

We shall use the following notations to denote some commonly used sets.

- $\mathbb{R}$: the set of all real numbers
- $\mathbb{R}^+$: the set of all positive real numbers
- $\mathbb{R}_0^+$: the set of all nonnegative real numbers (this is not a standard notation)

The following provides some basic facts involving inequalities.

Proposition 1.1. Let a, b, c, d be any real numbers. Then the following hold.

- If $a \geq b$ and $b \geq c$, then $a \geq c$.
- If $a \geq b$ and $c \geq d$, then $a + c \geq b + d$. In particular, $a \pm c \geq b \pm c$.
- If $a \geq b$ and $c \geq 0$, then $ac \geq bc$.
- If $a \geq b$ and $c \leq 0$, then $ac \leq bc$.
- If $a \geq b \geq 0$ and $c \geq d \geq 0$, then $ac \geq bd$. In particular, $ab \geq 0$ and $a^2 \geq b^2$.
- We have $a^2 \geq b^2$ iff $|a| \geq |b|$.

Proposition 1.1(a) suggests the transitivity of the order relation. One should note that $a \geq c$ and $b \geq c$ do not provide any information about the relation between a and b . For (b), (c) and (d), these are the elementary operations involving inequalities. Be careful that multiplication and division should not be done in the same way as the case of equations. One needs to take into account the sign of the constant. But in most cases, we shall only handle nonnegative real numbers, so that the operations are still simple. A generalization of (e) shows that $a \geq b \geq 0$ implies $a^k \geq b^k \geq 0$ for any positive integer k . Also, note that $a^2 \geq b^2$ does not necessarily yield $a \geq b$ as the bases may not be nonnegative. The correct version is given in (f).

In addition, we have the following important fact.

Proposition 1.2. Let a be any real number. Then $a^2 \geq 0$, with equality $a = 0$.

This is an important property of the real numbers. Indeed, every inequality about real numbers essentially comes from this fact, and possibly together with the definition $a \in \mathbb{R}_0^+$ iff $a \geq 0$. For the word ‘equality’, we means that the two sides of the inequality $a^2 \geq 0$ are equal iff $a = 0$. Sometimes we may be asked to give all or some of the equality cases, and looking for the equality cases may provide hints to the solutions.

1.2 Inequalities in One Variable

Proving inequalities involving a single variable is the first step to learn as these inequalities should be easier to handle. Although it is almost certain that you will never come across any inequality in one variable in Olympiad, this kind of inequality may arise in the final part of the solution. Therefore, it is important to know how to handle such inequalities.

As suggested in subsection 1.1, the most fundamental inequality we can apply is $a^2 \geq 0$ for any $a \in \mathbb{R}$. One should also be familiar with the many different forms of this inequality, such as

$$a^2 - 2a + 1 \geq 0, \quad b^2 - 4b + 4 \geq 0, \quad c + \frac{1}{c} \geq 2.$$

These inequalities are true for any $a, b \in \mathbb{R}$ and $c \in \mathbb{R}^+$, since they are the same as

$$(a - 1)^2 \geq 0, \quad (b - 2)^2 \geq 0, \quad \left(\sqrt{c} - \frac{1}{\sqrt{c}} \right)^2 \geq 0$$

respectively. The method of grouping various terms together to form a square of an expression is called the *method of completing square* (配方法), which should be

a useful trick as suggested above. Indeed, we have the following theoretical result (which is not needed for our purpose).

Proposition 1.3. Let $P(x)$ be a real polynomial in the variable x . Then $P(x) \geq 0$ for any $x \in \mathbb{R}$ iff $P(x) = Q(x)^2 + R(x)^2$ for some real polynomials $Q(x)$ and $R(x)$.

Proof. The ‘if’ part is trivial. In the following, we only prove the ‘only if’ part.

Suppose $P(x) \geq 0$ for any $x \in \mathbb{R}$. We may assume $P(x)$ is not the zero polynomial since otherwise it is trivial. By the fundamental theorem of algebra and a property of real polynomials ($z \in \mathbb{C}$ is a root of $P(x)$ iff $\bar{z}$ is a root of $P(x)$), we can write

$$P(x) = c \prod_{j=1}^n (x - a_j)^{i_j} \prod_{j=1}^m ((x - z_j)(x - \bar{z}_j))$$

for some $c \in \mathbb{R}$, distinct $a_1, a_2, \dots, a_n \in \mathbb{R}$ and non-real numbers $z_1, z_2, \dots, z_m \in \mathbb{C}$. This implies

$$P(x) = c \prod_{j=1}^n (x - a_j)^{i_j} \prod_{j=1}^m (x^2 - 2\operatorname{Re}(z_j)x + |z_j|^2).$$

For each $j = 1, 2, \dots, m$, we have

$$x^2 - 2\operatorname{Re}(z_j)x + |z_j|^2 = (x - \operatorname{Re}(z_j))^2 + (|z_j|^2 - \operatorname{Re}(z_j)^2) > 0$$

since $|z_j| \geq |\operatorname{Re}(z_j)|$ and equality only holds when $z_j \in \mathbb{R}$ (rejected). Thus, the condition becomes

$$c(x - a_1)^{i_1}(x - a_2)^{i_2} \dots (x - a_n)^{i_n} \geq 0$$

for any $x \in \mathbb{R}$. For sufficiently large $x \in \mathbb{R}$, this clearly implies $c > 0$. Also, each i_j must be even. For example, if i_1 is odd, then $P(a_1 + \varepsilon)$ and $P(a_1 - \varepsilon)$ have different signs (and are nonzero) for sufficiently small $\varepsilon > 0$ since $(a_1 + \varepsilon - a_1)^{i_1} > 0$ and $(a_1 - \varepsilon - a_1)^{i_1} < 0$, while the other factors in the two expressions have the same signs. This contradicts the assumption that both of them are nonnegative.

Therefore, by writing $c = (\sqrt{c})^2$ and $|z_j|^2 - \operatorname{Re}(z_j)^2 = \sqrt{|z_j|^2 - \operatorname{Re}(z_j)^2}^2$, we can now rewrite $P(x)$ as

$$P(x) = P_1(x)^2(Q_1(x)^2 + R_1(x)^2)(Q_2(x)^2 + R_2(x)^2) \dots (Q_m(x)^2 + R_m(x)^2)$$

for some real polynomials P_1, Q_j, R_j . We can also write $P_1(x)^2$ as $P_1(x)^2 + 0^2$. Now, it remains to use the identity

$$(A^2 + B^2)(C^2 + D^2) = (AC - BD)^2 + (AD + BC)^2$$

repeatedly to combine the product of all sums of two squares to a single sum of two squares as desired. $\square$

If $P(x) \geq 0$ for any $x \in \mathbb{R}$, we say that $P(x)$ is a *nonnegative polynomial*. Also, if $P(x) > 0$ for any $x \in \mathbb{R}$, we say that $P(x)$ is a *positive polynomial*.

Problem 1.1. Prove that $t^4 - 8t^3 + 20t^2 - 16t + 4 \geq 0$ for any real number t .

Analysis. In view of [proposition 1.3](#), we may try to rewrite the left-hand side as a sum of two squares and apply [proposition 1.2](#). The first trial is to see whether it is a perfect square or not. This step of completing square can always be done by inspection, especially when both the leading coefficient and the constant term are square numbers. Alternatively, we provide a more general method below.

Solution. Consider

$$t^4 - 8t^3 + 20t^2 - 16t + 4 = (at^2 + bt + c)^2.$$

By expanding the right-hand side and comparing coefficients, we need

$$\begin{cases} a^2 = 1, \\ 2ab = -8, \\ b^2 + 2ac = 20, \\ 2bc = -16, \\ c^2 = 4. \end{cases}$$

It is easy to find either of the solutions $(a, b, c) = (1, -4, 2), (-1, 4, -2)$. For example, we may take the former triple to get

$$t^4 - 8t^3 + 20t^2 - 16t + 4 = (t^2 - 4t + 2)^2 \geq 0.$$

□

Problem 1.2. Prove that $t^4 + 8t^3 + 30t^2 + 56t + 41 \geq 0$ for any real number t .

Analysis. It would be great if we could spot the identity

$$t^4 + 8t^3 + 30t^2 + 56t + 41 = \frac{1}{2}((t+1)^4 + (t+3)^4).$$

But it is not reasonable to expect everyone to be so sensitive to numbers like this. Indeed, as we find that the leading coefficient is a square (of course we can do so by dividing the leading coefficient from every term), we may try to complete square as follows:

$$t^4 + 8t^3 + 30t^2 + 56t + 41 = (t^2 + 4t)^2 + 14t^2 + 56t + 41.$$

But it is a pity that the remaining terms $14t^2 + 56t + 41$ do not necessarily be non-negative. This suggests using a stronger square based on this. The following gives a quite standard way to do so instead of trial and error.

Solution. Consider

$$t^4 + 8t^3 + 30t^2 + 56t + 41 = (t^2 + 4t + k)^2 + (14 - 2k)t^2 + (56 - 8k)t + (41 - k^2).$$

We aim at finding a k such that

$$(14 - 2k)t^2 + (56 - 8k)t + (41 - k^2) \geq 0$$

for any $t \in \mathbb{R}$. In [proposition 1.4](#) of subsection [1.3](#), we shall show that this is true if $14 - 2k > 0$ and $\Delta \leq 0$, where

$$\begin{aligned} \Delta &= (56 - 8k)^2 - 4(14 - 2k)(41 - k^2) \leq 0 \\ \Leftrightarrow \quad &3136 - 896k + 64k^2 - 2296 + 328k + 56k^2 - 8k^3 \leq 0 \\ \Leftrightarrow \quad &8k^3 - 120k^2 + 568k - 840 \geq 0 \\ \Leftrightarrow \quad &k^3 - 15k^2 + 71k - 105 \geq 0. \end{aligned}$$

It is not hard to see that $k = 3$ (or $k = 4, 5$) satisfies both conditions. Indeed, the last inequality is equivalent to

$$(k - 3)(k - 5)(k - 7) \geq 0,$$

so that we can choose any k between 3 and 5. Now we can write down a very neat solution that

$$\begin{aligned} t^4 + 8t^3 + 30t^2 + 56t + 41 &= (t^2 + 4t + 3)^2 + 8t^2 + 32t + 32 \\ &= (t^2 + 4t + 3)^2 + 8(t + 2)^2 \\ &\geq 0. \end{aligned}$$

□

Remarks. Note that in the above problem, there is no equality case. So there should be a better lower bound for the polynomial $t^4 + 8t^3 + 30t^2 + 56t + 41$. One should distinguish between two kinds of problem, namely, proving an inequality and finding the minimum/maximum value of an expression.

Problem 1.3. Prove that $2t^3 - 7t^2 + 4t + 4 \geq 0$ for any nonnegative real number t .

Analysis. We usually come across inequalities where the variables are nonnegative real numbers. In that case, the polynomial might not be a sum of squares (which is obviously not the case for odd degree polynomials). However, we can often factor out linear terms from the polynomial to obtain a trivial result.

Solution. By trial and error, it is not hard to see that $t = 2$ is a root to the equation $2t^3 - 7t^2 + 4t + 4 = 0$. Hence, we have

$$2t^3 - 7t^2 + 4t + 4 = (t - 2)(2t^2 - 3t - 2),$$

which further becomes $(t - 2)^2(2t + 1)$. This is clearly nonnegative as $t \in \mathbb{R}_0^+$. □

Remarks. Although we may not have a perfect square, the linear term $t - a$ (where $a \in \mathbb{R}^+$) always occurs an even number of times. This fact is essentially explained in the proof of [proposition 1.3](#). Here, as we find that $t - 2$ is a factor of $2t^3 - 7t^2 + 4t + 4$, we know that $(t - 2)^2$ must be a factor of the polynomial. This saves us some time in spotting some of the roots especially when the polynomial has large degree.

It seems not a good method as we need to spot the root $t = 2$ in the above solution. However, if this is an inequality arising from the original problem, then the root should not be difficult to observe, as it just indicates the equality case. Also, by the rational root theorem, any rational root $\frac{u}{v}$ (in lowest term) to the polynomial equation

$$a_n t^n + a_{n-1} t^{n-1} + \cdots + a_1 t + a_0 = 0 \quad (a_n \neq 0)$$

with integer coefficients $a_n, a_{n-1}, \dots, a_1, a_0$ must satisfy $v \mid a_n$ and $u \mid a_0$. So there is only a finite number of choices for the rational root.

Problem 1.4. Prove that $314t^6 + 1200t^5 - 1234t^4 - 765t^3 + 1357t^2 + 88t + 8 \geq 0$ for any nonnegative real number t .

Analysis. In many cases, when an inequality is reduced to a single variable one, it is a very weak one (if it is obviously not a perfect square and has no obvious factor). The only trouble is that the coefficients are large, and so one has to be patient in handling it.

The only negative terms are $-1234t^4$ and $-765t^3$. To eliminate t^4 , one may complete square with the terms t^5, t^3 (like $t^5 - 2t^4 + t^3 \geq 0$), or with t^6, t^2 . The former one may generate more negative terms of t^3 , and so we shall try the latter. With a glance on these terms, we can find that the ratio of coefficients of the terms t^6, t^4, t^2 is approximately $1 : (-4) : 4$. This suggests using the related result $a^2 - 4a + 4 \geq 0$. Similarly, the ratio of coefficients of the terms t^5, t^3, t is approximately $12 : (-8) : 1$, which is not enough to be a square. We may try to use a close ratio which corresponds to a perfect square such as $9 : (-6) : 1 = 12 : (-8) : \frac{4}{3}$, and see how to tackle the remaining terms.

Solution. Using $t^6 - 4t^4 + 4t^2 = t^2(t^2 - 2)^2$ and $9t^5 - 6t^3 + t = t(3t^2 - 1)^2$, we get

$$\begin{aligned} & 314t^6 + 1200t^5 - 1234t^4 - 765t^3 + 1357t^2 + 88t + 8 \\ &= 309(t^6 - 4t^4 + 4t^2) + 128(9t^5 - 6t^3 + t) + 5t^6 + 48t^5 + 2t^4 + 3t^3 + 121t^2 - 40t + 8 \\ &= 309t^2(t^2 - 2)^2 + 128t(3t^2 - 1)^2 + 5t^6 + 48t^5 + 2t^4 + 3t^3 + 41t^2 + 3 + (80t^2 - 40t + 5) \\ &= 309t^2(t^2 - 2)^2 + 128t(3t^2 - 1)^2 + 5t^6 + 48t^5 + 2t^4 + 3t^3 + 41t^2 + 3 + 5(4t - 1)^2 \\ &\geq 0. \end{aligned}$$

□

Remarks. There are a lot of ways to solve this problem. For convenience, we usually start by using some simple squares like $(a - 2)^2$ and $(3a - 1)^2$. If the remaining expression is too weak to be nonnegative, we then try stronger ones. But usually the former choices are enough to deal with most problems.

1.3 Quadratic Inequalities

Quadratic inequalities refer to those inequalities which are polynomials of degree 2. It does not restrict to inequalities in a single variable since we may focus on one of the variables and regard all other variables as constants to deduce useful results. As the polynomial has degree 2, the most useful tool is still the completing square method. This motivates the use of discriminants.

Definition 1.1. Let $P(x) = ax^2 + bx + c$ be a quadratic polynomial where a, b, c are real constants with $a \neq 0$. The *discriminant* (判别式) of $P(x)$, denoted by Δ , is defined by $\Delta = b^2 - 4ac$.

From the identity $P(x) = a \left(x + \frac{b}{2a} \right)^2 - \frac{\Delta}{4a}$, we see that Δ plays an important role of the behaviour of $P(x)$. Indeed, we have the following result.

Proposition 1.4. Let $P(x) = ax^2 + bx + c$ be a quadratic polynomial where a, b, c are real constants with $a > 0$. Then the following hold.

- (a) If $\Delta \leq 0$, then $P(x) \geq 0$ for any real number x . Equality holds iff $\Delta = 0$ and $x = -\frac{b}{2a}$.
- (b) If $\Delta > 0$, then $P(x) \geq 0$ iff $x \geq \frac{-b + \sqrt{\Delta}}{2a}$ or $x \leq \frac{-b - \sqrt{\Delta}}{2a}$, while $P(x) \leq 0$ iff $\frac{-b - \sqrt{\Delta}}{2a} \leq x \leq \frac{-b + \sqrt{\Delta}}{2a}$.

Proof.

- (a) If $\Delta \leq 0$, the inequality holds because

$$P(x) = a \left(x + \frac{b}{2a} \right)^2 - \frac{\Delta}{4a} \geq 0 + 0 = 0.$$

The equality case also follows readily.

- (b) If $\Delta > 0$, then $P(x) \geq 0$ iff $\left(x + \frac{b}{2a} \right)^2 \geq \frac{\Delta}{4a^2}$. Since both sides are nonnegative, this holds iff $x + \frac{b}{2a} \geq \frac{\sqrt{\Delta}}{2a}$ or $x + \frac{b}{2a} \leq -\frac{\sqrt{\Delta}}{2a}$. This gives the first result.

Similarly, $P(x) \leq 0$ iff $\left(x + \frac{b}{2a} \right)^2 \leq \frac{\Delta}{4a^2}$ iff $-\frac{\sqrt{\Delta}}{2a} \leq x + \frac{b}{2a} \leq \frac{\sqrt{\Delta}}{2a}$. This gives the second result.

□

One may also see from the completing square step that $P(x)$ attains its minimum value when $x = -\frac{b}{2a}$. The condition $a > 0$ in the proposition is not a big deal. Indeed, if $P(x)$ has a negative leading coefficient, it suffices to consider $-P(x)$, which has a positive leading coefficient, to check the positivity for different values of x .

An example of using the discriminant has been given in [problem 1.2](#) of subsection 1.2. We shall see some more examples in the remainder of this subsection.

Problem 1.5. Prove that $a^2 + 5b^2 + 5c^2 \geq 3ab + 3ac + bc$ for any real numbers a, b, c .

Analysis. The inequality becomes obvious once we have found that

$$a^2 + 5b^2 + 5c^2 - 3ab - 3ac - bc = \frac{1}{2}((a - 3b)^2 + (a - 3c)^2 + (b - c)^2).$$

However, this kind of ‘clever’ solution may not always work. It is better to learn more general methods that can be used to solve more problems.

Solution. Rewrite the inequality as a polynomial in a , i.e.

$$a^2 - 3(b + c)a + (5b^2 - bc + 5c^2) \geq 0.$$

By [proposition 1.4](#), this is true iff $\Delta \leq 0$. Indeed, we have

$$\begin{aligned} \Delta &= 9(b + c)^2 - 4(5b^2 - bc + 5c^2) \\ &= -11b^2 + 22bc - 11c^2 \\ &= -11(b - c)^2 \leq 0. \end{aligned}$$

The result then follows. □

Remarks. One may argue that we still need to do a completing square step at the end to derive $\Delta \leq 0$, so the method still requires some inspection. However, the fact is that we have successfully reduced the problem from 3 variables (the original question) to 2 variables (proving $-11b^2 + 22bc - 11c^2 \leq 0$). Even if one cannot spot the perfect square, one can still apply [proposition 1.4](#) again to further reduce the number of variables. At some stage, one should obtain an inequality which is sufficiently obvious.

Problem 1.6. (CMO 1988 Problem 4(1)) Let a, b, c be three positive real numbers satisfying

$$(a^2 + b^2 + c^2)^2 > 2(a^4 + b^4 + c^4).$$

Prove that a, b, c must be the side lengths of a triangle.

Analysis. To show that a, b, c are the side lengths of a triangle, one can only apply the triangle inequality. One way to deduce the result is to prove by contradiction. Though it seems to be an inequality of degree 4, it is indeed a quadratic polynomial of c^2 , for example.

Solution. Suppose on the contrary that a, b, c are not the side lengths of a triangle. Without loss of generality (WLOG), assume $c \geq a + b$. Note that the given condition is the same as

$$(c^2)^2 - 2(a^2 + b^2)c^2 + (a^4 - 2a^2b^2 + b^4) < 0.$$

By completing square, we get

$$(c^2 - a^2 - b^2)^2 - 4a^2b^2 < 0.$$

Note that $c \geq a + b$ implies $c^2 \geq (a + b)^2 > a^2 + b^2$. It follows that

$$(c^2 - a^2 - b^2)^2 \geq ((a + b)^2 - a^2 - b^2)^2 = 4a^2b^2.$$

This yields the contradiction

$$0 > (c^2 - a^2 - b^2)^2 - 4a^2b^2 \geq 4a^2b^2 - 4a^2b^2 = 0.$$

So the assumption is incorrect, and a, b, c must be the side lengths of a triangle. $\square$

Problem 1.7. Prove that for any real numbers a and b , the inequalities

$$\begin{aligned} a^2 + b^2 &< \frac{3}{7}, \\ a^2 + b^2 - 2b + \frac{3}{7} &< 0, \\ a^2 + b^2 - 2\sqrt{3}a + \frac{9}{7} &< 0 \end{aligned}$$

cannot hold simultaneously.

Analysis. This is a problem arisen when solving CMO 1996 Problem 6. Again, we shall prove by contradiction to prevent handling too many cases. We can find the range of the variables from these inequalities and seek for a contradiction.

Solution. Assume there exist $a, b \in \mathbb{R}$ satisfying

$$a^2 + b^2 < \frac{3}{7}, \tag{1}$$

$$a^2 + b^2 - 2b + \frac{3}{7} < 0, \tag{2}$$

$$a^2 + b^2 - 2\sqrt{3}a + \frac{9}{7} < 0. \tag{3}$$

Firstly, (2) can be rewritten as $b^2 - 2b + \left(a^2 + \frac{3}{7}\right) < 0$. This yields

$$1 - \sqrt{\frac{4}{7} - a^2} < b < 1 + \sqrt{\frac{4}{7} - a^2}. \tag{4}$$

We put (4) in (1) and (3) respectively. Then we find that

$$\begin{aligned}
& \frac{3}{7} > a^2 + b^2 > a^2 + \left(1 - \sqrt{\frac{4}{7} - a^2}\right)^2 \\
\Rightarrow & \frac{3}{7} > \frac{11}{7} - 2\sqrt{\frac{4}{7} - a^2} \\
\Rightarrow & \sqrt{\frac{4}{7} - a^2} > \frac{4}{7} \\
\Rightarrow & a^2 < \frac{12}{49}
\end{aligned}$$

and

$$\begin{aligned}
& 0 > a^2 + b^2 - 2\sqrt{3}a + \frac{9}{7} > a^2 - 2\sqrt{3}a + \frac{9}{7} + \left(1 - \sqrt{\frac{4}{7} - a^2}\right)^2 \\
\Rightarrow & 0 > -2\sqrt{3}a + \frac{20}{7} - 2\sqrt{\frac{4}{7} - a^2} \\
\Rightarrow & \sqrt{3}a + \sqrt{\frac{4}{7} - a^2} > \frac{10}{7} \\
\Rightarrow & 3a^2 + \frac{4}{7} - a^2 + 2\sqrt{3}a\sqrt{\frac{4}{7} - a^2} > \frac{100}{49} \\
\Rightarrow & \sqrt{3}a\sqrt{\frac{4}{7} - a^2} > \frac{36}{49} - a^2. \tag{5}
\end{aligned}$$

Since $a^2 < \frac{12}{49}$, both sides of (5) are positive. Hence, (5) implies

$$3a^2 \left(\frac{4}{7} - a^2\right) > \left(\frac{36}{49} - a^2\right)^2.$$

Now,

$$\begin{aligned}
& 3a^2 \left(\frac{4}{7} - a^2\right) > \left(\frac{36}{49} - a^2\right)^2 \\
\Rightarrow & \frac{12}{7}a^2 - 3a^4 > a^4 - \frac{72}{49}a^2 + \frac{1296}{2401} \\
\Rightarrow & 0 > a^4 - \frac{39}{49}a^2 + \frac{324}{2401} \\
\Rightarrow & 0 > \left(a^2 - \frac{12}{49}\right)\left(a^2 - \frac{27}{49}\right).
\end{aligned}$$

The last inequality is the same as $\frac{12}{49} < a^2 < \frac{27}{49}$, which contradicts $a^2 < \frac{12}{49}$ as found above. This suggests the assumption is incorrect, and not all of (1), (2), (3) can be true simultaneously. $\square$

Remarks. The last factorization should not be difficult to guess. As we only obtain the bound $a^2 < \frac{12}{49}$ and the bound should be quite strong, $a^2 - \frac{12}{49}$ should probably be a factor of the degree 4 polynomial. Indeed, if one tries out the original problem in this direction, one should have found out the equality case well before. In any case, we may still treat it as a quadratic inequality in a^2 and solve it.

Another reminder is that it is necessary to check that both sides of (5) are nonnegative before squaring both sides. For example, $3 > -5$ does not yield $(3)^2 > (-5)^2$. One should always recall the facts given in proposition 1.1.

Finally, we summarize some basic tricks and concepts we have used to prove certain inequalities.

- completing square (problems 1.1, 1.2, 1.4)
- factorization (problem 1.3)
- discriminant (problems 1.2, 1.5)
- quadratic polynomials (problems 1.6, 1.7)

2 Classical Inequalities

In this section, we shall introduce some classical inequalities and the simple usages of them. Most MO problems can be solved using these inequalities but sometimes the solutions are quite tricky.

2.1 The AM-GM Inequality and the Power Mean Inequality

Theorem 2.1. (AM-GM inequality(均值不等式)) Let $n \geq 2$. For any positive real numbers $a_1, a_2, \dots, a_n$, we have

$$\frac{a_1 + a_2 + \dots + a_n}{n} \geq \sqrt[n]{a_1 a_2 \dots a_n}.$$

Equality holds iff $a_1 = a_2 = \dots = a_n$.

The left-hand side is called the *arithmetic mean*(算術平均) of $a_1, a_2, \dots, a_n$, while the right-hand side is called their *geometric mean*(幾何平均). The result shows that the arithmetic mean of some positive real numbers is no less than their geometric mean. Note that the result also holds if some numbers are zero.

Proof. Let $b_j = \frac{a_j}{\sqrt[n]{a_1 a_2 \dots a_n}}$ for each $j = 1, 2, \dots, n$. Then we need to prove

$$b_1 + b_2 + \dots + b_n \geq n$$

under the constraint

$$b_1 b_2 \dots b_n = \frac{a_1 a_2 \dots a_n}{\left(\sqrt[n]{a_1 a_2 \dots a_n}\right)^n} = 1.$$

We shall prove this by induction. For the base case $n = 2$, we have $b_1 b_2 = 1$. Then $b_1 + b_2 \geq 2$ iff $(b_1 + b_2)^2 \geq 4 = 4b_1 b_2$, which is the same as the trivial inequality $(b_1 - b_2)^2 \geq 0$, with equality $b_1 = b_2$.

Suppose the result is true for the case $n - 1$. Consider $b_1 b_2 \dots b_n = 1$. It is not possible that all variables are larger than 1, or all variables are smaller than 1. WLOG assume $b_1 \geq 1 \geq b_2$. Then

$$(b_1 - 1)(1 - b_2) \geq 0,$$

which yields $b_1 + b_2 \geq 1 + b_1 b_2$. Then

$$b_1 + b_2 + \dots + b_n \geq 1 + b_1 b_2 + b_3 + b_4 + \dots + b_n.$$

Since the product of the $n - 1$ numbers $b_1 b_2, b_3, b_4, \dots, b_n$ is 1, we have

$$b_1 b_2 + b_3 + b_4 + \dots + b_n \geq n - 1$$

by the inductive hypothesis. Therefore, $b_1 + b_2 + \cdots + b_n \geq n$ as desired. Equality holds iff $b_1 = 1$ or $b_2 = 1$, together with $b_1 b_2 = b_3 = b_4 = \cdots = b_n = 1$. In any case, all b_i 's have to be equal to 1. This means all a_i 's are equal. $\square$

Indeed, there is a generalization of the AM-GM inequality given below.

Theorem 2.2. (**Weighted AM-GM inequality** (加權平均不等式)) Let $n \geq 2$. For any positive real numbers $a_1, a_2, \dots, a_n, w_1, w_2, \dots, w_n$, we have

$$\frac{w_1 a_1 + w_2 a_2 + \cdots + w_n a_n}{w_1 + w_2 + \cdots + w_n} \geq (a_1^{w_1} a_2^{w_2} \cdots a_n^{w_n})^{\frac{1}{w_1 + w_2 + \cdots + w_n}}.$$

Equality holds iff $a_1 = a_2 = \cdots = a_n$.

The AM-GM inequality is the special case where $w_1 = w_2 = \cdots = w_n$. The result can be explained easily if $w_1, w_2, \dots, w_n$ are positive integers. Indeed, in that case the result can be obtained by applying the AM-GM inequality to the $w_1 + w_2 + \cdots + w_n$ terms $\underbrace{a_1, a_1, \dots, a_1}_{w_1 \text{ copies}}, \dots, \underbrace{a_n, a_n, \dots, a_n}_{w_n \text{ copies}}$ directly. The proof to the general case is given in section 5.

Problem 2.1. Let a, b, c be positive real numbers satisfying $a + 2b + 3c = 15$. Find the maximum value of $a^2 b^2 c$.

Analysis. The AM-GM inequality is useful when we need to relate the sum and the product of several terms. A direct application using the AM-GM inequality only enables us to find the maximum value of abc . To obtain two a 's and two b 's in the product, we should make a little change to the sum. A simple way is to break up a into $\frac{1}{2}a + \frac{1}{2}a$, etc. Note that we should break it into equal parts as equality only holds when all terms are equal.

Solution. By the AM-GM inequality, we have

$$a + 2b + 3c = \frac{1}{2}a + \frac{1}{2}a + b + b + 3c \geq 5 \sqrt[5]{\frac{3}{4} a^2 b^2 c}.$$

Hence, $a^2 b^2 c \leq 324$. Equality holds iff $\frac{1}{2}a = b = 3c$. Together with $a + 2b + 3c = 15$, this happens when $(a, b, c) = (6, 3, 1)$. Therefore, the maximum value of $a^2 b^2 c$ is 324. $\square$

Remarks. Note that the same can be done by using the weighted AM-GM inequality.

For a problem asking for the optimum value, one needs to prove that the bound (e.g. 324 in this example) is true, as well as the equality can hold. Otherwise, although it might be obvious that $a^2 b^2 c \leq (15)^5$, the equality cannot hold and we cannot say that $(15)^5$ is the maximum value.

Problem 2.2. Let a, b, c, d be positive real numbers satisfying $3a^2 + b + c + d^2 = 10$. Find the maximum value of a^3b^2cd .

Analysis. Similar techniques can be applied to this problem. But this time we should consider a^2, b, c, d^2 as variables. For simplicity, we may first square the term.

Solution. It suffices to find the maximum value of $a^6b^4c^2d^2 = (a^3b^2cd)^2$. So we need to generate three copies of a^2 , four copies of b , two copies of c , and one copy of d^2 . By the weighted AM-GM inequality, we have

$$\begin{aligned} 3a^2 + b + c + d^2 &= 3(a^2) + 4\left(\frac{1}{4}b\right) + 2\left(\frac{1}{2}c\right) + (d^2) \\ &\geq (3 + 4 + 2 + 1) \left((a^2)^3 \left(\frac{1}{4}b\right)^4 \left(\frac{1}{2}c\right)^2 (d^2) \right)^{\frac{1}{3+4+2+1}} \\ &= 10 \sqrt[10]{\frac{1}{2^{10}} a^6 b^4 c^2 d^2} = 5 \sqrt[5]{a^3 b^2 c d}. \end{aligned}$$

As it is given that $3a^2 + b + c + d^2 = 10$, this yields $a^3b^2cd \leq 32$. Equality holds iff $a^2 = \frac{1}{4}b = \frac{1}{2}c = d^2$. Also, from the condition $3a^2 + b + c + d^2 = 10$, equality holds iff $(a, b, c, d) = (1, 4, 2, 1)$. Hence, the maximum value of a^3b^2cd is 32. $\square$

Problem 2.3. (IMO HK TST 2007 Test 1 Problem 3) Let ABC be a triangle with $AB = c$, $BC = a$ and $CA = b$. Show that

$$\frac{\cos A}{a^3} + \frac{\cos B}{b^3} + \frac{\cos C}{c^3} \geq \frac{3}{2abc}.$$

Analysis. The first step is to convert all cosines to the variables a, b, c . Afterwards, we have to prove an inequality in the positive variables a, b, c . We shall demonstrate how a simple use of the AM-GM inequality can solve this selection test problem easily.

Solution. By the cosine formula, we have

$$\begin{aligned} &\frac{\cos A}{a^3} + \frac{\cos B}{b^3} + \frac{\cos C}{c^3} \geq \frac{3}{2abc} \\ \Leftrightarrow &\frac{b^2 + c^2 - a^2}{2a^3bc} + \frac{c^2 + a^2 - b^2}{2ab^3c} + \frac{a^2 + b^2 - c^2}{2abc^3} \geq \frac{3}{2abc} \\ \Leftrightarrow &\frac{b^2 + c^2 - a^2}{a^2} + \frac{c^2 + a^2 - b^2}{b^2} + \frac{a^2 + b^2 - c^2}{c^2} \geq 3 \\ \Leftrightarrow &\frac{b^2}{a^2} + \frac{c^2}{a^2} + \frac{c^2}{b^2} + \frac{a^2}{b^2} + \frac{a^2}{c^2} + \frac{b^2}{c^2} \geq 6. \end{aligned}$$

This is true by applying the AM-GM inequality directly to the 6 positive terms on the left. Equality holds iff $a = b = c$. $\square$

Remarks. In most cases where the variables involved are sides of a triangle, we need to use the triangle inequality to get further information. However, this is not needed in this problem.

Problem 2.4. Let a, b, c be positive real numbers. Prove that

$$a^5b^3 + b^5c^3 + c^5a^3 \geq a^4b^3c + b^4c^3a + c^4a^3b.$$

Analysis. The inequality has a cyclic symmetry. Therefore, we may first see how a particular term such as a^4b^3c can be obtained from some of a^5b^3 , b^5c^3 and c^5a^3 . In other words, we aim at proving

$$\alpha(a^5b^3) + \beta(b^5c^3) + \gamma(c^5a^3) \geq a^4b^3c$$

for some appropriate choices of α, β, γ . If this can be done, we can sum up similar inequalities to obtain

$$(\alpha + \beta + \gamma)(a^5b^3 + b^5c^3 + c^5a^3) \geq a^4b^3c + b^4c^3a + c^4a^3b.$$

If we further have $\alpha + \beta + \gamma = 1$, then we are done. Other than trial and error, we shall provide a more general method below to deduce the values of α, β, γ .

Solution. By the weighted AM-GM inequality, we have

$$\frac{\alpha(a^5b^3) + \beta(b^5c^3) + \gamma(c^5a^3)}{\alpha + \beta + \gamma} \geq (a^{5\alpha+3\gamma}b^{3\alpha+5\beta}c^{3\beta+5\gamma})^{\frac{1}{\alpha+\beta+\gamma}}$$

for any nonnegative real numbers α, β, γ (which are not all zero). We would like the term on the right to be a^4b^3c . This means we need

$$\begin{cases} 5\alpha + 3\gamma = 4\alpha + 4\beta + 4\gamma, \\ 3\alpha + 5\beta = 3\alpha + 3\beta + 3\gamma, \\ 3\beta + 5\gamma = \alpha + \beta + \gamma. \end{cases}$$

Equivalently, we need

$$\begin{cases} \alpha - 4\beta - \gamma = 0, \\ 2\beta - 3\gamma = 0, \\ -\alpha + 2\beta + 4\gamma = 0. \end{cases}$$

It is not hard to find a solution $(\alpha, \beta, \gamma) = (14, 3, 2)$ (where the general solution is $(\alpha, \beta, \gamma) = (14t, 3t, 2t)$). In other words, we can apply the weighted AM-GM inequality as follows.

$$\frac{14a^5b^3 + 3b^5c^3 + 2c^5a^3}{19} \geq \sqrt[19]{a^{76}b^{57}c^{19}} = a^4b^3c.$$

Summing up similar inequalities, we obtain the desired inequality

$$a^5b^3 + b^5c^3 + c^5a^3 \geq a^4b^3c + b^4c^3a + c^4a^3b.$$

Equality holds when $a = b = c$. □

Remarks. The above simultaneous equation must have nontrivial solutions (solutions for which the variables are not all zero). However, it may not necessarily have a nonnegative solution. In that case, we cannot prove the inequality in this way.

Problem 2.5. Prove that

$$\frac{a}{b+c} + \frac{b}{c+d} + \frac{c}{d+a} + \frac{d}{a+b} \geq 2$$

for any positive real numbers a, b, c, d .

Analysis. For a cyclic inequality in four variables, a usual way to retain its symmetry is to group the first and the third terms together, and the other two terms together. Afterwards, in order to avoid doing expansion, we may try to apply the AM-GM inequality so as to make the denominator the same.

Solution. Note that

$$\begin{aligned} & \frac{a}{b+c} + \frac{b}{c+d} + \frac{c}{d+a} + \frac{d}{a+b} \geq 2 \\ \Leftrightarrow & \left(\frac{a}{b+c} + \frac{c}{d+a} \right) + \left(\frac{b}{c+d} + \frac{d}{a+b} \right) \geq 2 \\ \Leftrightarrow & \frac{a^2 + c^2 + ad + bc}{(b+c)(d+a)} + \frac{b^2 + d^2 + ab + cd}{(c+d)(a+b)} \geq 2. \end{aligned}$$

By the AM-GM inequality, we have $(b+c)(d+a) \leq \left(\frac{b+c+d+a}{2} \right)^2$. In a similar manner, $(c+d)(a+b) \leq \left(\frac{c+d+a+b}{2} \right)^2$. Therefore, it suffices to prove (note that inequality signs are reversed when the products are put in the denominators)

$$\frac{a^2 + c^2 + ad + bc}{\left(\frac{b+c+d+a}{2} \right)^2} + \frac{b^2 + d^2 + ab + cd}{\left(\frac{c+d+a+b}{2} \right)^2} \geq 2.$$

This is equivalent to $2(a^2 + c^2 + ad + bc + b^2 + d^2 + ab + cd) \geq (a+b+c+d)^2$, i.e.

$$a^2 + b^2 + c^2 + d^2 \geq 2ac + 2bd.$$

This is obvious as $a^2 + c^2 \geq 2ac$ and $b^2 + d^2 \geq 2bd$ by the AM-GM inequality or by completing squares. Equality holds iff $a = c$ and $b = d$. $\square$

Remarks. Note that we will not apply the AM-GM inequality like

$$(b+c)(d+a) \geq (2\sqrt{bc})(2\sqrt{da}) = 4\sqrt{abcd}$$

to equate the two denominators, since the sign of the desired inequality is reversed. One should take care of the inequality sign to see how to use some inequalities.

Problem 2.6. Prove that

$$\frac{1}{1+a^2} + \frac{1}{1+b^2} + \frac{1}{1+c^2} + \frac{1}{1+d^2} \geq 2$$

for any nonnegative real numbers a, b, c, d satisfying $a + b + c + d = 4$.

Analysis. If we try to apply the AM-GM inequality to simplify the denominators, we will get something like $\frac{1}{1+a^2} \leq \frac{1}{2a}$. However, the inequality sign is reversed. A useful way to get rid of this is to impose a negative sign before the expression, for example, by writing $\frac{1}{1+a^2}$ as $x - \frac{y}{1+a^2}$ for some suitable choices of x and y . Then it is at least $x - \frac{y}{2a}$, and the inequality sign is correct this time. Also, if we try to eliminate the term a in the denominator, then y should probably be chosen such that y has a factor a . This suggests the choice in the solution below.

Solution. Note that

$$\frac{1}{1+a^2} = 1 - \frac{a^2}{1+a^2} \geq 1 - \frac{a^2}{2a} = 1 - \frac{a}{2}$$

by the AM-GM inequality. Summing up similar inequalities, we get

$$\frac{1}{1+a^2} + \frac{1}{1+b^2} + \frac{1}{1+c^2} + \frac{1}{1+d^2} \geq 4 - \frac{a}{2} - \frac{b}{2} - \frac{c}{2} - \frac{d}{2} = 2$$

as desired. Equality holds iff $a = b = c = d = 1$. □

Remarks. When we apply the AM-GM inequality (or any other inequalities) in the denominator, we must ensure that the numerator is nonnegative. Otherwise, we cannot tell whether there is a sign change or not. For example, the term a^2 in $-\frac{a^2}{1+a^2}$ is nonnegative in the above solution. On the other hand,

$$\frac{1-2a}{a^2+1} \leq \frac{1-2a}{2a}$$

is not always true for $a \in \mathbb{R}^+$. For example, it does not hold when $a = 2$.

Problem 2.7. (IMO Shortlist 1998 A3) Let x, y and z be positive real numbers such that $xyz = 1$. Prove that

$$\frac{x^3}{(1+y)(1+z)} + \frac{y^3}{(1+z)(1+x)} + \frac{z^3}{(1+x)(1+y)} \geq \frac{3}{4}.$$

Analysis. To clear the denominators, one may try to apply the AM-GM inequality as follows:

$$\frac{x^3}{(1+y)(1+z)} + (1+y) + (1+z) \geq 3x.$$

However, we will not obtain the desired result. One simple reason is that the equality does not hold. Indeed, it is obvious that $x = y = z = 1$ is an equality case. In that case, the terms $\frac{x^3}{(1+y)(1+z)}$, $1+y$, $1+z$ are not all equal, which should be the case under the equality of the AM-GM inequality. Therefore, we make a more suitable choice in the solution by adding a coefficient.

Solution. By the AM-GM inequality, we have

$$\frac{x^3}{(1+y)(1+z)} + \frac{1+y}{8} + \frac{1+z}{8} \geq \frac{3}{4}x.$$

Adding similar inequalities, it suffices to prove

$$\frac{3}{4}(x+y+z) \geq \frac{3}{4} + \frac{1+x}{4} + \frac{1+y}{4} + \frac{1+z}{4},$$

i.e. $x+y+z \geq 3$. This is obvious as $x+y+z \geq 3\sqrt[3]{xyz} = 3$ by the AM-GM inequality. Equality holds iff $x = y = z = 1$. $\square$

Remarks. This problem suggests looking for the equality cases is important when applying the AM-GM inequality (and also any other inequalities).

Next, we shall go through a generalization of the AM-GM inequality.

Theorem 2.3. (**Power mean inequality**(幂平均不等式)) Let $n \geq 2$. Let $a_1, a_2, \dots, a_n$ be positive real numbers, and let $r < s$ be real numbers. Define

$$M^t(a_1, a_2, \dots, a_n) = \begin{cases} \left(\frac{a_1^t + a_2^t + \dots + a_n^t}{n} \right)^{\frac{1}{t}} & \text{if } t \neq 0, \\ \sqrt[n]{a_1 a_2 \dots a_n} & \text{if } t = 0. \end{cases}$$

Then $M^r(a_1, a_2, \dots, a_n) \leq M^s(a_1, a_2, \dots, a_n)$. Equality holds iff $a_1 = a_2 = \dots = a_n$.

The special value of $M^t(a_1, a_2, \dots, a_n)$ assigned to $t = 0$ is done by taking limit. Indeed, consider $y = \left(\frac{a_1^t + a_2^t + \dots + a_n^t}{n} \right)^{\frac{1}{t}}$ so that $\ln y = \frac{1}{t} \ln \left(\frac{a_1^t + a_2^t + \dots + a_n^t}{n} \right)$. By L'Hôpital's rule (see section 5),

$$\begin{aligned} \lim_{t \rightarrow 0} \ln y &= \lim_{t \rightarrow 0} \frac{n}{a_1^t + a_2^t + \dots + a_n^t} \cdot \frac{a_1^t \ln a_1 + a_2^t \ln a_2 + \dots + a_n^t \ln a_n}{n} \\ &= \frac{\ln a_1 + \ln a_2 + \dots + \ln a_n}{n} \\ &= \ln (a_1 a_2 \dots a_n)^{\frac{1}{n}}. \end{aligned}$$

Then $\lim_{t \rightarrow 0} y = \sqrt[n]{a_1 a_2 \dots a_n}$ as desired.

Under this definition, note that $t = 1$ and $t = 0$ correspond to the arithmetic mean and the geometric mean of $a_1, a_2, \dots, a_n$ respectively. So the power mean inequality is a generalization of the AM-GM inequality. In addition, the case $t = -1$ corresponds to the *harmonic mean*(調和平均) of the variables.

We also have a further generalization given below.

Theorem 2.4. (Weighted power mean inequality(加權幂平均不等式)) Let $n \geq 2$. Let $a_1, a_2, \dots, a_n, w_1, w_2, \dots, w_n$ be positive real numbers, and let $r < s$ be real numbers. Define

$$M_{\mathbf{w}}^t(a_1, a_2, \dots, a_n) = \begin{cases} \left(\frac{w_1 a_1^t + w_2 a_2^t + \dots + w_n a_n^t}{w_1 + w_2 + \dots + w_n} \right)^{\frac{1}{t}} & \text{if } t \neq 0, \\ (a_1^{w_1} a_2^{w_2} \dots a_n^{w_n})^{\frac{1}{w_1 + w_2 + \dots + w_n}} & \text{if } t = 0. \end{cases}$$

Then $M_{\mathbf{w}}^r(a_1, a_2, \dots, a_n) \leq M_{\mathbf{w}}^s(a_1, a_2, \dots, a_n)$. Equality holds iff $a_1 = a_2 = \dots = a_n$.

This is reduced to the power mean inequality when $w_1 = w_2 = \dots = w_n$. The proof of this result will be postponed to subsection 5.2.

Problem 2.8. Let x, y, z be real numbers satisfying $x^8 + y^8 + z^8 = 3$. Prove that

$$x^3 + y^3 + z^3 + 3xyz \leq 6.$$

Analysis. Since the condition and the inequality both involve high powers of the variables, the power mean inequality should be a good choice. Note that the power mean inequality only applies to positive real numbers, so we have to do a little bit trick.

Solution. By the power mean inequality, we have

$$\left(\frac{x^3 + y^3 + z^3}{3} \right)^{\frac{1}{3}} \leq \left(\frac{|x|^3 + |y|^3 + |z|^3}{3} \right)^{\frac{1}{3}} \leq \left(\frac{x^8 + y^8 + z^8}{3} \right)^{\frac{1}{8}} = 1.$$

This yields $x^3 + y^3 + z^3 \leq 3$. Therefore, it suffices to prove $xyz \leq 1$. Indeed, by the AM-GM inequality, we have

$$3 = x^8 + y^8 + z^8 \geq 3\sqrt[3]{x^8 y^8 z^8}.$$

Hence, $(xyz)^8 \leq 1$, i.e. $-1 \leq xyz \leq 1$ as desired. Equality holds iff $x = y = z = 1$. $\square$

Problem 2.9. Find all positive real numbers k such that

$$\frac{a^k + b^k}{a + b} + \frac{b^k + c^k}{b + c} + \frac{c^k + a^k}{c + a} \geq 3$$

for any positive real numbers a, b, c satisfying $abc = 1$.

Analysis. We can apply the power mean inequality to compare the k th power sum $a^k + b^k$ with the sum $a + b$. In view of the inequality sign, we can only do so for $k > 1$. The case $k = 1$ yields an identity, while for $0 < k < 1$, we should probably find some counterexamples to disprove the inequality.

Solution. For $k > 1$, by the power mean inequality, we have

$$\left(\frac{a^k + b^k}{2} \right)^{\frac{1}{k}} \geq \frac{a + b}{2}.$$

This gives

$$\frac{a^k + b^k}{a + b} \geq \left(\frac{a + b}{2} \right)^{k-1}.$$

Therefore,

$$\begin{aligned} \frac{a^k + b^k}{a + b} + \frac{b^k + c^k}{b + c} + \frac{c^k + a^k}{c + a} &\geq \left(\frac{a + b}{2} \right)^{k-1} + \left(\frac{b + c}{2} \right)^{k-1} + \left(\frac{c + a}{2} \right)^{k-1} \\ &\geq (\sqrt{ab})^{k-1} + (\sqrt{bc})^{k-1} + (\sqrt{ca})^{k-1} \\ &\geq 3 \sqrt[3]{(ab)^{\frac{k-1}{2}} (bc)^{\frac{k-1}{2}} (ca)^{\frac{k-1}{2}}} \\ &= 3 \end{aligned}$$

by successively applying the AM-GM inequality. Equality holds iff $a = b = c = 1$.

For $k = 1$, the inequality is just an identity. So this is always true, and equality holds for any a, b, c satisfying the condition.

For $0 < k < 1$, to construct a counterexample, we start with something simple. For example, we consider a triple $(a, b, c) = (t, t, t^{-2})$ having two equal terms, where $t > 0$. They clearly satisfy $abc = 1$. We compute

$$\begin{aligned} \frac{a^k + b^k}{a + b} + \frac{b^k + c^k}{b + c} + \frac{c^k + a^k}{c + a} &= \frac{2t^k}{2t} + 2 \cdot \frac{t^k + t^{-2k}}{t + t^{-2}} \\ &= \frac{1}{t^{1-k}} + 2 \cdot \frac{1 + t^{-3k}}{t^{1-k} + t^{-k-2}}. \end{aligned}$$

When t tends to infinity, this expression tends to zero. In particular, it cannot be always greater than or equal to 3.

Therefore, the answer is $k \geq 1$. □

Remarks. It is always useful to consider some extreme cases like $t \rightarrow \infty$ and $t \rightarrow 0^+$. More examples will be given in sections 5 and 6.

2.2 Cauchy-Schwarz Inequality and Hölder's Inequality

Theorem 2.5. (Cauchy-Schwarz inequality(柯西不等式)) Let $n \geq 2$. For any real numbers $a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n$, we have

$$(a_1^2 + a_2^2 + \dots + a_n^2)(b_1^2 + b_2^2 + \dots + b_n^2) \geq (a_1b_1 + a_2b_2 + \dots + a_nb_n)^2.$$

Equality holds iff $a_ib_j = a_jb_i$ for any $i \neq j$.

Proof. The assertion is obvious when $a_1 = a_2 = \dots = a_n = 0$. Otherwise, consider

$$f(x) = (a_1x - b_1)^2 + (a_2x - b_2)^2 + \dots + (a_nx - b_n)^2.$$

Clearly, $f(x) \geq 0$ for any $x \in \mathbb{R}$. Rewrite f as a quadratic polynomial

$$f(x) = (a_1^2 + a_2^2 + \dots + a_n^2)x^2 - 2(a_1b_1 + a_2b_2 + \dots + a_nb_n)x + (b_1^2 + b_2^2 + \dots + b_n^2).$$

The leading coefficient is nonzero (hence positive) by assumption. Therefore, by proposition 1.4, we can have $f(x) \geq 0$ for all $x \in \mathbb{R}$ iff $\Delta \leq 0$. In other words,

$$(-2(a_1b_1 + a_2b_2 + \dots + a_nb_n))^2 - 4(a_1^2 + a_2^2 + \dots + a_n^2)(b_1^2 + b_2^2 + \dots + b_n^2) \leq 0.$$

This is simply the desired inequality. Equality holds iff $\Delta = 0$, in which case $f(x) = 0$ for some particular x . Therefore, we must have $a_1x - b_1 = a_2x - b_2 = \dots = a_nx - b_n = 0$. It is not hard to see that this means $a_ib_j = a_jb_i$ for any $i \neq j$. $\square$

Note that the equality case can be interpreted as $a_1 : a_2 : \dots : a_n = b_1 : b_2 : \dots : b_n$. But in this notation, we use the convention that $a_j = 0$ iff $b_j = 0$ for any $1 \leq j \leq n$. The advantage of using the Cauchy-Schwarz inequality over the AM-GM inequality is that the former can be applied to any real numbers while the latter is restricted to positive (or nonnegative) real numbers. So there is a wider range of application of the Cauchy-Schwarz inequality.

Before demonstrating the use of the Cauchy-Schwarz inequality, we first state a few useful corollaries in the following proposition.

Proposition 2.1. Let $n \geq 2$, and let $a, a_1, a_2, \dots, a_n, b, b_1, b_2, \dots, b_n$ be positive real numbers. Then the following hold.

- (a) $\frac{a_1}{b_1} + \frac{a_2}{b_2} + \dots + \frac{a_n}{b_n} \geq \frac{(a_1 + a_2 + \dots + a_n)^2}{a_1b_1 + a_2b_2 + \dots + a_nb_n}$
- (b) $\frac{a_1^2}{b_1} + \frac{a_2^2}{b_2} + \dots + \frac{a_n^2}{b_n} \geq \frac{(a_1 + a_2 + \dots + a_n)^2}{b_1 + b_2 + \dots + b_n}$
- (c) $n(a_1^2 + a_2^2 + \dots + a_n^2) \geq (a_1 + a_2 + \dots + a_n)^2$
- (d) $\sqrt{n(a_1 + a_2 + \dots + a_n)} \geq \sqrt{a_1} + \sqrt{a_2} + \dots + \sqrt{a_n}$

$$(e) \sqrt{(a^2 + 1)(b^2 + 1)} \geq ab + 1$$

$$(f) \sqrt{(a^2 + 1)(b^2 + 1)} \geq a + b$$

$$(g) \sqrt{a_1^2 + b_1^2} + \sqrt{a_2^2 + b_2^2} + \cdots + \sqrt{a_n^2 + b_n^2} \geq \sqrt{(a_1 + a_2 + \cdots + a_n)^2 + (b_1 + b_2 + \cdots + b_n)^2}$$

Proof.

(a) We apply the Cauchy-Schwarz inequality as follows.

$$\begin{aligned} & (a_1b_1 + a_2b_2 + \cdots + a_nb_n) \left(\frac{a_1}{b_1} + \frac{a_2}{b_2} + \cdots + \frac{a_n}{b_n} \right) \\ &= (\sqrt{a_1b_1}^2 + \sqrt{a_2b_2}^2 + \cdots + \sqrt{a_nb_n}^2) \left(\sqrt{\frac{a_1}{b_1}}^2 + \sqrt{\frac{a_2}{b_2}}^2 + \cdots + \sqrt{\frac{a_n}{b_n}}^2 \right) \\ &\geq (a_1 + a_2 + \cdots + a_n)^2 \end{aligned}$$

Dividing both sides by $a_1b_1 + a_2b_2 + \cdots + a_nb_n$ yields (a).

(b) Similar to (a), we apply the Cauchy-Schwarz inequality to get

$$\begin{aligned} & (b_1 + b_2 + \cdots + b_n) \left(\frac{a_1^2}{b_1} + \frac{a_2^2}{b_2} + \cdots + \frac{a_n^2}{b_n} \right) \\ &= (\sqrt{b_1}^2 + \sqrt{b_2}^2 + \cdots + \sqrt{b_n}^2) \left(\sqrt{\frac{a_1^2}{b_1}}^2 + \sqrt{\frac{a_2^2}{b_2}}^2 + \cdots + \sqrt{\frac{a_n^2}{b_n}}^2 \right) \\ &\geq (a_1 + a_2 + \cdots + a_n)^2. \end{aligned}$$

This yields (b) by dividing both sides by $b_1 + b_2 + \cdots + b_n$.

(c) This can be proved by the power mean inequality. For a proof using the Cauchy-Schwarz inequality, we just need to note that

$$\begin{aligned} n(a_1^2 + a_2^2 + \cdots + a_n^2) &= (1^2 + 1^2 + \cdots + 1^2)(a_1^2 + a_2^2 + \cdots + a_n^2) \\ &\geq (a_1 + a_2 + \cdots + a_n)^2. \end{aligned}$$

(d) Next, (d) is a simple consequence of (c). Indeed, we just need to replace each a_i by $\sqrt{a_i}$ in (c) and take the square root from both sides.

(e) Using the Cauchy-Schwarz inequality, we have

$$(a^2 + 1)(b^2 + 1) = (a^2 + 1^2)(b^2 + 1^2) \geq (ab + 1)^2,$$

which yields (e).

(f) Similarly,

$$(a^2 + 1)(b^2 + 1) = (a^2 + 1^2)(1^2 + b^2) \geq (a + b)^2.$$

This gives (f).

(g) Lastly, we shall prove (g) by induction on n . When $n = 2$, we have

$$\begin{aligned} & \sqrt{a_1^2 + b_1^2} + \sqrt{a_2^2 + b_2^2} \geq \sqrt{(a_1 + a_2)^2 + (b_1 + b_2)^2} \\ \Leftrightarrow & a_1^2 + b_1^2 + a_2^2 + b_2^2 + 2\sqrt{(a_1^2 + b_1^2)(a_2^2 + b_2^2)} \geq (a_1 + a_2)^2 + (b_1 + b_2)^2 \\ \Leftrightarrow & (a_1^2 + b_1^2)(a_2^2 + b_2^2) \geq (a_1 a_2 + b_1 b_2)^2, \end{aligned}$$

which is true by the Cauchy-Schwarz inequality.

Suppose the inequality is true for any $2 \leq n < k$ for some $k \geq 2$. When $n = k$, we use the inductive hypothesis to get

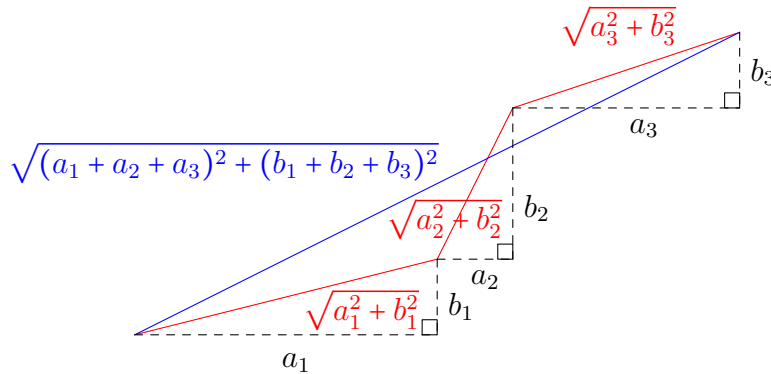
$$\begin{aligned} & \sqrt{a_1^2 + b_1^2} + \sqrt{a_2^2 + b_2^2} + \cdots + \sqrt{a_{n-1}^2 + b_{n-1}^2} + \sqrt{a_n^2 + b_n^2} \\ & \geq \sqrt{(a_1 + a_2 + \cdots + a_{n-1})^2 + (b_1 + b_2 + \cdots + b_{n-1})^2} + \sqrt{a_n^2 + b_n^2}. \end{aligned}$$

Note that this is exactly the same as the left-hand side of the case $n = 2$, with variables $(a_1 + a_2 + \cdots + a_{n-1})$, $(b_1 + b_2 + \cdots + b_{n-1})$, a_n, b_n . So we get

$$\begin{aligned} & \sqrt{(a_1 + a_2 + \cdots + a_{n-1})^2 + (b_1 + b_2 + \cdots + b_{n-1})^2} + \sqrt{a_n^2 + b_n^2} \\ & \geq \sqrt{(a_1 + a_2 + \cdots + a_n)^2 + (b_1 + b_2 + \cdots + b_n)^2}. \end{aligned}$$

Part (g) then follows by induction. □

Indeed, parts (c), (e), (f) and (g) hold for any real numbers. Note that part (g) is just the triangle inequality, as illustrated by the figure below (for the case $n = 3$).



Problem 2.10. Let a, b, c, d be real numbers satisfying

$$\begin{cases} a + b + c + d = 5, \\ a^2 + b^2 + c^2 + d^2 = 7. \end{cases}$$

Find the optimum values of a .

Analysis. This is a standard problem which can be solved by the Cauchy-Schwarz inequality. We just need to relate the terms a and a^2 .

Solution. By [proposition 2.1\(c\)](#), we get $3(b^2 + c^2 + d^2) \geq (b + c + d)^2$. Now,

$$\begin{aligned} & 3(b^2 + c^2 + d^2) \geq (b + c + d)^2 \\ \Rightarrow & 3(7 - a^2) \geq (5 - a)^2 \\ \Rightarrow & 2a^2 - 5a + 2 \leq 0 \\ \Rightarrow & \frac{1}{2} \leq a \leq 2. \end{aligned}$$

Note that equality holds iff $b = c = d$. The optimum values of a are $\frac{1}{2}$ and 2, which are attained when $(a, b, c, d) = \left(\frac{1}{2}, \frac{3}{2}, \frac{3}{2}, \frac{3}{2}\right)$ and $(a, b, c, d) = (2, 1, 1, 1)$ respectively. $\square$

Remarks. Due to symmetry, the values obtained are also the optimum values of b, c, d . This gives us one way to bound the variables.

Problem 2.11. Let x and y be real numbers such that $x, y \geq 1$. Prove that

$$\sqrt[4]{x^2 - 1} + \sqrt[4]{y^2 - 1} \leq \sqrt{2xy}.$$

Analysis. This is a standard problem involving radicals. We may try to apply various results given in [proposition 2.1](#) to get the desired result.

Solution. By [proposition 2.1](#), we have

$$\begin{aligned} \sqrt[4]{x^2 - 1} + \sqrt[4]{y^2 - 1} &= \sqrt{\sqrt{x^2 - 1}} + \sqrt{\sqrt{y^2 - 1}} \\ &\leq \left(2(\sqrt{x^2 - 1} + \sqrt{y^2 - 1})\right)^{\frac{1}{2}} \quad (\text{part (d)}) \\ &\leq \left(2\sqrt{(\sqrt{x^2 - 1}^2 + 1)(\sqrt{y^2 - 1}^2 + 1)}\right)^{\frac{1}{2}} \quad (\text{part (f)}) \\ &= \sqrt{2xy}. \end{aligned}$$

Equality holds iff $x = y = \sqrt{2}$. $\square$

Problem 2.12. (CGMO 2013 Problem 5) For any given positive real numbers $a_1, a_2, \dots, a_n$, prove that there exist positive real numbers $x_1, x_2, \dots, x_n$ satisfying $\sum_{i=1}^n x_i = 1$ such that for any positive real numbers $y_1, y_2, \dots, y_n$ with $\sum_{i=1}^n y_i = 1$, the inequality $\sum_{i=1}^n \frac{a_i x_i}{x_i + y_i} \geq \frac{1}{2} \sum_{i=1}^n a_i$ holds.

Analysis. To find $x_1, x_2, \dots, x_n$, we first consider $\sum_{i=1}^n \frac{a_i x_i}{x_i + y_i}$ as an inequality in the variables $y_1, y_2, \dots, y_n$ and find its lower bound. This can be done using the fraction expressions in [proposition 2.1](#). We will use part (b) instead of part (a) as the former yields a lower bound which does not involve any y_i 's.

Solution. By [proposition 2.1\(b\)](#), we have

$$\sum_{i=1}^n \frac{a_i x_i}{x_i + y_i} \geq \frac{(\sqrt{a_1 x_1} + \sqrt{a_2 x_2} + \dots + \sqrt{a_n x_n})^2}{x_1 + y_1 + x_2 + y_2 + \dots + x_n + y_n} = \frac{1}{2} (\sqrt{a_1 x_1} + \sqrt{a_2 x_2} + \dots + \sqrt{a_n x_n})^2.$$

This is indeed the minimum value of the expression (with fixed a_i, x_i) since the equality can hold. Hence, we need

$$(\sqrt{a_1 x_1} + \sqrt{a_2 x_2} + \dots + \sqrt{a_n x_n})^2 \geq a_1 + a_2 + \dots + a_n. \quad (1)$$

On the other hand, by the Cauchy-Schwarz inequality, we have

$$\begin{aligned} a_1 + a_2 + \dots + a_n &= (a_1 + a_2 + \dots + a_n)(x_1 + x_2 + \dots + x_n) \\ &= (\sqrt{a_1}^2 + \sqrt{a_2}^2 + \dots + \sqrt{a_n}^2)(\sqrt{x_1}^2 + \sqrt{x_2}^2 + \dots + \sqrt{x_n}^2) \\ &\geq (\sqrt{a_1 x_1} + \sqrt{a_2 x_2} + \dots + \sqrt{a_n x_n})^2. \end{aligned} \quad (2)$$

Therefore, in order that (1) holds, we must have equality for (2). In other words, $a_1 : a_2 : \dots : a_n = x_1 : x_2 : \dots : x_n$. Together with $x_1 + x_2 + \dots + x_n = 1$, the only choice is $x_j = \frac{a_j}{a_1 + a_2 + \dots + a_n}$ for each $j = 1, 2, \dots, n$.

Conversely, if $x_j = \frac{a_j}{a_1 + a_2 + \dots + a_n}$ for each j , the inequality $\sum_{i=1}^n \frac{a_i x_i}{x_i + y_i} \geq \frac{1}{2} \sum_{i=1}^n a_i$ holds

from above, with equality case $\frac{\sqrt{a_1 x_1}}{x_1 + y_1} = \frac{\sqrt{a_2 x_2}}{x_2 + y_2} = \dots = \frac{\sqrt{a_n x_n}}{x_n + y_n}$. In addition, by using

$x_j = \frac{a_j}{a_1 + a_2 + \dots + a_n}$ and $\sum_{i=1}^n y_i = 1$, this is equivalent to $y_j = \frac{a_j}{a_1 + a_2 + \dots + a_n}$ for each j . □

Problem 2.13. Let $\theta_1, \theta_2, \dots, \theta_n$ be real numbers satisfying $0 < \theta_i < \frac{\pi}{2}$ for each $i = 1, 2, \dots, n$ and such that $\tan \theta_1 + \tan \theta_2 + \dots + \tan \theta_n = n$. Prove that

$$\frac{1}{\cos \theta_1} + \frac{1}{\cos \theta_2} + \dots + \frac{1}{\cos \theta_n} \geq n\sqrt{2}.$$

Analysis. The difficulty of solving problems involving trigonometric functions is that there are implicit conditions imposed on the expressions, as those functions are related. One simple way is to use various identities in trigonometry. The one which relates $\tan \theta$ and $\frac{1}{\cos \theta} = \sec \theta$ is $\tan^2 \theta + 1 = \sec^2 \theta$.

Solution. Note that the left-hand side of the inequality is

$$\begin{aligned}\sec \theta_1 + \sec \theta_2 + \cdots + \sec \theta_n &= \sqrt{\sec^2 \theta_1} + \sqrt{\sec^2 \theta_2} + \cdots + \sqrt{\sec^2 \theta_n} \\ &= \sqrt{\tan^2 \theta_1 + 1} + \sqrt{\tan^2 \theta_2 + 1} + \cdots + \sqrt{\tan^2 \theta_n + 1}.\end{aligned}$$

Applying **proposition 2.1(g)** to the sequences $\tan \theta_1, \tan \theta_2, \dots, \tan \theta_n$ and $1, 1, \dots, 1$, we get

$$\begin{aligned}&\sqrt{\tan^2 \theta_1 + 1} + \sqrt{\tan^2 \theta_2 + 1} + \cdots + \sqrt{\tan^2 \theta_n + 1} \\ &\geq \sqrt{(\tan \theta_1 + \tan \theta_2 + \cdots + \tan \theta_n)^2 + (1 + 1 + \cdots + 1)^2} \\ &= n\sqrt{2}\end{aligned}$$

as desired. Equality holds iff $\theta_1 = \theta_2 = \cdots = \theta_n = \frac{\pi}{4}$. $\square$

Remarks. The same result can be easily obtained using Jensen's inequality introduced in subsection 5.2.

Problem 2.14. Let a, b, c be positive real numbers satisfying $\sqrt{ab} + \sqrt{bc} + \sqrt{ca} = 3$. Prove that

$$\sqrt{a+b+2} + \sqrt{b+c+2} + \sqrt{c+a+2} \geq 6.$$

Analysis. According to **proposition 2.1(d)**, it is easy to deduce that a sum of radicals does not exceed a certain expression. However, it cannot give us any lower bound. If we try to apply part (g), then we need to construct a sum of squares for each term. This might sometimes be tricky. Another way, as we shall use in the following solution, is to square both sides so that each radical consists of a product of two expressions, which looks like the Cauchy-Schwarz inequality.

Solution. Squaring both sides, it is the same as proving

$$\begin{aligned}&\sqrt{(a+b+2)(b+c+2)} + \sqrt{(b+c+2)(c+a+2)} + \sqrt{(c+a+2)(a+b+2)} \\ &\geq 15 - (a+b+c)\end{aligned}\tag{1}$$

after simplification. Now, for each term $\sqrt{(a+b+2)(b+c+2)}$, we apply the Cauchy-Schwarz inequality to get

$$\begin{aligned}\sqrt{(a+b+2)(b+c+2)} &= \sqrt{(\sqrt{a^2} + \sqrt{b^2} + 1^2 + 1^2)(\sqrt{b^2} + \sqrt{c^2} + 1^2 + 1^2)} \\ &\geq \sqrt{(\sqrt{ab} + \sqrt{bc} + 1 + 1)^2} \\ &= \sqrt{ab} + \sqrt{bc} + 2.\end{aligned}$$

Summing up similar inequalities, the left-hand side of (1) is at least

$$2(\sqrt{ab} + \sqrt{bc} + \sqrt{ca}) + 6 = 12.$$

Hence, it suffices to prove $a+b+c \geq 3$, which is the same as $a+b+c \geq \sqrt{ab} + \sqrt{bc} + \sqrt{ca}$. This is obvious from the identity

$$2(a+b+c) - 2(\sqrt{ab} + \sqrt{bc} + \sqrt{ca}) = (\sqrt{a} - \sqrt{b})^2 + (\sqrt{b} - \sqrt{c})^2 + (\sqrt{c} - \sqrt{a})^2.$$

So the inequality is true, with equality case $a = b = c = 1$. $\square$

Remarks. Once again, we break up the term 2 to 1+1 to apply the Cauchy-Schwarz inequality since we note that equality holds when $a = b = c = 1$. Actually, we can use the inequality as follows to get a better bound, namely,

$$\sqrt{(a+b+2)(b+c+2)} = \sqrt{(b+a+1+1)(b+c+1+1)} \geq b + \sqrt{ac} + 1 + 1.$$

Adding similar inequalities, the left-hand side of (1) is at least $a+b+c+9$ instead of 12.

Next, we shall give a generalization of the Cauchy-Schwarz inequality.

Theorem 2.6. (Hölder's inequality (赫爾德不等式)) For any positive real numbers $a_{1,1}, a_{1,2}, \dots, a_{1,n}, a_{2,1}, a_{2,2}, \dots, a_{2,n}, \dots, a_{m,1}, a_{m,2}, \dots, a_{m,n}$ and $w_1, w_2, \dots, w_m$ satisfying $w_1 + w_2 + \dots + w_m = 1$, we have

$$\begin{aligned} & (a_{1,1} + a_{1,2} + \dots + a_{1,n})^{w_1} (a_{2,1} + a_{2,2} + \dots + a_{2,n})^{w_2} \dots (a_{m,1} + a_{m,2} + \dots + a_{m,n})^{w_m} \\ & \geq a_{1,1}^{w_1} a_{2,1}^{w_2} \dots a_{m,1}^{w_m} + a_{1,2}^{w_1} a_{2,2}^{w_2} \dots a_{m,2}^{w_m} + \dots + a_{1,n}^{w_1} a_{2,n}^{w_2} \dots a_{m,n}^{w_m}. \end{aligned}$$

Equality holds iff $a_{1,1} : a_{1,2} : \dots : a_{1,n} = a_{2,1} : a_{2,2} : \dots : a_{2,n} = \dots = a_{m,1} : a_{m,2} : \dots : a_{m,n}$ or the trivial cases $m = 1$ or $n = 1$.

When $m = 2$ and $w_1 = w_2 = \frac{1}{2}$, we get the Cauchy-Schwarz inequality. Note that there are some forms of Hölder's inequality which are commonly used. For example,

$$\begin{aligned} & (a_1 + a_2 + \dots + a_n)^s (b_1 + b_2 + \dots + b_n)^{1-s} \geq a_1^s b_1^{1-s} + a_2^s b_2^{1-s} + \dots + a_n^s b_n^{1-s}, \\ & (a_1 + a_2 + a_3)(b_1 + b_2 + b_3)(c_1 + c_2 + c_3) \geq (\sqrt[3]{a_1 b_1 c_1} + \sqrt[3]{a_2 b_2 c_2} + \sqrt[3]{a_3 b_3 c_3})^3. \end{aligned}$$

A proof will be given in subsection 5.2 using Jensen's inequality.

Problem 2.15. Let a, b, c be positive real numbers. Prove that

$$(a^4 + a^2 + 1)(b^4 + b^2 + 1)(c^4 + c^2 + 1) \geq (ab + bc + ca)^3.$$

Analysis. The form of the inequality is quite similar to that of Hölder's inequality for $m = n = 3$. But directly applying the inequality would yield radicals. We should first construct cubic powers on the left-hand side.

Solution. By the AM-GM inequality, we have $a^4 + a^2 \geq 2a^3$. Hence, it suffices to prove

$$(a^3 + 1 + a^3)(b^3 + b^3 + 1)(1 + c^3 + c^3) \geq (ab + bc + ca)^3,$$

which is a direct consequence of Hölder's inequality. $\square$

Problem 2.16. For any positive real numbers a, b, c satisfying $a^2 + b^2 + c^2 = 1$, prove that

$$\frac{a}{(a+b)^2} + \frac{b}{(b+c)^2} + \frac{c}{(c+a)^2} > 1.$$

Analysis. Hölder's inequality, as a generalization of the Cauchy-Schwarz inequality, is also useful for clearing denominators. The usage is analogous to that in [proposition 2.1\(a\)](#). We can select some appropriate positive numbers s, x, y, z to deduce

$$(x+y+z)^s \left(\frac{a}{(a+b)^2} + \frac{b}{(b+c)^2} + \frac{c}{(c+a)^2} \right)^{1-s} \geq \frac{x^s a^{1-s}}{(a+b)^{2(1-s)}} + \frac{y^s b^{1-s}}{(b+c)^{2(1-s)}} + \frac{z^s c^{1-s}}{(c+a)^{2(1-s)}}.$$

Our goal is to select these numbers so that each expression $\frac{x^s a^{1-s}}{(a+b)^{2(1-s)}}$ is simple.

For example, we may set x to be a power of a times a power of $a+b$, so that the expression becomes a simple power of a under a simple choice of s . One possible choice is given in the solution below.

Solution. By Hölder's inequality, we have

$$(a(a+b) + b(b+c) + c(c+a))^{\frac{2}{3}} \left(\frac{a}{(a+b)^2} + \frac{b}{(b+c)^2} + \frac{c}{(c+a)^2} \right)^{\frac{1}{3}} \geq a+b+c.$$

This yields

$$\frac{a}{(a+b)^2} + \frac{b}{(b+c)^2} + \frac{c}{(c+a)^2} \geq \frac{(a+b+c)^3}{(a^2+b^2+c^2+ab+bc+ca)^2}.$$

It suffices to prove $(a+b+c)^3 > (a^2+b^2+c^2+ab+bc+ca)^2$. Let $a+b+c = s$ and $ab+bc+ca = t$. Then the condition is $s^2 - 2t = 1$, i.e. $t = \frac{1}{2}s^2 - \frac{1}{2}$. Now,

$$\begin{aligned} & (a+b+c)^3 > (a^2+b^2+c^2+ab+bc+ca)^2 \\ \Leftrightarrow & s^3 > (t+1)^2 \\ \Leftrightarrow & s^3 > \left(\frac{1}{2}s^2 + \frac{1}{2} \right)^2 \\ \Leftrightarrow & s^4 - 4s^3 + 2s^2 + 1 < 0 \\ \Leftrightarrow & (s-1)(s^3 - 3s^2 - s - 1) < 0. \end{aligned} \tag{1}$$

Note that $(a + b + c)^2 \geq 3(ab + bc + ca)$ implies $s^2 \geq 3t$. Putting $t = \frac{1}{2}s^2 - \frac{1}{2}$ into $\frac{1}{3}s^2 \geq t > 0$, we have $\sqrt{3} \geq s > 1$. It follows that

$$s^3 - 3s^2 - s - 1 = s^2(s - 3) - (s + 1) < 0$$

as $s - 3 < 0$ and $s + 1 > 0$. Together with $s - 1 > 0$, (1) clearly holds. This completes the proof. $\square$

Remarks. Consider $c = a^3$. The left-hand side of the inequality becomes

$$\frac{a}{(a+b)^2} + \frac{b}{(b+a^3)^2} + \frac{a}{(a^2+1)^2}.$$

When a (and hence c) tends to 0, we see from $a^2 + b^2 + c^2 = 1$ that b tends to 1. Then the left-hand side tends to 1. This shows 1 is the best bound, although the equality cannot hold.

2.3 The Rearrangement Inequality and Chebyshev's Inequality

Theorem 2.7. (**Rearrangement inequality**(排序不等式)) Let $n \geq 2$. For any real numbers $a_1 \geq a_2 \geq \dots \geq a_n$ and $b_1 \geq b_2 \geq \dots \geq b_n$, and for any permutation $(i_1, i_2, \dots, i_n)$ of $(1, 2, \dots, n)$, we have

$$a_1b_1 + a_2b_2 + \dots + a_nb_n \geq a_1b_{i_1} + a_2b_{i_2} + \dots + a_nb_{i_n} \geq a_1b_n + a_2b_{n-1} + \dots + a_nb_1.$$

Proof. We shall prove the assertion by induction on n .

When $n = 2$, we only need to prove $a_1b_1 + a_2b_2 \geq a_1b_2 + a_2b_1$. This is the same as

$$(a_1 - a_2)(b_1 - b_2) \geq 0,$$

which is obvious from the conditions.

Assume all cases $n \leq k$ are true. When $n = k + 1$, as both parts are similar, we shall only prove

$$a_1b_1 + a_2b_2 + \dots + a_{k+1}b_{k+1} \geq a_1b_{i_1} + a_2b_{i_2} + \dots + a_{k+1}b_{i_{k+1}}$$

for any permutation $(i_1, i_2, \dots, i_{k+1})$ of $(1, 2, \dots, k + 1)$. If $i_{k+1} = k + 1$, then we can eliminate the term $a_{k+1}b_{k+1}$ on both sides so that the result follows readily by the inductive hypothesis. If $i_{k+1} = t \neq k + 1$, let $i_s = k + 1$ for some $s \leq k$. Note that

$$a_sb_t + a_{k+1}b_{k+1} \geq a_sb_{k+1} + a_{k+1}b_t = a_sb_{i_s} + a_{k+1}b_{i_{k+1}}$$

since $a_s \geq a_{k+1}$ and $b_t \geq b_{k+1}$. Therefore, it remains to prove

$$a_1b_1 + a_2b_2 + \dots + a_kb_k \geq a_sb_t + \sum_{\substack{j=1 \\ j \neq s}}^k a_jb_{i_j}.$$

If we define $j_\ell = i_\ell$ for $\ell \neq s$ and $j_s = t$, then we need to prove

$$a_1b_1 + a_2b_2 + \cdots + a_kb_k \geq a_1b_{j_1} + a_2b_{j_2} + \cdots + a_kb_{j_k}.$$

Clearly, $(j_1, j_2, \dots, j_k)$ is a permutation of $(1, 2, \dots, k)$, so this is true by the inductive hypothesis. $\square$

The sums $a_1b_1 + a_2b_2 + \cdots + a_nb_n$ and $a_1b_n + a_2b_{n-1} + \cdots + a_nb_1$ are called the *direct sum*(順序和) and *reverse sum*(逆序和) respectively, as suggested from the order of the indices. The sum $a_1b_{i_1} + a_2b_{i_2} + \cdots + a_nb_{i_n}$ is called a *random sum*(亂序和).

Since the expressions in the rearrangement inequality are not symmetric, it does not have much application in ordinary inequalities. Sometimes it can be used to simplify the arguments as we can impose a certain order to the variables (or otherwise the inequality is weaker in view of the rearrangement inequality). On the other hand, there is a more friendly formulation of the rearrangement inequality given below.

Theorem 2.8. (*Chebyshev's inequality*(契比雪夫不等式)) Let $n \geq 2$. For any real numbers $a_1 \geq a_2 \geq \cdots \geq a_n$ and $b_1 \geq b_2 \geq \cdots \geq b_n$, we have

$$a_1b_1 + a_2b_2 + \cdots + a_nb_n \geq \frac{(a_1 + a_2 + \cdots + a_n)(b_1 + b_2 + \cdots + b_n)}{n}.$$

Equality holds iff $a_1 = a_2 = \cdots = a_n$ or $b_1 = b_2 = \cdots = b_n$.

It is also common to call this result *Chebyshev's sum inequality* (契比雪夫總和不等式) to avoid any confusion with the more famous Chebyshev's inequality in probability theory. This can be proved by summing up all possible permutations $(i_1, i_2, \dots, i_n)$ of the indices $(1, 2, \dots, n)$ in the rearrangement inequality. Alternatively, this follows from the generalized Chebyshev's inequality given at the end of this subsection. In that theorem, we provide an alternative proof which easily yields the equality case.

If $a_1 \geq a_2 \geq \cdots \geq a_n$ and $b_1 \leq b_2 \leq \cdots \leq b_n$, then $-b_n \geq -b_{n-1} \geq \cdots \geq -b_1$. Therefore, by Chebyshev's inequality, we obtain

$$a_1b_1 + a_2b_2 + \cdots + a_nb_n \leq \frac{(a_1 + a_2 + \cdots + a_n)(b_1 + b_2 + \cdots + b_n)}{n}.$$

In other words, the inequality sign is reversed in this case. Indeed, note that the left-hand side of this inequality is essentially the reverse sum of the two sequences.

Problem 2.17. Let a, b, c, d be positive real numbers such that $a^2 + b^2 + c^2 + d^2 = 4$. Prove that

$$\frac{a^2}{b+c+d} + \frac{b^2}{c+d+a} + \frac{c^2}{d+a+b} + \frac{d^2}{a+b+c} \geq \frac{4}{3}.$$

Analysis. Chebyshev's inequality is useful when the inequality is of the form

$$\frac{a_1}{b_1} + \frac{a_2}{b_2} + \cdots + \frac{a_n}{b_n} \geq C$$

where C is a constant. The following solution provides a standard method in proving such an inequality. Note that we can always rearrange the variables in any order if it is a symmetric expression.

Solution. Due to symmetry, we may assume $a \geq b \geq c \geq d$. Then $a^2 \geq b^2 \geq c^2 \geq d^2$ and $\frac{1}{b+c+d} \geq \frac{1}{c+d+a} \geq \frac{1}{d+a+b} \geq \frac{1}{a+b+c}$. By Chebyshev's inequality, we have

$$\begin{aligned} & \frac{a^2}{b+c+d} + \frac{b^2}{c+d+a} + \frac{c^2}{d+a+b} + \frac{d^2}{a+b+c} \\ & \geq \frac{1}{4}(a^2 + b^2 + c^2 + d^2) \left(\frac{1}{b+c+d} + \frac{1}{c+d+a} + \frac{1}{d+a+b} + \frac{1}{a+b+c} \right). \end{aligned}$$

It suffices to prove

$$\frac{1}{b+c+d} + \frac{1}{c+d+a} + \frac{1}{d+a+b} + \frac{1}{a+b+c} \geq \frac{4}{3}.$$

By [proposition 2.1\(a\)](#), we have

$$\begin{aligned} & \frac{1}{b+c+d} + \frac{1}{c+d+a} + \frac{1}{d+a+b} + \frac{1}{a+b+c} \\ & \geq \frac{(1+1+1+1)^2}{(b+c+d) + (c+d+a) + (d+a+b) + (a+b+c)} \\ & = \frac{16}{3(a+b+c+d)}. \end{aligned}$$

Hence, we need to prove $4 \geq a+b+c+d$. This is true since

$$\frac{a+b+c+d}{4} \leq \sqrt{\frac{a^2+b^2+c^2+d^2}{4}} = 1$$

by the power mean inequality. □

Problem 2.18. Let a, b, c be positive real numbers such that $a+b+c=3$. Prove that

$$\frac{1}{a^2+b+c} + \frac{1}{b^2+c+a} + \frac{1}{c^2+a+b} \leq 1.$$

Analysis. Note that Chebyshev's inequality also applies to negative numbers. Sometimes it may be simpler for us to put everything on the same side, and prove that a sum of several terms is nonnegative. In particular, if $a_1 \geq a_2 \geq \cdots \geq a_n$ and $b_1 \geq b_2 \geq \cdots \geq b_n$ are real numbers such that $a_1 + a_2 + \cdots + a_n = 0$, then

$$a_1 b_1 + a_2 b_2 + \cdots + a_n b_n \geq \frac{(a_1 + a_2 + \cdots + a_n)(b_1 + b_2 + \cdots + b_n)}{n} = 0$$

by Chebyshev's inequality. We shall use this idea in the following solution.

Solution. Rewrite the inequality as

$$\left(\frac{1}{a^2 - a + 3} - \frac{1}{3}\right) + \left(\frac{1}{b^2 - b + 3} - \frac{1}{3}\right) + \left(\frac{1}{c^2 - c + 3} - \frac{1}{3}\right) \leq 0,$$

which is the same as

$$\frac{a(1-a)}{a^2 - a + 3} + \frac{b(1-b)}{b^2 - b + 3} + \frac{c(1-c)}{c^2 - c + 3} \leq 0.$$

Now, WLOG assume $a \geq b \geq c$. Then $1-a \leq 1-b \leq 1-c$. We can also prove that

$$\frac{a}{a^2 - a + 3} \geq \frac{b}{b^2 - b + 3} \geq \frac{c}{c^2 - c + 3}. \text{ Indeed,}$$

$$\begin{aligned} & \frac{x}{x^2 - x + 3} \geq \frac{y}{y^2 - y + 3} \\ \Leftrightarrow & x(y^2 - y + 3) \geq y(x^2 - x + 3) \\ \Leftrightarrow & (x-y)(3-xy) \geq 0. \end{aligned}$$

This is true for $x \geq y$ as $xy \leq \frac{1}{4}(x+y)^2 < \frac{9}{4} < 3$ in the range of a, b, c (as $a+b+c=3$).

Then, by Chebyshev's inequality, we have

$$\begin{aligned} & \frac{a(1-a)}{a^2 - a + 3} + \frac{b(1-b)}{b^2 - b + 3} + \frac{c(1-c)}{c^2 - c + 3} \\ & \leq \frac{1}{3}((1-a) + (1-b) + (1-c)) \left(\frac{a}{a^2 - a + 3} + \frac{b}{b^2 - b + 3} + \frac{c}{c^2 - c + 3} \right) \\ & = 0. \end{aligned}$$

Equality holds iff $a = b = c = 1$. □

Remarks. As suggested, we are looking for a sequence whose sum is zero. This is one reason why we set one of the sequences to be $1-a, 1-b, 1-c$ instead of some other sequences such as $a(1-a), b(1-b), c(1-c)$. Thus, putting everything to the same side of the inequality may give us an easier way to guess the sequences.

Finally, we give a generalization of Chebyshev's inequality below without providing any application.

Theorem 2.9. (**Generalized Chebyshev's inequality**) Let $n \geq 2$. For any real numbers $a_1 \geq a_2 \geq \dots \geq a_n$ and $b_1 \geq b_2 \geq \dots \geq b_n$, and any positive real numbers $w_1, w_2, \dots, w_n$, we have

$$w_1 a_1 b_1 + w_2 a_2 b_2 + \dots + w_n a_n b_n \geq \frac{(w_1 a_1 + w_2 a_2 + \dots + w_n a_n)(w_1 b_1 + w_2 b_2 + \dots + w_n b_n)}{w_1 + w_2 + \dots + w_n}.$$

Equality holds iff $a_1 = a_2 = \dots = a_n$ or $b_1 = b_2 = \dots = b_n$.

Proof. Firstly, note that $w_i w_j (a_i - a_j)(b_i - b_j) \geq 0$ for any indices i and j . Summing over all pairs $i, j = 1, 2, \dots, n$, we obtain

$$\begin{aligned} 0 &\leq \sum_{i=1}^n \sum_{j=1}^n w_i w_j (a_i - a_j)(b_i - b_j) \\ &= \sum_{i=1}^n \sum_{j=1}^n (w_i w_j a_i b_i + w_i w_j a_j b_j - w_i w_j a_i b_j - w_i w_j a_j b_i) \\ &= 2 \left(\sum_{i=1}^n w_i a_i b_i \right) \left(\sum_{j=1}^n w_j \right) - 2 \left(\sum_{i=1}^n w_i a_i \right) \left(\sum_{j=1}^n w_j b_j \right). \end{aligned}$$

The result follows readily. Equality holds when $a_i = a_j$ or $b_i = b_j$ for each pair of i, j . This means $a_1 = a_2 = \dots = a_n$ or $b_1 = b_2 = \dots = b_n$. $\square$

Chebyshev's inequality is the special case when $w_1 = w_2 = \dots = w_n$.

Here is a summary of the inequalities and some related tricks covered in this section.

- (weighted) AM-GM inequality
 - basic usage ([problems 2.1, 2.2, 2.3, 2.4](#))
 - breaking up a term into several equal terms ([problems 2.1, 2.2, 2.4](#))
 - simplifying fractions ([problems 2.5, 2.6](#))
 - changing the signs of some expressions ([problem 2.6](#))
 - considering the equality case ([problem 2.7](#))
- (weighted) power mean inequality ([problems 2.8, 2.9](#))
- Cauchy-Schwarz inequality
 - basic usage ([problem 2.10](#))
 - handling radicals ([problems 2.11, 2.13, 2.14](#))
 - handling fractions ([problems 2.12, 2.16](#))
- (weighted) Hölder's inequality ([problems 2.15, 2.16](#))
- rearrangement inequality
- (generalized) Chebyshev's inequality ([problems 2.17, 2.18](#))

3 Homogeneous Inequalities

This section deals with homogeneous inequalities. A function $f(x_1, x_2, \dots, x_n)$ is *homogeneous* (齊次) if there exists an integer k such that

$$f(rx_1, rx_2, \dots, rx_n) = r^k f(x_1, x_2, \dots, x_n)$$

for any real numbers $x_1, x_2, \dots, x_n$ and any $r \neq 0$. For example, $x^2 + 3yz + 2z^2$ is a homogeneous polynomial because every term has total degree 2. Also, the function $\frac{x+y}{x^2+y^2} + \frac{y+z}{y^2+z^2} + \frac{z+x}{z^2+x^2}$ in the positive variables x, y, z is also homogeneous (of degree -1).

We focus on this kind of inequalities because many famous inequalities such as the AM-GM inequality are homogeneous. However, such inequalities are also considered as some more classical problems, and they seldom show up in MO problems nowadays. But still it is necessary to know some standard tricks in solving such problems.

Many inequalities we come across are of symmetric or cyclic forms. Therefore, we first introduce two notations so as to simplify some expressions. Firstly, we define the following *symmetric summation sign* in 3 variables.

$$\sum_{\text{sym}} f(x, y, z) = f(x, y, z) + f(x, z, y) + f(y, x, z) + f(y, z, x) + f(z, x, y) + f(z, y, x)$$

In other words, we sum up all similar expressions to $f(x, y, z)$ by taking all permutations of x, y, z . From the definition, each summation sign must consist of 6 terms no matter what the function f is. The following gives some of the examples.

$$\begin{aligned} \sum_{\text{sym}} x^2 y &= x^2 y + x^2 z + y^2 x + y^2 z + z^2 x + z^2 y \\ \sum_{\text{sym}} x^2 &= x^2 + x^2 + y^2 + y^2 + z^2 + z^2 = 2(x^2 + y^2 + z^2) \\ \sum_{\text{sym}} xyz &= xyz + xzy + yxz + yzx + zxy + zyx = 6xyz \end{aligned}$$

One should realize that the expressions $\sum_{\text{sym}} x^2 y$, $\sum_{\text{sym}} xy^2$, $\sum_{\text{sym}} y^2 z$ are the same by definition. This suggests how its name ‘symmetric’ comes from, since every two variables in the expression can be interchanged. For convenience, we usually assign the largest power to the first variable (e.g. x) and so on, and write the polynomial in descending order of degree as usual, e.g. $\sum_{\text{sym}} x^4 y^2 z$, $\sum_{\text{sym}} x^3 y^2 z^2$, $\sum_{\text{sym}} x^5 y^2$.

The same notation can be generalized to n variables, where there are $n!$ terms for each symmetric summation sign. But it is rarely used as it is probably not ideal to handle so many terms as an expanded sum.

The sum and the difference of two symmetric summation signs can be done by combining like terms, except that we should recall that $\sum_{\text{sym}} x^2 y = \sum_{\text{sym}} xy^2$, for example.

To find the product of two symmetric summation signs, one should first write down all the terms of one of them and multiply them directly. The following gives an example.

$$\begin{aligned}
\left(\sum_{\text{sym}} x^2\right)\left(\sum_{\text{sym}} x^2 y\right) &= 2(x^2 + y^2 + z^2)\left(\sum_{\text{sym}} x^2 y\right) \\
&= 2 \sum_{\text{sym}} (x^2 + y^2 + z^2)x^2 y \\
&= 2 \sum_{\text{sym}} (x^4 y + x^2 y^3 + x^2 y z^2) \\
&= 2 \sum_{\text{sym}} (x^4 y + x^3 y^2 + x^2 y^2 z)
\end{aligned}$$

As shown, it would be easier to break down $\sum_{\text{sym}} x^2$ instead of $\sum_{\text{sym}} x^2 y$, since the former only has three distinct terms, and this simplifies the algebraic steps.

The *cyclic summation sign* is also commonly used for cyclic expressions. Its definition is given below.

$$\sum_{\text{cyc}} f(x, y, z) = f(x, y, z) + f(y, z, x) + f(z, x, y)$$

In other words, the variables x, y, z can change in a cyclic order. Here, every sign consists of 3 terms only. A few examples are listed as follows.

$$\begin{aligned}
\sum_{\text{cyc}} x^2 y &= x^2 y + y^2 z + z^2 x \\
\sum_{\text{cyc}} x y^2 &= x y^2 + y z^2 + z x^2 \neq \sum_{\text{cyc}} x^2 y \\
\sum_{\text{cyc}} x^2 &= x^2 + y^2 + z^2 \\
\sum_{\text{cyc}} x y z &= x y z + y z x + z x y = 3 x y z
\end{aligned}$$

We are going to state Muirhead's theorem, which is a classical result about symmetric homogeneous polynomial inequalities. Before that, we have to introduce the concept of *majorization* (蓋). We say that a sequence $x_1 \geq x_2 \geq \dots \geq x_n$ *majorizes* (優於) another sequence $y_1 \geq y_2 \geq \dots \geq y_n$ if they satisfy the following conditions:

$$\begin{aligned}
x_1 &\geq y_1, \\
x_1 + x_2 &\geq y_1 + y_2, \\
&\dots \\
x_1 + x_2 + \dots + x_{n-1} &\geq y_1 + y_2 + \dots + y_{n-1}, \\
x_1 + x_2 + \dots + x_n &= y_1 + y_2 + \dots + y_n.
\end{aligned}$$

We write $(x_1, x_2, \dots, x_n) > (y_1, y_2, \dots, y_n)$ or $(y_1, y_2, \dots, y_n) < (x_1, x_2, \dots, x_n)$ in that case. These are not the inequality signs as the symbols bend a little bit. Note that the sums of the two sequences must be equal in view of the definition.

Theorem 3.1. (Muirhead's theorem(米爾黑德定理)) Let $n \geq 2$. For any positive real numbers $a_1, a_2, \dots, a_n$ and nonnegative real numbers $x_1, x_2, \dots, x_n, y_1, y_2, \dots, y_n$ satisfying $(x_1, x_2, \dots, x_n) > (y_1, y_2, \dots, y_n)$, we have

$$\sum_{\text{sym}} a_1^{x_1} a_2^{x_2} \cdots a_n^{x_n} \geq \sum_{\text{sym}} a_1^{y_1} a_2^{y_2} \cdots a_n^{y_n}.$$

Equality holds iff $a_1 = a_2 = \cdots = a_n$ or $(x_1, x_2, \dots, x_n) = (y_1, y_2, \dots, y_n)$.

Proof. We first consider the case when all exponents x_i, y_i 's are integers. We shall prove the inequality by induction on n . When $n = 2$, we need to prove

$$a_1^{x_1} a_2^{x_2} + a_1^{x_2} a_2^{x_1} \geq a_1^{y_1} a_2^{y_2} + a_1^{y_2} a_2^{y_1} \quad (1)$$

subjected to the conditions $x_1 \geq y_1 \geq y_2 \geq x_2$ and $x_1 + x_2 = y_1 + y_2$. Indeed, this can be proved using the weighted AM-GM inequality. For $w_1, w_2 \geq 0$, we have

$$\frac{w_1(a_1^{x_1} a_2^{x_2}) + w_2(a_1^{x_2} a_2^{x_1})}{w_1 + w_2} \geq (a_1^{w_1 x_1 + w_2 x_2} a_2^{w_1 x_2 + w_2 x_1})^{\frac{1}{w_1 + w_2}}.$$

We aim at choosing w_1, w_2 such that

$$\begin{cases} w_1 x_1 + w_2 x_2 = (w_1 + w_2) y_1, \\ w_1 x_2 + w_2 x_1 = (w_1 + w_2) y_2, \end{cases}$$

to generate the term $a_1^{y_1} a_2^{y_2}$. The equation is equivalent to

$$\begin{cases} w_1(x_1 - y_1) + w_2(x_2 - y_1) = 0, \\ w_1(x_2 - y_2) + w_2(x_1 - y_2) = 0. \end{cases}$$

The two equations are actually the same since $x_1 + x_2 = y_1 + y_2$. So we may simply take $w_1 = y_1 - x_2$ and $w_2 = x_1 - y_1$. Note that both w_1, w_2 are nonnegative. For this choice of w_1, w_2 , we have

$$\sum_{\text{sym}} a_1^{x_1} a_2^{x_2} = \sum_{\text{sym}} \frac{w_1(a_1^{x_1} a_2^{x_2}) + w_2(a_1^{x_2} a_2^{x_1})}{w_1 + w_2} \geq \sum_{\text{sym}} a_1^{y_1} a_2^{y_2},$$

yielding (1).

Now, assume the inequality holds for all n such that $2 \leq n \leq k$ for some $k \geq 2$. To proceed with the inductive step $n = k + 1$, we first observe the following fact.

Claim. If $x_i = y_i$ for some $1 \leq i \leq k + 1$, then the case $n = k + 1$ holds by the inductive hypothesis.

To prove this, it suffices to notice that whenever $(x_1, x_2, \dots, x_{k+1}) > (y_1, y_2, \dots, y_{k+1})$, we also have $(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_{k+1}) > (y_1, \dots, y_{i-1}, y_{i+1}, \dots, y_{k+1})$. This is because

we can always eliminate the equal terms x_i and y_i from both sides of the conditions to verify those inequalities and equation. Thus, by the inductive hypothesis, we have

$$\begin{aligned} \sum_{\text{sym}} a_1^{x_1} a_2^{x_2} \cdots a_{k+1}^{x_{k+1}} &= \sum_{j=1}^{k+1} \left(a_j^{x_j} \sum'_{\text{sym}} a_1^{x_1} a_2^{x_2} \cdots a_{k+1}^{x_{k+1}} \right) \\ &\geq \sum_{j=1}^{k+1} \left(a_j^{x_j} \sum'_{\text{sym}} a_1^{y_1} a_2^{y_2} \cdots a_{k+1}^{y_{k+1}} \right) \\ &= \sum_{j=1}^{k+1} \left(a_j^{y_j} \sum'_{\text{sym}} a_1^{y_1} a_2^{y_2} \cdots a_{k+1}^{y_{k+1}} \right) \\ &= \sum_{\text{sym}} a_1^{y_1} a_2^{y_2} \cdots a_{k+1}^{y_{k+1}}. \end{aligned}$$

Here, the first summation $\sum'_{\text{sym}} a_1^{x_1} a_2^{x_2} \cdots a_{k+1}^{x_{k+1}}$ means that we sum up all symmetric expressions with respect to (w.r.t.) the variables $a_1, \dots, a_{j-1}, a_{j+1}, \dots, a_{k+1}$ (i.e. only the term a_j is missing), while the exponents are fixed to be $x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_{k+1}$. As there are only k variables in this summation sign, we can apply the inductive hypothesis as above. This proves the [claim](#).

We go back to the inductive step. Suppose $(x_1, x_2, \dots, x_{k+1}) > (y_1, y_2, \dots, y_{k+1})$. It is given that $x_1 \geq y_1$. Also, from $\sum_{i=1}^k x_i \geq \sum_{i=1}^k y_i$ and $\sum_{i=1}^{k+1} x_i = \sum_{i=1}^{k+1} y_i$, we must have $x_{k+1} \leq y_{k+1}$. We may assume $x_1 > y_1$ and $x_{k+1} < y_{k+1}$, since otherwise we can apply the [claim](#) to prove the inequality.

Observe that the sequence $x_1 - 1, x_2, \dots, x_k, x_{k+1} + 1$ (with suitable ordering) majorizes $y_1, y_2, \dots, y_k, y_{k+1}$. Let $x'_1, x'_2, \dots, x'_{k+1}$ be the permutation of $x_1 - 1, x_2, \dots, x_k, x_{k+1} + 1$ in descending order. Then $(x_1, x_2, \dots, x_{k+1}) > (x'_1, x'_2, \dots, x'_{k+1})$, so that

$$\sum_{\text{sym}} a_1^{x_1} a_2^{x_2} \cdots a_{k+1}^{x_{k+1}} \geq \sum_{\text{sym}} a_1^{x'_1} a_2^{x'_2} \cdots a_{k+1}^{x'_{k+1}}$$

by the [claim](#) as the two sequences have at least one common term. Hence, it suffices to prove

$$\sum_{\text{sym}} a_1^{x'_1} a_2^{x'_2} \cdots a_{k+1}^{x'_{k+1}} \geq \sum_{\text{sym}} a_1^{y_1} a_2^{y_2} \cdots a_{k+1}^{y_{k+1}}.$$

The same procedure can be carried out until $x_1 = y_1$ or $x_{k+1} = y_{k+1}$, since the difference between the largest exponent and the smallest exponent is monotonic decreasing, and must decrease after sufficiently many steps. The following gives an example to illustrate the procedure. Suppose we need to prove

$$\sum_{\text{sym}} a^9 b^8 c \geq \sum_{\text{sym}} a^7 b^6 c^5$$

using the inductive hypothesis on the case $n = 2$. We get a chain of inequalities

$$\sum_{\text{sym}} a^9 b^8 c \geq \sum_{\text{sym}} a^8 b^8 c^2 \geq \sum_{\text{sym}} a^7 b^8 c^3 = \sum_{\text{sym}} a^8 b^7 c^3 \geq \sum_{\text{sym}} a^7 b^7 c^4.$$

Lastly, the sequences $(7, 7, 4)$ and $(7, 6, 5)$ have the same leading term 7, so that we are done by the [claim](#). Thus, the inductive step is complete, and we have solved the case when all exponents are integers.

Next, if all exponents x_i, y_i 's are rational numbers, we can write $x_i = \frac{x'_i}{D}$ and $y_i = \frac{y'_i}{D}$ for some integers x'_i, y'_i, D . From above, we have

$$\sum_{\text{sym}} b_1^{x'_1} b_2^{x'_2} \dots b_n^{x'_n} \geq \sum_{\text{sym}} b_1^{y'_1} b_2^{y'_2} \dots b_n^{y'_n}$$

for any nonnegative real numbers $b_1, b_2, \dots, b_n$. By setting $b_i = a_i^{\frac{1}{D}}$ for each i , we obtain

$$\sum_{\text{sym}} a_1^{x_1} a_2^{x_2} \dots a_n^{x_n} \geq \sum_{\text{sym}} a_1^{y_1} a_2^{y_2} \dots a_n^{y_n}$$

as desired.

Finally, for the general case when x_i, y_i 's are real numbers, we can always find rational numbers $x_{i1} \geq x_{i2} \geq \dots \geq x_{in}$ and $y_{i1} \geq y_{i2} \geq \dots \geq y_{in}$ for each i such that $(x_{i1}, x_{i2}, \dots, x_{in}) > (y_{i1}, y_{i2}, \dots, y_{in})$ and $\lim_{i \rightarrow \infty} x_{ij} = x_j$, $\lim_{i \rightarrow \infty} y_{ij} = y_j$ for $j = 1, 2, \dots, n$. (Loosely speaking, we are finding sequences of rational numbers which converge to $x_1, x_2, \dots, x_n, y_1, y_2, \dots, y_n$ respectively while keeping the majorization condition. This can be done mainly because of the denseness of $\mathbb{R}$. We omit the technical details here.) From above, we have

$$\sum_{\text{sym}} a_1^{x_{i1}} a_2^{x_{i2}} \dots a_n^{x_{in}} \geq \sum_{\text{sym}} a_1^{y_{i1}} a_2^{y_{i2}} \dots a_n^{y_{in}}$$

for each i . Since exponential functions are continuous, we can take the limit $i \rightarrow \infty$ and use the continuity to deduce

$$\sum_{\text{sym}} a_1^{x_1} a_2^{x_2} \dots a_n^{x_n} \geq \sum_{\text{sym}} a_1^{y_1} a_2^{y_2} \dots a_n^{y_n}.$$

This proves the general case of Muirhead's theorem. We omit the details of tracing the equality cases. $\square$

As suggested from the proof, the results of Muirhead's theorem can always be obtained by using the AM-GM inequality. For example, Muirhead's theorem implies $\sum_{\text{sym}} x^3 y z \geq \sum_{\text{sym}} x^2 y^2 z$. Alternatively, this can be obtained from

$$\sum_{\text{sym}} x^3 y z = \sum_{\text{sym}} \left(\frac{x^3 y z + x y^3 z}{2} \right) \geq \sum_{\text{sym}} \sqrt{x^4 y^4 z^2} = \sum_{\text{sym}} x^2 y^2 z.$$

Usually, it is not difficult to figure out the coefficients of the terms, or we can use the same idea as the solution to [problem 2.4](#) to work out the coefficients.

Note that whenever we write $(x_1, x_2, \dots, x_n) > (y_1, y_2, \dots, y_n)$, we implicitly assume $x_1 \geq x_2 \geq \dots \geq x_n$ and $y_1 \geq y_2 \geq \dots \geq y_n$. Therefore, when we apply Muirhead's theorem, we must arrange the exponents in descending order.

Theorem 3.2. (Schur's inequality(舒爾不等式)) For any nonnegative real numbers a, b, c and $r > 0$, we have

$$a^r(a-b)(a-c) + b^r(b-c)(b-a) + c^r(c-a)(c-b) \geq 0.$$

Equality holds iff $a = b = c$, or one of a, b, c is zero while the other two are equal.

Proof. WLOG assume $a \geq b \geq c$. Then

$$\begin{aligned} & a^r(a-b)(a-c) + b^r(b-c)(b-a) + c^r(c-a)(c-b) \\ &= (a-b)(a^r(a-c) - b^r(b-c)) + c^r(a-c)(b-c) \\ &= (a-b)(a^r(a-b+b-c) - b^r(b-c)) + c^r(a-c)(b-c) \\ &= (a-b)(a^r(a-b) + (a^r - b^r)(b-c)) + c^r(a-c)(b-c) \\ &\geq 0 \end{aligned}$$

since $a \geq b \geq c$ and $r > 0$. It is easy to see that equality holds iff $a = b = c$, or $a = b$, $c = 0$. $\square$

From the equality case, we see that Schur's inequality is a rather strong inequality. Therefore, it is often used when Muirhead's theorem is not sufficient to prove the inequality. The following gives an equivalent form of Schur's inequality which can be obtained via expansion.

$$\sum_{\text{sym}} a^{r+2} + \sum_{\text{sym}} a^r bc \geq 2 \sum_{\text{sym}} a^{r+1} b$$

Clearly, we cannot use Muirhead's theorem directly to prove this. Indeed, we only have $(r+2, 0, 0) > (r+1, 1, 0)$, but at the same time $(r, 1, 1) < (r+1, 1, 0)$.

We usually apply Schur's inequality with $r = 1$, where

$$a^3 + b^3 + c^3 + 3abc \geq a^2b + a^2c + b^2a + b^2c + c^2a + c^2b.$$

We first give one example which can be solved by direct expansion, with the use of Muirhead's theorem and Schur's inequality.

Problem 3.1. (Iran TST 1996/CHKMO 2010 Problem 4) For any positive real numbers a, b, c , prove that

$$(ab + bc + ca) \left(\frac{1}{(a+b)^2} + \frac{1}{(b+c)^2} + \frac{1}{(c+a)^2} \right) \geq \frac{9}{4}.$$

Analysis. There are many 'easy' solutions using classical inequalities and/or various substitutions. But sometimes it is hard to think of such solutions. One natural idea is to expand everything and apply some well-known inequalities directly. This may need a lot of hard work and careful computations, but could always give a complete proof to some homogeneous polynomial inequalities. Another advantage of doing so is that we do not need to think of how to start the problem.

Solution. Clearing denominators, the inequality is equivalent to

$$(ab + bc + ca) \sum_{\text{cyc}} (b + c)^2 (c + a)^2 \geq \frac{9}{4} (a + b)^2 (b + c)^2 (c + a)^2. \quad (1)$$

Now, the left-hand side of (1) is

$$\begin{aligned} & (ab + bc + ca) \sum_{\text{cyc}} (b^2 + 2bc + c^2)(a^2 + 2ac + c^2) \\ &= (ab + bc + ca) \sum_{\text{cyc}} (a^2b^2 + 2a^2bc + a^2c^2 + 2ab^2c + 4abc^2 + 2ac^3 + b^2c^2 + 2bc^3 + c^4) \\ &= (ab + bc + ca) \sum_{\text{sym}} \left(\frac{1}{2}a^4 + 2a^3b + \frac{3}{2}a^2b^2 + 4a^2bc \right) \\ & \quad (\text{notice how we convert between cyclic sums and symmetric sums}) \\ &= \sum_{\text{sym}} \left(\frac{1}{2}a^5b + 2a^4b^2 + \frac{3}{2}a^3b^3 + 4a^3b^2c + \frac{1}{2}a^4bc + 2a^3b^2c + \frac{3}{2}a^2b^3c + 4a^2b^2c^2 \right. \\ & \quad \left. + \frac{1}{2}a^5c + 2a^4bc + \frac{3}{2}a^3b^2c + 4a^3bc^2 \right) \\ &= \sum_{\text{sym}} \left(a^5b + 2a^4b^2 + \frac{5}{2}a^4bc + \frac{3}{2}a^3b^3 + 13a^3b^2c + 4a^2b^2c^2 \right) \end{aligned}$$

while the right-hand side is

$$\begin{aligned} & \frac{9}{4} \left(\sum_{\text{sym}} a^2b + 2abc \right)^2 \\ &= \frac{9}{4} \left(\left(\sum_{\text{sym}} a^2b \right)^2 + 4abc \sum_{\text{sym}} a^2b + 4a^2b^2c^2 \right) \\ &= \frac{9}{4} \left(\sum_{\text{sym}} (a^4b^2 + a^4bc + a^3b^3 + a^2b^3c + a^3bc^2 + a^2b^2c^2) + 4 \sum_{\text{sym}} a^3b^2c + 4a^2b^2c^2 \right) \\ &= \frac{9}{4} \sum_{\text{sym}} \left(a^4b^2 + a^4bc + a^3b^3 + 6a^3b^2c + \frac{5}{3}a^2b^2c^2 \right). \end{aligned}$$

Therefore, the inequality is equivalent to

$$\sum_{\text{sym}} \left(a^5b - \frac{1}{4}a^4b^2 + \frac{1}{4}a^4bc - \frac{3}{4}a^3b^3 - \frac{1}{2}a^3b^2c + \frac{1}{4}a^2b^2c^2 \right) \geq 0. \quad (2)$$

Now, note that the last term $\sum_{\text{sym}} a^2b^2c^2$ is the weakest term (in the sense of majorization) among the 6th degree terms. Therefore, we cannot use Muirhead's theorem alone to prove (2). One idea is to use Schur's inequality to remove this term. For example, since $\sum_{\text{sym}} (a^3 - 2a^2b + abc) \geq 0$ by Schur's inequality, we have

$$\frac{1}{4}abc \sum_{\text{sym}} (a^3 - 2a^2b + abc) \geq 0,$$

which means

$$\sum_{\text{sym}} \left(\frac{1}{4}a^4bc - \frac{1}{2}a^3b^2c + \frac{1}{4}a^2b^2c^2 \right) \geq 0.$$

This generates the term $\frac{1}{4}a^2b^2c^2$. Comparing this with (2), it remains to prove

$$\sum_{\text{sym}} \left(a^5b - \frac{1}{4}a^4b^2 - \frac{3}{4}a^3b^3 \right) \geq 0.$$

But this is obvious by Muirhead's theorem since we have $(5, 1, 0) \succ (4, 2, 0)$ and $(5, 1, 0) \succ (3, 3, 0)$. This proves the inequality. Equality holds when $a = b = c$. $\square$

Remarks. There is not much skill for the expansion steps. One may see that using symmetric and cyclic sums can reduce a lot of work. But one needs to be very careful when changing the two signs. For example, we have $\sum_{\text{cyc}} a^2 = \sum_{\text{sym}} \frac{1}{2}a^2$. To test whether it is correct or not, the simplest way is to count the total number of terms before and after each step of expansion. This is the same as putting $a = b = c = 1$ to see if the two values are consistent or not. Always remember that a symmetric sum has 6 terms while a cyclic sum has 3 terms.

It is almost meaningless to talk about expansion only, since this is a routine job and everyone should be able to do that. The only issues are the pace and the correctness, which can only be trained up via practices. Therefore, in the remaining part, we are more interested in some additional tricks related to the homogeneity of the inequalities.

Sometimes we may not be working on homogeneous inequalities, but we can make the inequalities homogeneous via a process called *homogenization* (齊次化). This can usually be done using some given equality constraints. Here are some examples.

- If it is given that $a + b + c = k$ (with $a, b, c > 0$) for some constant k , we can multiply each term with a suitable power of $\frac{a+b+c}{k}$ to increase its degree.
- If it is given that $abc = k$ (with $a, b, c > 0$) for some constant k , one way to homogenize the inequality is to multiply each term by a suitable power of $\sqrt[3]{\frac{abc}{k}}$. But this usually produces clumsy expressions.

When $abc = 1$ (while the general case is similar), a simpler way is to apply the substitution $y = \frac{x}{a}$ and $z = cx$ for some $x > 0$. This is the same as $a = \frac{x}{y}$, $b = \frac{y}{z}$, $c = \frac{z}{x}$. However, this substitution is a cyclic one, which may distort the symmetry of the original inequality. One way to preserve the symmetry is to set

$$y = x\sqrt[3]{\frac{b}{a}} \text{ and } z = x\sqrt[3]{\frac{1}{a^2b}} \text{ for some } x > 0, \text{ which corresponds to } a = \frac{x^2}{yz}, b = \frac{y^2}{zx}, \\ c = \frac{z^2}{xy}.$$

Once an inequality becomes a homogeneous one, some constraints like $a+b+c=1$ can be removed as it does not provide any further information. Indeed, if $f(a,b,c) \geq 0$ for some homogeneous function f , then $f(ra,rb,rc) = r^k f(a,b,c) \geq 0$ for any $r > 0$. This gives a different sum $ra+rb+rc=r$ of the variables, which is arbitrary. In view of this, the inequality actually holds for whatever sum of the variables.

Problem 3.2. (IMO Shortlist 2000 A1/Problem 2) A, B, C are positive reals with product 1. Prove that

$$\left(A - 1 + \frac{1}{B}\right)\left(B - 1 + \frac{1}{C}\right)\left(C - 1 + \frac{1}{A}\right) \leq 1.$$

Analysis. We are given that $ABC = 1$. Therefore, a standard way is to use the substitution $A = \frac{x}{y}$, $B = \frac{y}{z}$, $C = \frac{z}{x}$. Since the original inequality and our substitution are cyclic, sometimes they may be combined to give a symmetric inequality.

Solution. Put $A = \frac{x}{y}$, $B = \frac{y}{z}$, $C = \frac{z}{x}$. Then

$$\begin{aligned} & \left(A - 1 + \frac{1}{B}\right)\left(B - 1 + \frac{1}{C}\right)\left(C - 1 + \frac{1}{A}\right) \leq 1 \\ \Leftrightarrow & \left(\frac{x}{y} - 1 + \frac{z}{y}\right)\left(\frac{y}{z} - 1 + \frac{x}{z}\right)\left(\frac{z}{x} - 1 + \frac{y}{x}\right) \leq 1 \\ \Leftrightarrow & (x - y + z)(y - z + x)(z - x + y) \leq xyz \\ \Leftrightarrow & \sum_{\text{sym}} \left(-\frac{1}{2}x^3 + x^2y - \frac{1}{3}xyz\right) \leq \sum_{\text{sym}} \frac{1}{6}xyz \\ \Leftrightarrow & \sum_{\text{sym}} \left(\frac{1}{2}x^3 - x^2y + \frac{1}{2}xyz\right) \geq 0, \end{aligned}$$

which is just Schur's inequality. □

Remarks. Note that when one expands the expression $(x-y+z)(y-z+x)(z-x+y)$, it is not necessary to expand everything out. For example, to find the coefficients of the symmetric terms $x^2y, x^2z, y^2x, y^2z, z^2x, z^2y$, one can just calculate the coefficient of x^2y , and then add a symmetric sign. In other words, we just need to find the coefficients of x^3, x^2y, xyz in the expansion, which is rather easy.

Problem 3.3. (CSMO 2006 Problem 6) Find the minimum real number m such that for any positive real numbers a, b, c satisfying $a+b+c=1$,

$$m(a^3 + b^3 + c^3) \geq 6(a^2 + b^2 + c^2) + 1.$$

Analysis. One may easily guess the answer by taking $a = b = c = \frac{1}{3}$, which is usually the equality case of symmetric inequalities. In that case, we should have $m = 27$. This time we can use the condition $a + b + c = 1$ to homogenize the inequality and prove it.

Solution. We shall prove that

$$27(a^3 + b^3 + c^3) \geq 6(a^2 + b^2 + c^2) + 1.$$

Using $a + b + c = 1$ to homogenize the inequality, we have

$$\begin{aligned} & 27(a^3 + b^3 + c^3) \geq 6(a^2 + b^2 + c^2) + 1 \\ \Leftrightarrow & \sum_{\text{sym}} \frac{27}{2} a^3 \geq 3(a + b + c) \sum_{\text{sym}} a^2 + (a + b + c)^3 \\ \Leftrightarrow & \sum_{\text{sym}} \frac{27}{2} a^3 \geq 3 \sum_{\text{sym}} (a^3 + a^2b + a^2c) + \sum_{\text{sym}} \left(\frac{1}{2} a^3 + 3a^2b + abc \right) \\ \Leftrightarrow & \sum_{\text{sym}} \frac{27}{2} a^3 \geq 3 \sum_{\text{sym}} (a^3 + 2a^2b) + \sum_{\text{sym}} \left(\frac{1}{2} a^3 + 3a^2b + abc \right) \\ \Leftrightarrow & \sum_{\text{sym}} 10a^3 \geq \sum_{\text{sym}} (9a^2b + abc). \end{aligned}$$

This is obvious by Muirhead's theorem as $(3, 0, 0) > (2, 1, 0)$ and $(3, 0, 0) > (1, 1, 1)$.

By taking $a = b = c = \frac{1}{3}$, we need $m \geq 27$. So the minimum constant m which satisfies the condition is 27. $\square$

Remarks. Through more practices, one will be familiar with the expansion of expressions like $(a + b + c)^k$.

It is not always necessary to expand everything even if one intends to use Muirhead's theorem. Sometimes it is more convenient for us to apply some standard results before expansion like those theorems introduced in section 2. This reduces the amount of tedious work, but may weaken the bound at the same time. So one should make a balance between these.

For example, recall from proposition 2.1(a) that

$$\frac{a_1}{b_1} + \frac{a_2}{b_2} + \cdots + \frac{a_n}{b_n} \geq \frac{(a_1 + a_2 + \cdots + a_n)^2}{a_1b_1 + a_2b_2 + \cdots + a_nb_n}$$

for any positive real numbers $a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n$. This helps us clear denominators in an easier way, compared to multiplying both sides by $b_1b_2 \cdots b_n$. We may also consider other applications of the Cauchy-Schwarz inequality and Hölder's inequality for the same purpose.

Apart from clearing denominators, there are also simpler ways to clear radicals, in addition to those given in proposition 2.1. Here are two more results.

Proposition 3.1. Let $a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n$ be positive real numbers. Then the following hold.

- (a) $\frac{a_1}{\sqrt[k]{b_1}} + \frac{a_2}{\sqrt[k]{b_2}} + \dots + \frac{a_n}{\sqrt[k]{b_n}} \geq \sqrt[k]{\frac{(a_1 + a_2 + \dots + a_n)^{k+1}}{a_1 b_1 + a_2 b_2 + \dots + a_n b_n}}$ for any $k > 0$
- (b) $a_1 \sqrt[k]{b_1} + a_2 \sqrt[k]{b_2} + \dots + a_n \sqrt[k]{b_n} \leq \sqrt[k]{(a_1 + a_2 + \dots + a_n)^{k-1} (a_1 b_1 + a_2 b_2 + \dots + a_n b_n)}$ for any $k > 1$

Proof.

(a) By the weighted power mean inequality, we have

$$\left(\frac{a_1 b_1^{-\frac{1}{k}} + a_2 b_2^{-\frac{1}{k}} + \dots + a_n b_n^{-\frac{1}{k}}}{a_1 + a_2 + \dots + a_n} \right)^{-k} \leq \frac{a_1 b_1 + a_2 b_2 + \dots + a_n b_n}{a_1 + a_2 + \dots + a_n}$$

since $-\frac{1}{k} < 1$. Rearranging the terms, this means exactly

$$\frac{a_1}{\sqrt[k]{b_1}} + \frac{a_2}{\sqrt[k]{b_2}} + \dots + \frac{a_n}{\sqrt[k]{b_n}} \geq \sqrt[k]{\frac{(a_1 + a_2 + \dots + a_n)^{k+1}}{a_1 b_1 + a_2 b_2 + \dots + a_n b_n}}.$$

(b) By the weighted power mean inequality, we have

$$\left(\frac{a_1 b_1^{\frac{1}{k}} + a_2 b_2^{\frac{1}{k}} + \dots + a_n b_n^{\frac{1}{k}}}{a_1 + a_2 + \dots + a_n} \right)^k \leq \frac{a_1 b_1 + a_2 b_2 + \dots + a_n b_n}{a_1 + a_2 + \dots + a_n}$$

since $\frac{1}{k} < 1$. Rearranging the terms, this means exactly

$$a_1 \sqrt[k]{b_1} + a_2 \sqrt[k]{b_2} + \dots + a_n \sqrt[k]{b_n} \leq \sqrt[k]{(a_1 + a_2 + \dots + a_n)^{k-1} (a_1 b_1 + a_2 b_2 + \dots + a_n b_n)}.$$

□

Problem 3.4. Let x, y, z be positive real numbers such that $x + y + z = 1$. Prove that

$$\frac{x}{y^2 + z} + \frac{y}{z^2 + x} + \frac{z}{x^2 + y} \geq \frac{9}{4}.$$

Analysis. We use this simple problem to demonstrate how to apply [proposition 2.1\(a\)](#) together with homogenization. All the steps are standard.

Solution. By [proposition 2.1\(a\)](#), we have

$$\frac{x}{y^2+z} + \frac{y}{z^2+x} + \frac{z}{x^2+y} \geq \frac{(x+y+z)^2}{\sum_{\text{cyc}} (xy^2+xz)} = \frac{\sum_{\text{cyc}} (x^2+2xy)}{\sum_{\text{cyc}} (xy^2+xy)}.$$

Therefore, it suffices to prove

$$\frac{\sum_{\text{cyc}} (x^2+2xy)}{\sum_{\text{cyc}} (xy^2+xy)} \geq \frac{9}{4}.$$

Using $x+y+z=1$, we have

$$\begin{aligned} & \frac{\sum_{\text{cyc}} (x^2+2xy)}{\sum_{\text{cyc}} (xy^2+xy)} \geq \frac{9}{4} \\ \Leftrightarrow & 4(x+y+z) \sum_{\text{cyc}} (x^2+2xy) \geq 9 \sum_{\text{cyc}} (xy^2+xy(x+y+z)) \\ \Leftrightarrow & 4 \sum_{\text{cyc}} (x^3+x^2y+x^2z+2x^2y+2xy^2+2xyz) \geq 9 \sum_{\text{cyc}} (xy^2+x^2y+xy^2+xyz) \\ \Leftrightarrow & \sum_{\text{cyc}} (4x^3+12x^2y+12xy^2+8xyz) \geq \sum_{\text{cyc}} (9x^2y+18xy^2+9xyz) \\ \Leftrightarrow & \sum_{\text{cyc}} (4x^3+3x^2y-6xy^2-xyz) \geq 0. \end{aligned}$$

Now, since this is cyclic, we can only rely on the AM-GM inequality. We just need to see how to generate the terms xy^2 and xyz from x^3, x^2y and their cyclic permutations of the variables. For example, we have

$$\begin{aligned} 3 \sum_{\text{cyc}} (x^3+x^2y) &= 6 \sum_{\text{cyc}} \frac{y^3+x^2y}{2} \geq 6 \sum_{\text{cyc}} \sqrt{x^2y^4} = 6 \sum_{\text{cyc}} xy^2, \\ \sum_{\text{cyc}} x^3 &\geq 3xyz = \sum_{\text{cyc}} xyz. \end{aligned}$$

Adding these, we obtain the desired inequality. Equality holds iff $x=y=z=\frac{1}{3}$. $\square$

Problem 3.5. (IMO Shortlist 2001 A6/Problem 2) Prove that

$$\frac{a}{\sqrt{a^2+8bc}} + \frac{b}{\sqrt{b^2+8ca}} + \frac{c}{\sqrt{c^2+8ab}} \geq 1$$

for all positive real numbers a, b and c .

Analysis. As the radicals appear in the denominators and the inequality sign is $\geq$, we should probably apply [proposition 3.1\(a\)](#).

Solution. By [proposition 3.1\(a\)](#), we have

$$\frac{a}{\sqrt{a^2 + 8bc}} + \frac{b}{\sqrt{b^2 + 8ca}} + \frac{c}{\sqrt{c^2 + 8ab}} \geq \left(\frac{(a+b+c)^3}{\sum_{\text{cyc}} (a^3 + 8abc)} \right)^{\frac{1}{2}}.$$

Therefore, it suffices to prove $\frac{(a+b+c)^3}{\sum_{\text{cyc}} (a^3 + 8abc)} \geq 1$. Indeed,

$$\begin{aligned} & \frac{(a+b+c)^3}{\sum_{\text{cyc}} (a^3 + 8abc)} \geq 1 \\ \Leftrightarrow & (a+b+c)^3 \geq \sum_{\text{cyc}} (a^3 + 8abc) \\ \Leftrightarrow & \sum_{\text{sym}} \left(\frac{1}{2}a^3 + 3a^2b + abc \right) \geq \sum_{\text{sym}} \left(\frac{1}{2}a^3 + 4abc \right) \\ \Leftrightarrow & 3 \sum_{\text{sym}} a^2b \geq 3 \sum_{\text{sym}} abc, \end{aligned}$$

which is obviously true by Muirhead's theorem. Equality holds iff $a = b = c$. $\square$

Problem 3.6. (IMO Shortlist 2004 A5) If a, b, c are three positive real numbers such that $ab + bc + ca = 1$, prove that

$$\sqrt[3]{\frac{1}{a} + 6b} + \sqrt[3]{\frac{1}{b} + 6c} + \sqrt[3]{\frac{1}{c} + 6a} \leq \frac{1}{abc}.$$

Analysis. Let's see one more similar example. This time we shall use the other inequality for radicals as the inequality sign is reversed.

Solution. By [proposition 3.1\(b\)](#), we have

$$\begin{aligned} \sqrt[3]{\frac{1}{a} + 6b} + \sqrt[3]{\frac{1}{b} + 6c} + \sqrt[3]{\frac{1}{c} + 6a} & \leq \sqrt[3]{(1+1+1)^2 \sum_{\text{cyc}} \left(\frac{1}{a} + 6b \right)} \\ & = \sqrt[3]{9 \sum_{\text{sym}} \left(3a + \frac{1}{2a} \right)}. \end{aligned}$$

Therefore, it suffices to prove

$$\sqrt[3]{9 \sum_{\text{sym}} \left(3a + \frac{1}{2a} \right)} \leq \frac{1}{abc}.$$

We have

$$\begin{aligned}
& \sqrt[3]{9 \sum_{\text{sym}} \left(3a + \frac{1}{2a}\right)} \leq \frac{1}{abc} \\
\Leftrightarrow & \sum_{\text{sym}} \left(27a + \frac{9}{2a}\right) \leq \frac{1}{a^3 b^3 c^3} \\
\Leftrightarrow & \sum_{\text{sym}} \left(27a^4 b^3 c^3 + \frac{9}{2} a^2 b^3 c^3\right) \leq 1 \\
\Leftrightarrow & \sum_{\text{sym}} \left(27a^4 b^3 c^3 + \frac{9}{2} a^3 b^3 c^2\right) \leq 1.
\end{aligned}$$

For simplicity, we let $x = ab$, $y = bc$, $z = ca$. The given condition becomes $x + y + z = 1$ and we need to prove

$$\sum_{\text{sym}} \left(27x^2 y^2 z + \frac{9}{2} x^2 y z\right) \leq 1. \quad (1)$$

We homogenize the inequality to the 5th degree. The left-hand side of (1) is

$$\sum_{\text{sym}} \left(27x^2 y^2 z + \frac{9}{2} x^2 y z(x + y + z)\right) = \sum_{\text{sym}} \left(27x^2 y^2 z + \frac{9}{2} x^3 y z + \frac{9}{2} x^2 y^2 z + \frac{9}{2} x^2 y z^2\right),$$

and the right-hand side is

$$(x + y + z)^5 = \sum_{\text{sym}} \left(\frac{1}{2} x^5 + 5x^4 y + 10x^3 y^2 + 10x^3 y z + 15x^2 y^2 z\right).$$

Comparing the two sides, it remains to prove

$$\sum_{\text{sym}} \left(\frac{1}{2} x^5 + 5x^4 y + 10x^3 y^2 + \frac{11}{2} x^3 y z\right) \geq 21 \sum_{\text{sym}} x^2 y^2 z.$$

This is true by Muirhead's theorem as $(2, 2, 1)$ is majorized by all other triples of exponents. Equality holds iff $x = y = z = \frac{1}{3}$, i.e. $a = b = c = \frac{1}{\sqrt{3}}$. $\square$

Next, we work on another problem of a different style.

Problem 3.7. (APMO 2022 Problem 5) Let a, b, c, d be real numbers such that $a^2 + b^2 + c^2 + d^2 = 1$. Determine the minimum value of $(a - b)(b - c)(c - d)(d - a)$ and determine all values of (a, b, c, d) such that the minimum value is achieved.

Analysis. This is a cyclic inequality, and the equality case is not the case when all variables are equal. It is not hard to see that the minimum value should be negative, so it is possible for us to focus on a particular order of the variables, say $a > b > c > d$. Under this assumption, we may let $a = b + x$, $b = c + y$ and $c = d + z$ so that $x, y, z > 0$. Although this distorts the cyclic symmetry of the question, this simplifies the expression to minimize.

Solution. It suffices to consider the case when $(a-b)(b-c)(c-d)(d-a)$ is negative. WLOG assume $a > b, c, d$. As $a-b > 0 > d-a$, either both $b-c$ and $c-d$ are positive, or both are negative. Equivalently, either $a > b > c > d$ or $b < c < d < a$. The two cases are the same up to symmetry. Therefore, in the following we assume $a > b > c > d$.

Let $a = b + x$, $b = c + y$ and $c = d + z$ so that $x, y, z > 0$. The constraint becomes

$$(d+x+y+z)^2 + (d+y+z)^2 + (d+z)^2 + d^2 = 1, \quad (1)$$

and we need to minimize $-xyz(x+y+z)$, i.e. to maximize $xyz(x+y+z)$. Since this is independent of d , we would like to modify the constraint (1) in some ways to remove the variable d , hence obtaining a constraint on x, y, z only. There are several simple ways to do so. For example, by [proposition 2.1\(c\)](#), we have

$$\alpha^2 + \beta^2 \geq \frac{(\alpha + \beta)^2}{2}$$

for any real numbers α and β (indeed, this is equivalent to $(\alpha - \beta)^2 \geq 0$). Using this, we obtain

$$\begin{aligned} 1 &= (d+x+y+z)^2 + (d+y+z)^2 + (d+z)^2 + d^2 \\ &= [(d+x+y+z)^2 + (-d)^2] + [(d+y+z)^2 + (-d-z)^2] \\ &\geq \frac{[(d+x+y+z) + (-d)]^2}{2} + \frac{[(d+y+z) + (-d-z)]^2}{2} \\ &= \frac{(x+y+z)^2 + y^2}{2}. \end{aligned}$$

Here, we choose to combine the smallest and the largest terms (d and $d+x+y+z$), and also the other two terms because this is the only possibility that the equality may hold (for instance, it is impossible that $d+x+y+z = -(d+z)$ and $d+y+z = -d$ hold simultaneously as $x, y, z > 0$). Now, we have obtained a constraint

$$(x+y+z)^2 + y^2 \leq 2$$

which is independent of d . To maximize $xyz(x+y+z)$, it is not hard to see that x and z should be equal by symmetry. More rigorously, we use the AM-GM inequality to deduce

$$xyz(x+y+z) = (xz)y(x+y+z) \leq \frac{(x+z)^2}{4}y(x+y+z) = \frac{y(x+z)^2(y+x+z)}{4}.$$

Letting $s = x + z$, it remains to maximize $ys^2(y+s)$ given $(y+s)^2 + y^2 \leq 2$. This is a rather standard question and there are many possible solutions. For example, noticing that

$$(y+s)^2 + y^2 = 2y^2 + 2ys + s^2 = 2y(y+s) + s^2,$$

we use the AM-GM inequality to obtain

$$ys^2(y+s) = \frac{[2y(y+s)]s^2}{2} \leq \frac{[2y(y+s) + s^2]^2}{8} \leq \frac{1}{2}.$$

Combining all these, we see that

$$(a-b)(b-c)(c-d)(d-a) = -xyz(x+y+z) \geq -\frac{ys^2(y+s)}{4} \geq -\frac{1}{8}.$$

Equality holds when $d+x+y+z = -d$, $d+y+z = -d-z$, $x = z$ and $2y(y+x+z) = (x+z)^2$.

Equivalently, this means $x = z$, $y = (\sqrt{3}-1)x$ (since $x, y > 0$) and $d = -\frac{\sqrt{3}+1}{2}x$, i.e.

$c = \frac{1-\sqrt{3}}{2}x$, $b = \frac{\sqrt{3}-1}{2}x$ and $a = \frac{\sqrt{3}+1}{2}x$. Together with $a^2 + b^2 + c^2 + d^2 = 1$, we obtain the equality case

$$(a, b, c, d) = \left(\frac{\sqrt{3}+1}{4}, \frac{\sqrt{3}-1}{4}, \frac{1-\sqrt{3}}{4}, -\frac{\sqrt{3}+1}{4} \right).$$

This shows the minimum value $-\frac{1}{8}$ can be attained. In general, the minimum value is attained when

$$(a, b, c, d) = \left(\frac{\sqrt{3}+1}{4}, \frac{\sqrt{3}-1}{4}, \frac{1-\sqrt{3}}{4}, -\frac{\sqrt{3}+1}{4} \right), \left(\frac{\sqrt{3}+1}{4}, -\frac{\sqrt{3}+1}{4}, \frac{1-\sqrt{3}}{4}, \frac{\sqrt{3}-1}{4} \right)$$

or their cyclic permutations. $\square$

Remarks. The substitutions $a = b + x$, $b = c + y$, $c = d + z$ used in this example can be used to solve a number of difficult problems (see **Expansion Methods in Inequalities 1.0**).

In contrast to homogenization, we have the process of *normalization* (標準化). If we need to prove a homogeneous inequality $f(a, b, c) \geq 0$ (very often with $a, b, c > 0$), we may impose a normalized condition $g(a, b, c) = k$ where $g(a, b, c)$ is a homogeneous polynomial (e.g. $a + b + c$) and k is a constant whose sign agrees with $g(a, b, c)$ for some a, b, c (for example, we cannot impose the condition $a + b + c = -1$ if $a, b, c > 0$). Again, this can be done because $f(a, b, c) \geq 0$ iff $f(ra, rb, rc) \geq 0$ for any $r > 0$ as f is homogeneous. In particular, we can always choose a suitable r so that $g(ra, rb, rc) = r^d g(a, b, c)$ is equal to the desired value k (where d is the degree of g).

Indeed, we have already seen one application of normalization in subsection 2.1 when we prove the AM-GM inequality. Recall that to prove

$$\frac{a_1 + a_2 + \cdots + a_n}{n} \geq \sqrt[n]{a_1 a_2 \cdots a_n}, \quad (1)$$

we can transform the question so that it suffices to prove

$$b_1 + b_2 + \cdots + b_n \geq n$$

under the constraint $b_1 b_2 \cdots b_n = 1$. Indeed, we need not define the variables b_i 's. We may simply say that we impose the normalized condition $a_1 a_2 \cdots a_n = 1$, and so (1) is

equivalent to $a_1 + a_2 + \cdots + a_n \geq n$. This simplifies the problem statement as we no longer need to handle the radical.

Let's see some more examples of using normalization.

Problem 3.8. Let a, b, c be positive real numbers. Prove that

$$\sqrt{\frac{ab+bc+ca}{3}} \leq \sqrt[3]{\frac{(a+b)(b+c)(c+a)}{8}}.$$

Analysis. The inequality is homogeneous as both sides are of degree 1. We can remove the radicals by imposing the normalized condition $ab+bc+ca = 3$, for example (we choose 3 instead of 1 because the equality case is $a = b = c = 1$, which looks nicer). Then it remains to prove $(a+b)(b+c)(c+a) \geq 8$ under this constraint. We may use some elementary symmetric polynomials (i.e. $a+b+c$, $ab+bc+ca$ and abc) to relate the two polynomial expressions via some identities.

Solution. WLOG assume $ab+bc+ca = 3$. Then it remains to prove

$$(a+b)(b+c)(c+a) \geq 8.$$

Note that

$$\begin{aligned} (a+b)(b+c)(c+a) &= \sum_{\text{sym}} a^2b + 2abc \\ &= (a+b+c)(ab+bc+ca) - abc \\ &= 3(a+b+c) - abc. \end{aligned}$$

To find a lower bound for this term, one simple way is to give a lower bound for $a+b+c$ and an upper bound for abc . It turns out these can be done using the constraint $ab+bc+ca = 3$.

- Firstly, it is not hard to prove that $(a+b+c)^2 \geq 3(ab+bc+ca) = 9$, and so $a+b+c \geq 3$.
- Secondly, by the AM-GM inequality, we have $3 = ab+bc+ca \geq 3\sqrt[3]{a^2b^2c^2}$, and so $abc \leq 1$.

Combining these, we obtain $(a+b)(b+c)(c+a) \geq 3(3) - (1) = 8$. This proves the inequality. Equality holds when $a = b = c$ (note that it needs not be $a = b = c = 1$ since the normalized condition should be removed). $\square$

Problem 3.9. Let a, b, c be positive real numbers. Prove that

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} + \frac{3\sqrt[3]{abc}}{2(a+b+c)} \geq 2.$$

Analysis. Again, the inequality is homogeneous as both sides have degree 0. There are different normalized conditions that we may impose. The main goal is to simplify the expression for consideration (note that this process will not simplify the problem because we are working on an equivalent problem). For example, setting $abc = 1$ is a possible choice.

Again, we try to express everything in terms of $a + b + c$ and $ab + bc + ca$. To simplify the steps, we may first apply the Cauchy-Schwarz inequality to combine the first three fractions.

Solution. WLOG assume $abc = 1$. By [proposition 2.1\(a\)](#), we have

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{(a+b+c)^2}{a(b+c) + b(c+a) + c(a+b)}.$$

Letting $s = a + b + c$ and $t = ab + bc + ca$, it suffices to prove

$$\frac{s^2}{2t} + \frac{3}{2s} \geq 2.$$

The next step is to maximize t in terms of s and $abc = 1$. This can be done by Schur's inequality. Indeed, Schur's inequality gives

$$\sum_{\text{cyc}} a^3 + 3abc \geq \sum_{\text{sym}} a^2b. \quad (1)$$

By using the identities

$$\begin{aligned} s^3 &= \sum_{\text{cyc}} a^3 + 3 \sum_{\text{sym}} a^2b + 6abc, \\ st &= \sum_{\text{sym}} a^2b + 3abc, \end{aligned}$$

it is not hard to rewrite (1) as $s^3 + 9abc \geq 4st$. It follows that

$$t \leq \frac{s^3 + 9abc}{4s} = \frac{s^3 + 9}{4s}.$$

Thus, we have

$$\frac{s^2}{2t} + \frac{3}{2s} \geq \frac{2s^3}{s^3 + 9} + \frac{3}{2s}.$$

It remains to prove

$$\frac{2s^3}{s^3 + 9} + \frac{3}{2s} \geq 2.$$

After expansion and simplification, it is the same as $s^3 - 12s + 9 \geq 0$, i.e.

$$(s-3)(s^2 + 3s - 3) \geq 0. \quad (2)$$

Now, by the AM-GM inequality, it is obvious that $s \geq 3\sqrt[3]{abc} = 3$. So both factors on the left-hand side of (2) are nonnegative, and hence we are done. Equality holds when $a = b = c$. $\square$

To end this section, we state a few more theorems which are various generalizations of Schur's inequality. They are less useful for our purpose.

Theorem 3.3. ([Vornicu-Schur's inequality](#)) For any real numbers a, b, c with $a \geq b \geq c$ and nonnegative real numbers x, y, z with $x + z \geq y$, we have

$$x(a-b)(a-c) + y(b-c)(b-a) + z(c-a)(c-b) \geq 0.$$

Equality holds iff one of the following holds:

- $a = b = c$;
- $x = y = z = 0$;
- $a = b, z = 0$; or
- $b = c, x = 0$.

Theorem 3.4. ([Turkevich's inequality](#)) For any nonnegative real numbers a, b, c, d , we have

$$a^4 + b^4 + c^4 + d^4 + 2abcd \geq a^2b^2 + a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2 + c^2d^2.$$

Equality holds iff $a = b = c = d$, or one of a, b, c, d is zero while the other three are equal.

Theorem 3.5. ([Generalized Turkevich's inequality](#)) For any nonnegative real numbers $a_1, a_2, \dots, a_n$, we have

$$(n-1)(a_1^2 + a_2^2 + \dots + a_n^2) + n(a_1a_2 \dots a_n)^{\frac{2}{n}} \geq (a_1 + a_2 + \dots + a_n)^2.$$

Equality holds iff $a_1 = a_2 = \dots = a_n$, or one of $a_1, a_2, \dots, a_n$ is zero while the others are equal.

Theorem 3.6. ([Suranyi's inequality](#)) For any real numbers $a_1, a_2, \dots, a_n \geq 0$, we have

$$(n-1)(a_1^n + a_2^n + \dots + a_n^n) + na_1a_2 \dots a_n \geq (a_1 + a_2 + \dots + a_n)(a_1^{n-1} + a_2^{n-1} + \dots + a_n^{n-1}).$$

Equality holds iff $a_1 = a_2 = \dots = a_n$, or one of $a_1, a_2, \dots, a_n$ is zero while the others are equal.

Here are the results and tricks covered in this section.

- Muirhead's theorem ([problems 3.1, 3.3, 3.5, 3.6](#))

- Schur's inequality ([problems 3.1, 3.2, 3.9](#))
- symmetric and cyclic summation signs ([problems 3.1, 3.2, 3.3, 3.4, 3.5, 3.6](#))
- homogenization ([problems 3.2, 3.3, 3.4, 3.6](#))
- substitution ([problem 3.7](#))
- normalization ([problems 3.8, 3.9](#))

4 The Local Refinement Method

The *local refinement method* (局部調整法) is a technique used in inequality. When one applies this technique, one should keep parts of the variables constant and examine how the whole expression can attain its optimum value when changing one or two variables only. In this way, an inequality of many variables can be separated into a few inequalities of one or two variables, which are usually easier to deal with.

For example, such an idea has been used in [problem 3.7](#). In that problem, after transforming the question, we need to maximize $xyz(x+y+z)$ under the constraint

$$(d+x+y+z)^2 + (d+y+z)^2 + (d+z)^2 + d^2 = 1.$$

We first focus on the variable d and find some ways to remove this variable, and obtain another constraint

$$(x+y+z)^2 + y^2 \leq 2$$

on x, y, z . Next, we assume y is a constant, and maximize $xyz(x+y+z)$ in the variables x and z . We see that it is maximized when $x = z$, hence reducing the problem to an inequality in the two variables y and $s = x + z$. It turns out that we can do every step easily, and the whole problem is solved.

In this section, we are going to see some more examples, and we will also introduce a useful technique called mixing variables.

Problem 4.1. Let $n \geq 2$ be a positive integer and let $0 \leq x_1, x_2, \dots, x_n \leq 1$ be real numbers. Find the maximum value of

$$S = \sum_{1 \leq i < j \leq n} (x_i - x_j)^2.$$

Analysis. If we regard S as a function in x_1 , then it is a quadratic polynomial with leading coefficient $n-1$. So we can use the properties of quadratic polynomials to maximize S .

Solution. Note that

$$\begin{aligned} S &= (n-1)(x_1^2 + x_2^2 + \dots + x_n^2) - 2 \sum_{1 \leq i < j \leq n} x_i x_j \\ &= (n-1)x_1^2 - 2 \left(\sum_{j=2}^n x_j \right) x_1 + (n-1) \sum_{j=2}^n x_j^2 - 2 \sum_{2 \leq i < j \leq n} x_i x_j. \end{aligned}$$

This is a quadratic polynomial in x_1 with positive leading coefficient. Thus, its maximum value is attained at one of the endpoints, i.e. S is maximized when $x_1 = 0$ or $x_1 = 1$. By symmetry, this holds for each variable. Therefore, it suffices to consider the case when all variables are 0 or 1. Suppose there are a zeros and b ones where $a+b=n$. Then $S = ab$. This is maximized when a and b are as close to each other as possible.

- When n is even, the maximum value of S is $\left(\frac{n}{2}\right)^2 = \frac{n^2}{4}$. Equality holds when $\frac{n}{2}$ of the x_i 's are 0 and the others are 1.
- When n is odd, the maximum value of S is $\left(\frac{n-1}{2}\right)\left(\frac{n+1}{2}\right) = \frac{n^2-1}{4}$. Equality holds when $\frac{n-1}{2}$ of the x_i 's are 1 and the others are 0, or $\frac{n-1}{2}$ of the x_i 's are 0 and the others are 1.

□

Problem 4.2. (IMO Shortlist 2005 A3) Four real numbers p, q, r, s satisfy $p + q + r + s = 9$ and $p^2 + q^2 + r^2 + s^2 = 21$. Prove that there exists a permutation (a, b, c, d) of (p, q, r, s) such that $ab - cd \geq 2$.

Analysis. Before proving the desired inequality, we first find the range of the variables. We may rearrange p, q, r, s in ascending order so as to reduce the number of cases. Afterwards, we shall reduce the number of variables step by step.

Solution. By the Cauchy-Schwarz inequality, we have

$$(1 + 1 + 1)(q^2 + r^2 + s^2) \geq (q + r + s)^2.$$

Note that

$$\begin{aligned} & (1 + 1 + 1)(q^2 + r^2 + s^2) \geq (q + r + s)^2 \\ \Rightarrow & 3(21 - p^2) \geq (9 - p)^2 \\ \Rightarrow & 4p^2 - 18p + 18 \leq 0 \\ \Rightarrow & 2(p - 3)(2p - 3) \leq 0 \\ \Rightarrow & \frac{3}{2} \leq p \leq 3. \end{aligned}$$

The bounds also hold for q, r, s . WLOG assume $p \leq q \leq r \leq s$. We only need to prove $rs - pq \geq 2$. Keeping r, s as constants, we solve for p, q . Indeed, from

$$p + q = 9 - r - s$$

and

$$\begin{aligned} pq &= \frac{1}{2}((p + q)^2 - (p^2 + q^2)) \\ &= \frac{1}{2}((9 - r - s)^2 - (21 - r^2 - s^2)) \\ &= r^2 + rs + s^2 - 9r - 9s + 30, \end{aligned}$$

we know that p, q are the roots of the equation

$$x^2 - (9 - r - s)x + (r^2 + rs + s^2 - 9r - 9s + 30) = 0.$$

In other words, we have

$$p, q = \frac{(9 - r - s) \pm \sqrt{-3r^2 - 2rs - 3s^2 + 18r + 18s - 39}}{2}.$$

In particular, $r \geq q$ yields $3r + s - 9 \geq \sqrt{-3r^2 - 2rs - 3s^2 + 18r + 18s - 39}$. From

$$\begin{aligned} & 3r + s - 9 \geq \sqrt{-3r^2 - 2rs - 3s^2 + 18r + 18s - 39} \\ \Rightarrow & 9r^2 + 6rs + s^2 - 54r - 18s + 81 \geq -3r^2 - 2rs - 3s^2 + 18r + 18s - 39 \\ \Rightarrow & 3r^2 + (2s - 18)r + (s^2 - 9s + 30) \geq 0, \end{aligned}$$

we get $r \leq \frac{(9 - s) - \sqrt{-2s^2 + 9s - 9}}{3}$ or $r \geq \frac{(9 - s) + \sqrt{-2s^2 + 9s - 9}}{3}$. Note that we have

$3r + s \geq p + q + r + s = 9$. Hence, the case $r \leq \frac{(9 - s) - \sqrt{-2s^2 + 9s - 9}}{3}$ is rejected.

Now, $s \geq r \geq \frac{(9 - s) + \sqrt{-2s^2 + 9s - 9}}{3}$ implies $4s - 9 \geq \sqrt{-2s^2 + 9s - 9}$. We find that

$$\begin{aligned} & 4s - 9 \geq \sqrt{-2s^2 + 9s - 9} \\ \Rightarrow & 16s^2 - 72s + 81 \geq -2s^2 + 9s - 9 \\ \Rightarrow & 2s^2 - 9s + 10 \geq 0 \\ \Rightarrow & (2s - 5)(s - 2) \geq 0 \\ \Rightarrow & s \geq \frac{5}{2}. \end{aligned}$$

Note that $s \leq 2$ is rejected since $s \geq \frac{p + q + r + s}{4} = \frac{9}{4}$. Consider

$$rs - pq = rs - (r^2 + rs + s^2 - 9r - 9s + 30) = -r^2 + 9r - (s^2 - 9s + 30).$$

The quadratic polynomial has a maximum at $r = \frac{9}{2}$, increasing for $r < \frac{9}{2}$ and decreasing for $r > \frac{9}{2}$. From $\frac{(9 - s) + \sqrt{-2s^2 + 9s - 9}}{3} \leq r \leq 3$, the minimum of $rs - pq$ occurs

when $r = \frac{(9 - s) + \sqrt{-2s^2 + 9s - 9}}{3}$, and so we have

$$\begin{aligned} & rs - pq \\ \geq & - \left(\frac{(9 - s) + \sqrt{-2s^2 + 9s - 9}}{3} \right)^2 + 9 \left(\frac{(9 - s) + \sqrt{-2s^2 + 9s - 9}}{3} \right) - (s^2 - 9s + 30) \\ = & - \frac{s^2 - 18s + 81}{9} - \frac{2(9 - s)\sqrt{-2s^2 + 9s - 9}}{9} - \frac{-2s^2 + 9s - 9}{9} + (27 - 3s) \\ & + 3\sqrt{-2s^2 + 9s - 9} - (s^2 - 9s + 30) \\ = & -\frac{8}{9}s^2 + 7s - 11 + \frac{(2s + 9)\sqrt{-2s^2 + 9s - 9}}{9}. \end{aligned}$$

Hence, it suffices to prove

$$-\frac{8}{9}s^2 + 7s - 11 + \frac{(2s+9)\sqrt{-2s^2+9s-9}}{9} \geq 2,$$

i.e.

$$\frac{(2s+9)\sqrt{-2s^2+9s-9}}{9} \geq \frac{8}{9}s^2 - 7s + 13.$$

Since $\frac{(2s+9)\sqrt{-2s^2+9s-9}}{9} > 0$, we just need to prove

$$(2s+9)^2(-2s^2+9s-9) \geq (8s^2-63s+117)^2,$$

i.e.

$$-(2s+9)^2(s-3)(2s-3) \geq (s-3)^2(8s-39)^2.$$

As $s \leq 3$, it remains to prove $-(2s+9)^2(2s-3) \leq (s-3)(8s-39)^2$. We have

$$\begin{aligned} & -(2s+9)^2(2s-3) \leq (s-3)(8s-39)^2 \\ \Leftrightarrow & -8s^3 - 60s^2 - 54s + 243 \leq 64s^3 - 816s^2 + 3393s - 4563 \\ \Leftrightarrow & 72s^3 - 756s^2 + 3447s - 4806 \geq 0 \\ \Leftrightarrow & 8s^3 - 84s^2 + 383s - 534 \geq 0 \\ \Leftrightarrow & (2s-5)\left(4s^2 - 32s + \frac{223}{2}\right) + \frac{47}{2} \geq 0. \end{aligned}$$

As $s \geq \frac{5}{2}$, the first factor $2s-5$ is nonnegative. The second factor $4s^2 - 32s + \frac{223}{2}$ is positive since its discriminant is $(-32)^2 - 4(4)\left(\frac{223}{2}\right) = 1024 - 1784 < 0$. Thus, the inequality is proved. Equality holds when $(p, q, r, s) = (2, 2, 2, 3)$ or its permutation. $\square$

The technique of *mixing variables*, or *smoothing method*(磨光法) is always used together with the local refinement method. There are many variations of this method. For example, suppose we have a continuous function f in n variables. If we can prove that

$$f(x_1, x_2, \dots, x_i, \dots, x_j, \dots, x_n) \geq f\left(x_1, x_2, \dots, \frac{x_i + x_j}{2}, \dots, \frac{x_i + x_j}{2}, \dots, x_n\right)$$

for any x_i, x_j , then $f(x_1, x_2, \dots, x_n) \geq f(x, x, \dots, x)$, where $x = \frac{x_1 + x_2 + \dots + x_n}{n}$. The same holds true when all inequality signs are reversed. The function should be continuous as we probably need to take limit. Indeed, the whole process can be illustrated as follows.

Firstly, note that the claim is trivial in the case where $n = 2^k$ for some $k \geq 1$. For example, when $n = 4$, we have

$$\begin{aligned} f(a, b, c, d) &\geq f\left(\frac{a+b}{2}, \frac{a+b}{2}, c, d\right) \\ &\geq f\left(\frac{a+b}{2}, \frac{a+b}{2}, \frac{c+d}{2}, \frac{c+d}{2}\right) \\ &\geq f\left(s, \frac{a+b}{2}, s, \frac{c+d}{2}\right) \\ &\geq f(s, s, s, s) \end{aligned}$$

where $s = \frac{a+b+c+d}{4}$. In general, we can mix any 2^k variables to the same value in finitely many steps.

Now, suppose there are n variables $x_1, x_2, \dots, x_n$ with $2^k < n < 2^{k+1}$. We first mix the initial 2^k variables to get $\underbrace{a, a, \dots, a}_{2^k \text{ times}}, x_{2^k+1}, \dots, x_n$. Then, we mix the last 2^k variables

(including some but not all a 's) to $\underbrace{a, a, \dots, a}_{(n-2^k) \text{ times}}, \underbrace{b, b, \dots, b}_{2^k \text{ times}}$, so that all variables can be divided into two groups of equal values.

Next, for each step, we mix one a and one b until all a 's are used up. Then we will obtain $\frac{a+b}{2}, \frac{a+b}{2}, \dots, \frac{a+b}{2}, b, b, \dots, b$. Consider the difference between the two groups of numbers. Originally, the difference is $|b-a|$. After a sequence of steps, it becomes $\left|b - \frac{a+b}{2}\right| = \left|\frac{b-a}{2}\right|$. This means the difference is halved each time (this holds even if the b 's are first used up). As $\lim_{m \rightarrow \infty} \frac{1}{2^m} = 0$ and f is continuous, the difference tends to zero, and all variables are the same by taking limits. This proves $f(x_1, x_2, \dots, x_n) \geq f(x, x, \dots, x)$. Note that x is the arithmetic mean of the x_i 's since the sum of all variables is an invariant throughout the process.

One way to avoid taking limits is to construct a way of mixing variables that end in finitely many steps, such as replacing x_i, x_j by $x, x_i + x_j - x$ or by $x_i + \alpha, x_j - \alpha$ for some α . Apart from using the arithmetic mean, one may also use the geometric mean or any other expressions. However, the expression must be consistent to the constraint. For example, if the constraint is $x_1 x_2 \cdots x_n = 1$, then one should use the geometric mean or other ways which keep the product unchanged.

To demonstrate this idea, let's go through some standard examples.

Problem 4.3. (AM-GM inequality) Let $n \geq 2$. Let $a_1, a_2, \dots, a_n$ be positive real numbers. Prove that

$$\frac{a_1 + a_2 + \cdots + a_n}{n} \geq \sqrt[n]{a_1 a_2 \cdots a_n}.$$

Analysis. We start with the AM-GM inequality. This time we give a proof using the idea of mixing variables. For example, we may fix the sum of the variables in every step so that the left-hand side remains unchanged. Each time we can change at least one more variable to the arithmetic mean. This ensures the process must end in finitely many steps.

Solution. Consider

$$f(a_1, a_2, \dots, a_n) = \frac{a_1 + a_2 + \dots + a_n}{n} - \sqrt[n]{a_1 a_2 \dots a_n}.$$

Let $a = \frac{a_1 + a_2 + \dots + a_n}{n}$. We first claim that

$$f(a_1, \dots, a_i, \dots, a_j, \dots, a_n) \geq f(a_1, \dots, a, \dots, a_i + a_j - a, \dots, a_n)$$

whenever $a_i < a < a_j$ (or $a_j < a < a_i$). Indeed,

$$\begin{aligned} f(a_1, \dots, a_i, \dots, a_j, \dots, a_n) &\geq f(a_1, \dots, a, \dots, a_i + a_j - a, \dots, a_n) \\ \Leftrightarrow \sqrt[n]{a_1 \dots a_i \dots (a_i + a_j - a) \dots a_n} &\geq \sqrt[n]{a_1 \dots a_i \dots a_j \dots a_n} \\ \Leftrightarrow a(a_i + a_j - a) &\geq a_i a_j \\ \Leftrightarrow (a - a_i)(a_j - a) &\geq 0. \end{aligned}$$

This is true since $a_i < a < a_j$ (or $a_j < a < a_i$).

In each step, if the variables are not all equal (to a), then we can always find a_i, a_j such that $a_i < a < a_j$. By the mixing step, at least one of a_i, a_j becomes a , and the function value is reduced or unchanged. After a finite number of steps, all variables will become the arithmetic mean, and we have

$$f(a_1, a_2, \dots, a_n) \geq f(a, a, \dots, a) = 0.$$

Since the initial tuple $(a_1, a_2, \dots, a_n)$ is arbitrary, this means $f(a, a, \dots, a) = 0$ is the minimum value. $\square$

Remarks. There are many similar ways that can prove the AM-GM inequality using mixing variables. For example, one can use the mixing step

$$f(a_1, \dots, a_i, \dots, a_j, \dots, a_n) \geq f\left(a_1, \dots, \frac{a_i + a_j}{2}, \dots, \frac{a_i + a_j}{2}, \dots, a_n\right)$$

or

$$f(a_1, \dots, a_i, \dots, a_j, \dots, a_n) \geq f(a_1, \dots, \sqrt{a_i a_j}, \dots, \sqrt{a_i a_j}, \dots, a_n),$$

etc. Note that we need the continuity of f for these two options.

Problem 4.4. Let x, y, z be nonnegative real numbers such that no two of them can be zero simultaneously. Prove that

$$\frac{1}{(x+y)^2} + \frac{1}{(y+z)^2} + \frac{1}{(z+x)^2} + \frac{24}{(x+y+z)^2} \geq \frac{8}{xy + yz + zx}.$$

Analysis. One should immediately find that equality does not hold when $x = y = z$. As the condition for x, y, z is nonnegative, we guess that equality might hold when one of them is zero. This suggests we should mix one of the variables to zero. Thus, sometimes it is useful to make a guess on the equality case (or the features of the equality case) so as to decide which mixing step to use.

Note that we usually do not attempt to prove $f(x, y, z) \geq f(x, y, 0)$ (for some suitable function f). This is because it normally does not work (if we can replace each variable by 0 for free, then the inequality is too simple!). One possible mixing step is to replace y, z by $y + z, 0$. This preserves the sums $y + z$ and $x + y + z$, and so two common terms can be removed.

Solution. Consider

$$f(x, y, z) = \frac{1}{(x+y)^2} + \frac{1}{(y+z)^2} + \frac{1}{(z+x)^2} + \frac{24}{(x+y+z)^2} - \frac{8}{xy+yz+zx}.$$

We first try to prove that $f(x, y, z) \geq f(x, y+z, 0)$. The inequality

$$\begin{aligned} & \frac{1}{(x+y)^2} + \frac{1}{(y+z)^2} + \frac{1}{(z+x)^2} + \frac{24}{(x+y+z)^2} - \frac{8}{xy+yz+zx} \\ & \geq \frac{1}{(x+y+z)^2} + \frac{1}{(y+z)^2} + \frac{1}{x^2} + \frac{24}{(x+y+z)^2} - \frac{8}{x(y+z)} \end{aligned}$$

is equivalent to

$$\left(\frac{1}{(x+y)^2} - \frac{1}{(x+y+z)^2} \right) + \left(\frac{1}{(z+x)^2} - \frac{1}{x^2} \right) + \left(\frac{8}{x(y+z)} - \frac{8}{xy+yz+zx} \right) \geq 0.$$

Note that we group similar terms together as $z = 0$ is a factor. Now, this is the same as

$$\frac{z(2x+2y+z)}{(x+y)^2(x+y+z)^2} - \frac{z(2x+z)}{(z+x)^2x^2} + \frac{8yz}{x(y+z)(xy+yz+zx)} \geq 0.$$

For simplicity, by ignoring the first term, it suffices to have

$$\frac{8y}{(y+z)(xy+yz+zx)} \geq \frac{2x+z}{(z+x)^2x},$$

i.e.

$$8xy(z+x)^2 \geq (2x+z)(y+z)(xy+yz+zx). \quad (1)$$

Noting that the left-hand side has a term x^3y while the degree of x on the right-hand side is at most 2, we should first assume $x \geq y \geq z$ (indeed, this should be done at the first step). Then we have $2(z+x) \geq 2x+z$, $z+x \geq y+z$ and $4xy \geq xy+yz+zx$. Multiplying these, we obtain (1).

In other words, if $x \geq y \geq z$, then $f(x, y, z) \geq f(x, y+z, 0)$. Hence, we just need to

consider the case $z = 0$. Now,

$$\begin{aligned}
f(x, y, 0) &= \frac{1}{(x+y)^2} + \frac{1}{y^2} + \frac{1}{x^2} + \frac{24}{(x+y)^2} - \frac{8}{xy} \geq 0 \\
&\Leftrightarrow \frac{25}{(x+y)^2} + \frac{x^2 - 8xy + y^2}{x^2y^2} \geq 0 \\
&\Leftrightarrow 25x^2y^2 + (x+y)^2(x^2 - 8xy + y^2) \geq 0 \\
&\Leftrightarrow x^4 - 6x^3y + 11x^2y^2 - 6xy^3 + y^4 \geq 0 \\
&\Leftrightarrow (x^2 - 3xy + y^2)^2 \geq 0.
\end{aligned}$$

This is clearly true. Equality holds iff $(x, y, z) = \left(\frac{3+\sqrt{5}}{2}t, t, 0\right)$ or its permutation, where t is a positive constant. $\square$

Remarks. When we try to mix two of the variables, we first randomly choose two of them. If it works for any two variables, then the result follows immediately. If not, we can add extra conditions to the variables we have chosen (usually an ordering) so that this mixing step can still reduce the number of variables. The inequality we need to establish for the mixing step (i.e. $f(x, y, z) \geq f(x, y+z, 0)$) is usually weak. Therefore, we may drop some clumsy terms and use some trivial inequalities when proving it.

The above solution demonstrates how mixing variables help reduce the original problem to some simpler inequalities. Even if we cannot completely solve the problem after a mixing step, we can deduce some more useful information (like we may assume $z = 0$). So this method is rather flexible. In addition, we may notice that this problem is not easy to be solved using the classical inequalities alone because of its strange equality case. But it turns out that we do not even need to work out the exact equality case when proving this via mixing variables.

Problem 4.5. Let a, b, c, d be positive real numbers such that $a + b + c + d = 1$. Prove that

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} - 48(a^2 + b^2 + c^2 + d^2) \geq 4.$$

Analysis. Noting that the equality holds when $a = b = c = d = \frac{1}{4}$, we should mix two variables to equal value this time. From the condition $a + b + c + d = 1$, we should keep their sum constant so we should probably take their arithmetic mean. Indeed, this inequality has more than one equality case, though we do not need to realize it before finishing the proof!

Solution. Consider

$$f(a, b, c, d) = \frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} - 48(a^2 + b^2 + c^2 + d^2).$$

We try to prove that $f(a, b, c, d) \geq f\left(a, b, \frac{c+d}{2}, \frac{c+d}{2}\right)$. The inequality

$$\begin{aligned} & \frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} - 48(a^2 + b^2 + c^2 + d^2) \\ & \geq \frac{1}{a} + \frac{1}{b} + \frac{2}{c+d} + \frac{2}{c+d} - 48\left(a^2 + b^2 + \left(\frac{c+d}{2}\right)^2 + \left(\frac{c+d}{2}\right)^2\right) \end{aligned}$$

is equivalent to

$$\left(\frac{1}{c} + \frac{1}{d} - \frac{4}{c+d}\right) + 48\left(2\left(\frac{c+d}{2}\right)^2 - c^2 - d^2\right) \geq 0,$$

i.e.

$$\frac{(c-d)^2}{cd(c+d)} - 24(c-d)^2 \geq 0.$$

This is true if $cd(c+d) \leq \frac{1}{24}$. This suggests we should choose the small terms from a, b, c, d to apply this mixing step. Usually, for inequalities involving 4 variables, we would mix the 1st and 3rd largest, or the 2nd largest and the smallest terms. Therefore, we may first assume $a \geq c \geq b \geq d$.

In that case, we have $c+d \leq \frac{a+b+c+d}{2} = \frac{1}{2}$. By the AM-GM inequality,

$$cd(c+d) \leq \left(\frac{c+d}{2}\right)^2 (c+d) \leq \frac{1}{32} < \frac{1}{24}.$$

In this smoothing step, each time we choose the 2nd largest and the smallest variables and take their arithmetic mean. By the principle of mixing variables, we can mix the smallest 3 variables to equal values, and the value of f will not increase. Therefore, we may restrict to the case $0 < b = c = d = \frac{1-a}{3} \leq a$. Note that this gives $\frac{1}{4} \leq a < 1$.

We need to prove

$$\frac{1}{a} + \frac{9}{1-a} - 48\left(a^2 + 3\left(\frac{1-a}{3}\right)^2\right) \geq 4.$$

We have

$$\begin{aligned} & \frac{1}{a} + \frac{9}{1-a} - 48\left(a^2 + 3\left(\frac{1-a}{3}\right)^2\right) \geq 4 \\ \Leftrightarrow & \frac{1+8a}{a(1-a)} - 16(4a^2 - 2a + 1) \geq 4 \\ \Leftrightarrow & 1 + 8a - 16a(1-a)(4a^2 - 2a + 1) \geq 4a(1-a) \\ \Leftrightarrow & 64a^4 - 96a^3 + 48a^2 - 8a + 1 \geq -4a^2 + 4a \\ \Leftrightarrow & 64a^4 - 96a^3 + 52a^2 - 12a + 1 \geq 0. \end{aligned}$$

Since $a = \frac{1}{4}$ is an equality case, we can easily obtain the factorization. We have

$$\begin{aligned}
& 64a^4 - 96a^3 + 52a^2 - 12a + 1 \geq 0 \\
\Leftrightarrow & (4a - 1)(16a^3 - 20a^2 + 8a - 1) \geq 0 \\
\Leftrightarrow & (4a - 1)^2(4a^2 - 4a + 1) \geq 0 \\
\Leftrightarrow & (4a - 1)^2(2a - 1)^2 \geq 0.
\end{aligned}$$

This is obviously true. Equality holds iff $(a, b, c, d) = \left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right), \left(\frac{1}{2}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}\right)$ or their permutations. $\square$

Remarks. The smoothing method is extremely useful when the inequality mainly consists of isolated terms, such as $\frac{1}{a}, a^2$, since the terms can be cancelled out easily.

Problem 4.6. (CMO 1988 Problem 4(2)) Let $a_1, a_2, \dots, a_n$ be $n \geq 3$ positive real numbers satisfying the inequality

$$(a_1^2 + a_2^2 + \dots + a_n^2)^2 > (n-1)(a_1^4 + a_2^4 + \dots + a_n^4).$$

Prove that any three of these numbers are the side lengths of a triangle.

Analysis. WLOG we may assume $a_1 \leq a_2 \leq \dots \leq a_n$ and prove that $a_1 + a_2 > a_n$. To do so, it is natural to try to prove by contradiction. Since $a_3, a_4, \dots, a_{n-1}$ play similar roles, we guess equality holds when they are all equal. However, the same mixing step cannot be applied to a_1, a_2, a_n . This suggests the equality does not necessarily hold when $a_1 = a_2$. But in any case, we can reduce the problem to an inequality of 4 variables, which is simple enough to be solved by elementary methods.

Solution. WLOG assume $a_1 \leq a_2 \leq \dots \leq a_n$. Suppose on the contrary that $a_1 + a_2 \leq a_n$. Consider

$$f(a_1, a_2, \dots, a_n) = (a_1^2 + a_2^2 + \dots + a_n^2)^2 - (n-1)(a_1^4 + a_2^4 + \dots + a_n^4).$$

We shall derive the contradiction that $f(a_1, a_2, \dots, a_n) \leq 0$. For simplicity, we choose a mixing step which fixes the sum of squares of the variables. Hence, we shall prove that

$$f(a_1, \dots, a_i, \dots, a_j, \dots, a_n) \leq f\left(a_1, \dots, \sqrt{\frac{a_i^2 + a_j^2}{2}}, \dots, \sqrt{\frac{a_i^2 + a_j^2}{2}}, \dots, a_n\right)$$

for $3 \leq i < j \leq n-1$. By cancelling like terms on both sides, it is not hard to see that this is equivalent to

$$2\left(\frac{a_i^2 + a_j^2}{2}\right)^2 \leq a_i^4 + a_j^4,$$

i.e. $(a_i^2 - a_j^2)^2 \geq 0$. This is obviously true.

Note that the mixing step cannot be applied to a_1, a_2, a_n as the condition $a_1 + a_2 \leq a_n$ may no longer be satisfied using the mixing step. Therefore, we can only prove that

$$f(a_1, a_2, \dots, a_n) \leq f(a_1, a_2, a, a, \dots, a, a_n)$$

for some a with $a_2 \leq a \leq a_n$. We write this as a polynomial in a_n . Indeed,

$$\begin{aligned} & f(a_1, a_2, a, a, \dots, a, a_n) \\ &= (a_1^2 + a_2^2 + (n-3)a^2 + a_n^2)^2 - (n-1)(a_1^4 + a_2^4 + (n-3)a^4 + a_n^4) \\ &= (2-n)a_n^4 + 2(a_1^2 + a_2^2 + (n-3)a^2)a_n^2 + (a_1^2 + a_2^2 + (n-3)a^2)^2 \\ &\quad - (n-1)(a_1^4 + a_2^4 + (n-3)a^4). \end{aligned}$$

The graph of the quadratic polynomial in a_n^2 has a maximum (since $2-n < 0$) at $a_n^2 = \frac{a_1^2 + a_2^2 + (n-3)a^2}{n-2}$. From the conditions $a_n \geq \max\{a_1 + a_2, a\}$, we have

$$(n-2)a_n^2 \geq (a_1 + a_2)^2 + (n-3)a^2 \geq a_1^2 + a_2^2 + (n-3)a^2.$$

Therefore, f is decreasing in the range of a_n . In particular, as $a_n \geq a_1 + a_2$, it suffices to prove that $f \leq 0$ in the case $a_n = a_1 + a_2$. Now,

$$\begin{aligned} f &= (a_1^2 + a_2^2 + (n-3)a^2 + (a_1 + a_2)^2)^2 - (n-1)(a_1^4 + a_2^4 + (n-3)a^4 + (a_1 + a_2)^4) \\ &= 4(a_1^2 + a_1a_2 + a_2^2)^2 + 4(n-3)(a_1^2 + a_1a_2 + a_2^2)a^2 + (n-3)^2a^4 \\ &\quad - 2(n-1)(a_1^2 + a_1a_2 + a_2^2)^2 - (n-1)(n-3)a^4 \\ &= -2(n-3)(a_1^2 + a_1a_2 + a_2^2)^2 + 4(n-3)(a_1^2 + a_1a_2 + a_2^2)a^2 - 2(n-3)a^4 \\ &= -2(n-3)((a_1^2 + a_1a_2 + a_2^2) - a^2)^2 \\ &\leq 0, \end{aligned}$$

which contradicts the given condition. So any three of $a_1, a_2, \dots, a_n$ are the side lengths of a triangle. $\square$

Remarks. After the mixing step, we are also using the local refinement method by focusing on the variable a_n .

Note that there are strange equality cases like $(a_1, a_2, a_3, a_4) = (1, 2, \sqrt{7}, 3)$ when $n = 4$.

Problem 4.7. (CMO 2011 Problem 5) Given an integer $n \geq 4$, find the maximum

of $\frac{\sum_{i=1}^n a_i(a_i + b_i)}{\sum_{i=1}^n b_i(a_i + b_i)}$ for nonnegative real numbers $a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n$ satisfying

$$a_1 + a_2 + \dots + a_n = b_1 + b_2 + \dots + b_n > 0.$$

Analysis. Firstly, we transform the expression to a polynomial expression since polynomials are easier to handle. For example, letting k be the maximum value, it is the same as proving

$$f(a_1, \dots, a_n, b_1, \dots, b_n) = k \sum_{i=1}^n b_i(a_i + b_i) - \sum_{i=1}^n a_i(a_i + b_i) \geq 0.$$

We are going to use the local refinement method to minimize f step by step. There are several mixing steps that can preserve the sum of each sequence. Firstly, we can rearrange the order of the terms within each sequence. This allows us to fix the order of the variables so that $\{a_j\}$ is decreasing while $\{b_j\}$ is increasing.

Secondly, we can mix two variables in the same sequence in a way that preserves their sum. By examining some extreme cases (for $n = 4$) such as the following:

$$\begin{aligned} f(1, 1, 1, 1, 1, 1, 1, 1) &= 8k - 8, \\ f(4, 0, 0, 0, 1, 1, 1, 1) &= 8k - 20, \\ f(1, 1, 1, 1, 0, 0, 0, 4) &= 20k - 8, \\ f(4, 0, 0, 0, 0, 0, 0, 4) &= 16k - 16, \end{aligned}$$

we may have the sense that the sequence $\{a_j\}$ should contain more zeros to maximize k (for example, we need $k \geq \frac{5}{2}$ by considering $f(4, 0, 0, 0, 1, 1, 1, 1)$). Therefore, we may consider a mixing step that replaces a_1, a_i by $a_1 + a_i, 0$ where $i \geq 2$. Similarly, we can do similar things to study the structure of the sequence $\{b_j\}$. Eventually, we obtain a simple inequality to optimize.

Solution. Let k be the maximum value. In other words, we are going to prove

$$\frac{\sum_{i=1}^n a_i(a_i + b_i)}{\sum_{i=1}^n b_i(a_i + b_i)} \leq k \quad (1)$$

for a suitable choice of k such that the equality case exists. By considering $a_i = b_i = 1$ for all i , we see that $k \geq 1$. (If one is writing a solution, of course the best is to state the value of k at the beginning. But here we are going to illustrate how we can determine its value without knowing what it is.)

Rearranging the terms, the inequality is just $k \sum_{i=1}^n b_i(a_i + b_i) - \sum_{i=1}^n a_i(a_i + b_i) \geq 0$, which means

$$f(a_1, \dots, a_n, b_1, \dots, b_n) = k \sum_{i=1}^n b_i^2 + (k-1) \sum_{i=1}^n a_i b_i - \sum_{i=1}^n a_i^2 \geq 0. \quad (2)$$

If we permute the terms in $\{a_j\}$ and permute the terms in $\{b_j\}$, their sums as well as their sums of squares remain unchanged. Therefore, f is minimized when $a_1 b_1 + a_2 b_2 + \dots + a_n b_n$ is minimized (notice that we have used $k-1 \geq 0$ here). By the

rearrangement inequality, this is minimized when it is a reverse sum. WLOG assume $a_1 \geq a_2 \geq \dots \geq a_n$ and $b_1 \leq b_2 \leq \dots \leq b_n$.

Next, we are going to prove

$$f(a_1, \dots, a_i, \dots, a_n, b_1, \dots, b_n) \geq f(a_1 + a_i, \dots, 0, \dots, a_n, b_1, \dots, b_n)$$

for any $2 \leq i \leq n$. This clearly preserves the condition. Removing like terms, this is equivalent to

$$(k-1)(a_1 b_1 + a_i b_i) - (a_1^2 + a_i^2) \geq (k-1)(a_1 + a_i)b_1 - (a_1 + a_i)^2,$$

i.e.

$$(k-1)a_i(b_i - b_1) + 2a_1 a_i \geq 0.$$

This is true since $k \geq 1$, $b_i \geq b_1$ and $a_1, a_i \geq 0$. Thus, to prove (2), it suffices to consider the case when $a_2 = a_3 = \dots = a_n = 0$. We now drop these inputs in f and restate the question as follows. We are given $b_1 + b_2 + \dots + b_n = a_1 > 0$, and we want to prove

$$f(a_1, b_1, \dots, b_n) = k(b_1^2 + b_2^2 + \dots + b_n^2) + (k-1)a_1 b_1 - a_1^2 \geq 0.$$

By the power mean inequality, it is obvious that $b_2^2 + b_3^2 + \dots + b_n^2$ (and hence f) is minimized when $b_2 = b_3 = \dots = b_n$. Thus, we may assume $b_2 = b_3 = \dots = b_n = \frac{a_1 - b_1}{n-1}$ and write f as a function in a_1 and b_1 . Indeed, we now need to prove

$$f(a_1, b_1) = k b_1^2 + \frac{k(a_1 - b_1)^2}{n-1} + (k-1)a_1 b_1 - a_1^2 \geq 0.$$

This is a simple problem as it is a quadratic inequality in two variables (and as it is homogeneous, it is essentially a one variable quadratic inequality). For instance, we rewrite f as

$$f(a_1, b_1) = \frac{kn}{n-1} b_1^2 + \frac{(n-3)k - (n-1)}{n-1} a_1 b_1 + \frac{k-n+1}{n-1} a_1^2.$$

If $b_1 = 0$, then we need $f(a_1, 0) = \frac{k-n+1}{n-1} a_1^2 \geq 0$. In other words, we need $k \geq n-1$. On the other hand, whenever $k \geq n-1$, we see that

$$\frac{(n-3)k - (n-1)}{n-1} \geq \frac{(n-3)(n-1) - (n-1)}{n-1} = n-4 \geq 0,$$

and so the coefficients of $b_1^2, a_1 b_1, a_1^2$ are all nonnegative. It is then clear that $f \geq 0$. As $a_1 > 0$, equality holds when $k = n-1$ and $b_1 = 0$.

To conclude, we have proved the inequality (1) when $k = n-1$. Also, equality holds when $a_2 = a_3 = \dots = a_n = b_1 = 0$ and $b_2 = b_3 = \dots = b_n = \frac{1}{n-1} a_1$. This shows $n-1$ is the desired maximum value. $\square$

Remarks. This example perfectly demonstrates the idea of the local refinement method and provides some useful ideas for solving problems in this way.

This section is mainly about the local refinement method. We give the summary of several ideas used in some problems as follows. Of course in practice there are much more variations of these, which greatly depends on the expressions and the equality cases.

- regarding the expression as a function in one variable ([problems 4.1, 4.2, 4.6](#))
- reducing the number of variables ([problems 4.2, 4.4, 4.5, 4.6, 4.7](#))
- mixing variables
 - replacing x, y by $a, x + y - a$ ([problem 4.3](#))
 - replacing x, y by $\frac{x+y}{2}, \frac{x+y}{2}$ ([problem 4.5](#))
 - replacing x, y by $x+y, 0$ ([problems 4.4, 4.7](#))
 - replacing x, y by $\sqrt{\frac{x^2+y^2}{2}}, \sqrt{\frac{x^2+y^2}{2}}$ ([problem 4.6](#))
 - mixing part of the variables ([problems 4.4, 4.5, 4.6, 4.7](#))

5 Calculus

Calculus is the most powerful tool in dealing with inequalities. Indeed, the word ‘inequality’ in advanced mathematics is more or less regarded as calculus (and analysis). Calculus is usually not introduced to the beginners in the Olympiad because it is considered as a topic which is out of syllabus. However, there is a growing trend that more problems (especially harder problems) can be solved using this advanced tool, and one may have a clearer picture to find out the correct ideas to solve the problems with such insight.

Calculus is composed of two parts, differentiation and integration. Differentiation is more useful for our purpose. In particular, one should learn the concepts of convexity and concavity, and also the important Jensen’s inequality. More advanced tools such as partial derivatives and the method of Lagrange multiplier are covered in **Optimization of Multivariable Functions 1.0**.

5.1 Basic Differentiation

For the sake of completeness, we first provide some basic knowledge about differentiation. Those who are familiar with such knowledge may skip this part and jump to the problems directly.

In the following, we always deal with a real-valued function $f : I \rightarrow \mathbb{R}$ defined on an interval I such as (a, b) , $[a, \infty)$, $(-\infty, \infty)$, etc.

Definition 5.1. Let $f : I \rightarrow \mathbb{R}$ be a function where I is an interval in $\mathbb{R}$, and let c be an interior point of I . We say that the *limit* (極限) of f at c is $L \in \mathbb{R}$ if the following holds. For any $\varepsilon > 0$, there exists $\delta > 0$ such that if $0 < |x - c| < \delta$ and $x \in I$, then $|f(x) - L| < \varepsilon$.

In that case, we write

$$\lim_{x \rightarrow c} f(x) = L.$$

We can also say that f tends to L when x tends to c , or write $f(x) \rightarrow L$ when $x \rightarrow c$. Loosely speaking, when x is close to c but not equal to c , then $f(x)$ is close to L . Since x is always different from c , the value of $f(c)$ is irrelevant.

We have the following basic facts. Suppose $\lim_{x \rightarrow c} f(x) = L$ and $\lim_{x \rightarrow c} g(x) = L'$ where both f and g are defined on I .

- The limit L is uniquely determined.
- If $f(x) \leq g(x)$ for any $x \in I$, then $L \leq L'$.

- If $m \leq f(x) \leq M$ for any $x \in I$, then $m \leq L \leq M$.
- $\lim_{x \rightarrow c} k = k$ for any $k \in \mathbb{R}$
- $\lim_{x \rightarrow c} x = c$
- $\lim_{x \rightarrow c} (f + g)(x) = L + L'$
- $\lim_{x \rightarrow c} (fg)(x) = LL'$
- $\lim_{x \rightarrow c} \left(\frac{f}{g}\right)(x) = \frac{L}{L'}$ if $L' \neq 0$

Definition 5.2. Let $f : I \rightarrow \mathbb{R}$ be a function where I is an interval in $\mathbb{R}$, and let c be an interior point of I . We say that the *right-hand limit* (右極限) of f at c is $L \in \mathbb{R}$ if the following holds. For any $\varepsilon > 0$, there exists $\delta > 0$ such that if $c < x < c + \delta$ and $x \in I$, then $|f(x) - L| < \varepsilon$.

In that case, we write

$$\lim_{x \rightarrow c^+} f(x) = L.$$

One can define the *left-hand limit* (左極限) $\lim_{x \rightarrow c^-} f(x)$ in an analogous way. These are called *one-sided limits* (單側極限). These mean we only consider the behaviour of $f(x)$ when x approaches c from one side. One basic result is that

$$\lim_{x \rightarrow c} f(x) = L \quad \Leftrightarrow \quad \lim_{x \rightarrow c^+} f(x) = \lim_{x \rightarrow c^-} f(x) = L.$$

It is also possible to define one-sided limits if c is an endpoint of I (even if $c \notin I$). For example, if $I = (a, b)$ where $a, b \in \mathbb{R}$, we can define $\lim_{x \rightarrow a^+} f(x)$ and $\lim_{x \rightarrow b^-} f(x)$ analogously. Since only the right-hand limit or the left-hand limit at an endpoint can be defined, we can simply write $\lim_{x \rightarrow a} f(x)$ or $\lim_{x \rightarrow b} f(x)$ to mean these one-sided limits.

Definition 5.3. Let $f : I \rightarrow \mathbb{R}$ be a function where I is an interval of the form (a, ∞) , $[a, \infty)$ or $(-\infty, \infty)$. We say that f tends to $L \in \mathbb{R}$ when x tends to ∞ if the following holds. For any $\varepsilon > 0$, there exists $M > 0$ such that if $x > M$ and $x \in I$, then $|f(x) - L| < \varepsilon$.

In that case, we write

$$\lim_{x \rightarrow \infty} f(x) = L.$$

One can define $\lim_{x \rightarrow -\infty} f(x) = L$, $\lim_{x \rightarrow c} f(x) = \pm\infty$ and $\lim_{x \rightarrow \pm\infty} f(x) = \pm\infty$ in analogous ways. These are just more formal ways to express the intuitive understanding of limits.

Using the concept of limits, we can now define a continuous function rigorously.

Definition 5.4. Let $f : I \rightarrow \mathbb{R}$ be a function where I is an interval in $\mathbb{R}$, and let $c \in I$. We say that f is *continuous*(連續) at c if

$$\lim_{x \rightarrow c} f(x) = f(c).$$

Otherwise, we say that f is *discontinuous*(不連續) at c .

We say that f is a *continuous function*(連續函數) if it is continuous at every point.

Most common functions like polynomials (e.g. $f(x) = x^2$), exponential functions (e.g. $f(x) = 2^x$) and logarithm functions (e.g. $\log x$) are continuous. Also, it is known that if f and g are continuous, so are $f + g$, fg , $\frac{f}{g}$ and $g \circ f$.

Using these facts, we can easily compute some limits. For example, we have

$$\lim_{x \rightarrow 2} \frac{3x + \sqrt{2x}}{2^x - 1} = \frac{3 \lim_{x \rightarrow 2} x + \sqrt{2 \lim_{x \rightarrow 2} x}}{\lim_{x \rightarrow 2} 2^x - 1} = \frac{3(2) + \sqrt{2(2)}}{2^2 - 1} = \frac{8}{3}.$$

Definition 5.5. Let $f : I \rightarrow \mathbb{R}$ be a function where I is an open interval in $\mathbb{R}$, and let $c \in I$. We say that f is *differentiable*(可微分) at c if

$$\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$$

exists. In that case, this is called the *derivative*(導數) of f at c , and is denoted by $\frac{df}{dx}(c)$ or $f'(c)$.

We say that f is a *differentiable function*(可微函數) if it is differentiable at every point.

The derivative of a function measures the rate of change of the function. Indeed, if we look at the graph of $y = f(x)$, we find that the quantity $\frac{f(x) - f(c)}{x - c}$ is the slope between the two points $(x, f(x))$ and $(c, f(c))$ on the graph. When x tends to c , it approaches the slope of the tangent to the graph at c , if the limit exists.

Sometimes we can also define the differentiability at the endpoints (such as $f'(a)$ if $I = [a, b]$), but it is less useful for our purpose.

Let $f : I \rightarrow \mathbb{R}$ be a differentiable function. Then we can regard f' as another function defined on I . If it turns out that f' is also differentiable, we can further define the

derivative of f' , which is denoted by f'' and is called the *second derivative*(二階導數) of f .

The following lists out the derivatives of some common functions.

Proposition 5.1. The following results hold.

- (a) If $f(x) = k$ where k is a constant, then $f'(x) = 0$.
- (b) If $f(x) = x^k$ where $k \in \mathbb{R}$, then $f'(x) = kx^{k-1}$.
- (c) If $f(x) = \sin x$, then $f'(x) = \cos x$.
- (d) If $f(x) = \cos x$, then $f'(x) = -\sin x$.
- (e) If $f(x) = e^x$, then $f'(x) = e^x$.
- (f) If $f(x) = \ln x$, then $f'(x) = \frac{1}{x}$.

Although **proposition 5.1** only gives the derivatives of a few functions, they can be combined together to give a lot of results, based on the following facts.

Proposition 5.2. Let $f(x), g(x), h(x)$ be differentiable functions. Then the following hold.

- (a) $(kf)' = kf'$ where k is a constant
- (b) $(f \pm g)' = f' \pm g'$
- (c) $(fg)' = fg' + f'g$
- (d) $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$ where $g(x) \neq 0$
- (e) (**Chain rule**(鏈式法則)) $f'(x) = h'(g(x))g'(x)$ where $f(x) = h(g(x))$

The chain rule gives the derivative of a composition of functions. For instance, the function $f(x) = (x^2 + 1)^8$ can be regarded as the composition $h(g(x))$ where $g(x) = x^2 + 1$ and $h(x) = x^8$. Since $g'(x) = 2x$ and $h'(x) = 8x^7$, we can use the chain rule to deduce

$$f'(x) = h'(x^2 + 1)g'(x) = 8(x^2 + 1)^7(2x) = 16x(x^2 + 1)^7.$$

Another example of computing the derivative of a simple function is given below.

$$\begin{aligned}
 \frac{d}{dx} \left(\frac{\sqrt{2x+1}}{x} \right) &= \frac{x \frac{d}{dx} (2x+1)^{\frac{1}{2}} - (2x+1)^{\frac{1}{2}} \frac{dx}{dx}}{x^2} \\
 &= \frac{x \cdot \frac{1}{2} (2x+1)^{-\frac{1}{2}} \frac{d}{dx} (2x+1) - (2x+1)^{\frac{1}{2}}}{x^2} \\
 &= \frac{x \cdot \frac{1}{2} (2x+1)^{-\frac{1}{2}} \cdot 2 - (2x+1)^{\frac{1}{2}}}{x^2} \\
 &= -\frac{x+1}{x^2 \sqrt{2x+1}}
 \end{aligned}$$

Here gives a simpler way to compute the derivative of a product of functions such as $f(x) = \frac{(x+1)^3(2x-1)}{x^5}$ without using the product or quotient rules. Indeed, we have

$$\begin{aligned}
 f(x) &= \frac{(x+1)^3(2x-1)}{x^5} \\
 \Rightarrow \ln f(x) &= \ln \frac{(x+1)^3(2x-1)}{x^5} \\
 \Rightarrow \ln f(x) &= 3 \ln(x+1) + \ln(2x-1) - 5 \ln x \\
 \Rightarrow \frac{f'(x)}{f(x)} &= \frac{3}{x+1} + \frac{2}{2x-1} - \frac{5}{x} \quad (\text{by taking derivatives}) \\
 \Rightarrow \frac{f'(x)}{f(x)} &= \frac{-2x^2 - 6x + 5}{x(x+1)(2x-1)} \\
 \Rightarrow f'(x) &= \frac{(x+1)^2(-2x^2 - 6x + 5)}{x^6}.
 \end{aligned}$$

We now proceed to see how differentiation can be used in the topic of inequalities. Recall that differentiation is a measure of the rate of change of the function. In particular, it is easy to tell when a differentiable function is increasing or decreasing.

Proposition 5.3. Let $f(x)$ be a differentiable function defined on an open interval I . Then the following hold.

- (a) If $f'(x) > 0$ (or $f'(x) \geq 0$) for any $x \in I$, then f is strictly (monotonic) increasing on I .
- (b) If $f'(x) < 0$ (or $f'(x) \leq 0$) for any $x \in I$, then f is strictly (monotonic) decreasing on I .
- (c) If f has a local maximum (or minimum) at $c \in I$, then $f'(c) = 0$.

A point $c \in I$ satisfying $f'(c) = 0$ is called a **critical point**(駐點/臨界點). Note that $f'(c) = 0$ does not imply c is a local maximum or minimum, as (c) only says that the converse is true. The word ‘local’ means that the point is only an extremum point relative to the points which are sufficiently close to it. For instance, the function

$$f(x) = x^4 - x^2$$

has a local maximum at $x = 0$ because $f(x) = x^2(x^2 - 1) \leq 0 = f(0)$ for any $x \in (-1, 1)$, where $(-1, 1)$ is an open interval containing 0. However, f does not have any (global) maximum since $\lim_{x \rightarrow \pm\infty} f(x) = \infty$.

Thus, a function having local extrema may not necessarily have global extrema. Therefore, more should be justified before we claim that such a point is an extremum point on the whole interval I . Actually, if we are able to solve the equation $f'(x) = 0$, then normally we are able to solve the inequalities $f'(x) > 0$ and $f'(x) < 0$. Then we can use parts (a) and (b) to tell whether f has extremum points. For example, if f is decreasing on (a, c) and is increasing on (c, b) (and is continuous), then f must attain its minimum at c . Thus, these results are more useful than part (c).

Finally, the results of parts (a) and (b) can be slightly modified so that they still hold when I includes some of the endpoints (e.g. $I = [a, b]$). But part (c) is only true for interior points. For example, the function $f(x) = x^2$ defined on $[0, 1]$ clearly has a maximum value at $x = 1$, but we have $f'(1) = 2 \neq 0$. On the other hand, we can use $f'(x) = 2x \geq 0$ to justify that f is increasing, and hence $f(x) \leq f(1)$ for any $x \in [0, 1]$.

To prove (a) and (b), we need the following well-known theorem.

Theorem 5.1. (**Mean value theorem**(中值定理)). Let f be a differentiable function defined on an interval I . Let $a < b$ be two points in I . Then there exists $c \in (a, b)$ such that $f'(c) = \frac{f(b) - f(a)}{b - a}$.

Equivalently, this means there exists a point c between a and b such that the tangent line to the graph $y = f(x)$ at $x = c$ is parallel to the line joining the points $(a, f(a))$ and $(b, f(b))$ on the graph. The proof of the mean value theorem requires **proposition 5.3(c)** together with other deeper results. So we omit the proof here. In the following, we give a proof to **proposition 5.3**.

Proof.

(a) Suppose $f'(x) > 0$ for any $x \in I$. For any $a, b \in I$ with $a < b$, by the mean value theorem, there exists $c \in (a, b)$ (hence $c \in I$) such that $f'(c) = \frac{f(b) - f(a)}{b - a}$. Since $f'(c) > 0$ and $b - a > 0$, we must have $f(b) > f(a)$. As this holds for arbitrary a, b , this means f is strictly increasing.

(b) The proof is similar to that of part (a).

(c) Suppose c is a local maximum point. By definition, there exists $\delta > 0$ such that $f(c) \geq f(x)$ for any $x \in (c - \delta, c + \delta) \cap I$. Recall that

$$f'(c) = \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}.$$

For $x \in (c, c + \delta) \cap I$, we have $\frac{f(x) - f(c)}{x - c} \leq 0$ since $x > c$. This implies $f'(c) \leq 0$.

For $x \in (c - \delta, c) \cap I$, we have $\frac{f(x) - f(c)}{x - c} \geq 0$ since $x < c$. This implies $f'(c) \geq 0$. Combining these, we must have $f'(c) = 0$ as desired.

□

How do we use [proposition 5.3](#) to solve some inequality problems? For example, recall that in [problem 4.2](#), in the end we need to prove that

$$f(s) = 8s^3 - 84s^2 + 383s - 534 \geq 0$$

for any $s \in \left[\frac{5}{2}, 3\right]$. In that solution, we manually rewrite this as

$$f(s) = (2s - 5) \left(4s^2 - 32s + \frac{223}{2} \right) + \frac{47}{2}$$

to prove the assertion. However, sometimes it may not be easy or natural to rewrite the function in this way. Alternatively, we can compute

$$f'(s) = 24s^2 - 168s + 383.$$

It is not hard to check that $\Delta < 0$. This implies $f'(s) > 0$, and so f is increasing by [proposition 5.3\(a\)](#). Thus, we have

$$f(s) \geq f\left(\frac{5}{2}\right) > 125 - 21 \cdot 25 + 191 \cdot 5 - 534 > 0.$$

Let's see some more examples of inequality problems that can be solved using differentiation.

Problem 5.1. Prove that $t^{10} - 3t^3 + 2 \geq 0$ for any real number $t \geq 1$.

Analysis. Of course one solution is to observe the equality case $t = 1$ and obtain the factorization

$$t^{10} - 3t^3 + 2 = (t - 1)(t^9 + t^8 + t^7 + t^6 + t^5 + t^4 + t^3 - 2t^2 - 2t - 2).$$

However, sometimes it is troublesome to do factorization, especially when the degree is even higher. Here, we demonstrate how to use differentiation to prove the inequality.

Solution. Let $f(t) = t^{10} - 3t^3 + 2$. Then $f'(t) = 10t^9 - 9t^2$. For $t \geq 1$, we have

$$f'(t) = 10t^9 - 9t^2 \geq 10t^2 - 9t^2 = t^2 > 0.$$

By [proposition 5.3\(a\)](#), f is strictly increasing on $[1, \infty)$. Therefore,

$$f(t) \geq f(1) = 0$$

for any $t \in [1, \infty)$. □

Problem 5.2. Let a be a real number and b a positive real number. Find the real number r such that

$$r(x - a) \leq \sqrt{x^2 + b} - \sqrt{a^2 + b}$$

for any real number x .

Analysis. This is a simplified version of CMO 1988 Problem 1. As equality clearly holds when $x = a$. We may assume $x \neq a$ and consider the function

$$f(x) = \frac{\sqrt{x^2 + b} - \sqrt{a^2 + b}}{x - a}$$

We can use calculus to study the behaviour of the function f , and hence determine the value r .

Solution. Let $f(x) = \frac{\sqrt{x^2 + b} - \sqrt{a^2 + b}}{x - a}$ for $x \neq a$. Then

$$\begin{aligned} f'(x) &= \frac{(x - a) \frac{d}{dx}(\sqrt{x^2 + b}) - (\sqrt{x^2 + b} - \sqrt{a^2 + b}) \frac{d}{dx}(x - a)}{(x - a)^2} \\ &= \frac{(x - a) \frac{2x}{2\sqrt{x^2 + b}} - (\sqrt{x^2 + b} - \sqrt{a^2 + b})}{(x - a)^2} \\ &= \frac{\sqrt{a^2 + b}\sqrt{x^2 + b} - ax - b}{(x - a)^2\sqrt{x^2 + b}}. \end{aligned}$$

By the Cauchy-Schwarz inequality, we have

$$(a^2 + b)(x^2 + b) \geq (ax + b)^2.$$

Thus, we see that $f'(x) \geq 0$ for any $x \neq a$. By [proposition 5.3\(a\)](#), f is increasing everywhere on its domain.

Next, we want to study the behaviour of $f(x)$ when x approaches a . A standard way

is to rewrite f as follows:

$$\begin{aligned} f(x) &= \frac{\sqrt{x^2 + b} - \sqrt{a^2 + b}}{x - a} \\ &= \frac{(x^2 + b) - (a^2 + b)}{(x - a)(\sqrt{x^2 + b} + \sqrt{a^2 + b})} \\ &= \frac{x + a}{\sqrt{x^2 + b} + \sqrt{a^2 + b}}. \end{aligned}$$

In this form of the function, we are able to compute $f(a)$ (technically we have to extend the domain).

Now, for $x > a$, the inequality

$$r(x - a) \leq \sqrt{x^2 + b} - \sqrt{a^2 + b} \quad (1)$$

is true iff $r \leq f(x)$. From above, we have

$$\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^+} \frac{x + a}{\sqrt{x^2 + b} + \sqrt{a^2 + b}} = \frac{a}{\sqrt{a^2 + b}}.$$

This shows $r \leq \frac{a}{\sqrt{a^2 + b}}$. Conversely, for any such r , we have $r \leq \frac{a}{\sqrt{a^2 + b}} \leq f(x)$ since f is increasing.

Similarly, (1) is true for $x < a$ iff $r \geq f(x)$. Again, we have

$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^-} \frac{x + a}{\sqrt{x^2 + b} + \sqrt{a^2 + b}} = \frac{a}{\sqrt{a^2 + b}},$$

so that $r \geq \frac{a}{\sqrt{a^2 + b}}$. Conversely, for any such r , we have $r \geq \frac{a}{\sqrt{a^2 + b}} \geq f(x)$ since f is increasing.

Combining these, $r = \frac{a}{\sqrt{a^2 + b}}$ is the only possibility, and for this choice of r , the inequality holds from above. $\square$

In the solution of the above problem, we wish to compute

$$\lim_{x \rightarrow a} \frac{\sqrt{x^2 + b} - \sqrt{a^2 + b}}{x - a}.$$

We cannot replace x by a to compute the limit because we merely get a fraction of the form $\frac{0}{0}$, whose value cannot be determined. In the solution, we rewrite the function as a quotient of two expressions such that the denominator is nonzero at $x = a$. However, it is not always possible to do such a transformation. One useful way to compute such a limit is to use the following well-known result.

Theorem 5.2. (L'Hôpital's rule(羅必達法則)). Let f and g be differentiable functions defined on an interval I , and let $c \in I$. If $\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} g(x) = 0$ and

$$\lim_{x \rightarrow c} \frac{f'(x)}{g'(x)} = L,$$

then

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = L.$$

In short, to compute the limit of $\frac{f}{g}$ which is of an indeterminate form $\frac{0}{0}$, we just need to compute the limit of $\frac{f'}{g'}$. However, it is not quite correct to write

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}$$

because it is possible that $\lim_{x \rightarrow c} \frac{f(x)}{g(x)}$ exists but $\lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}$ does not exist. Therefore, in the theorem, we first require that $\lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}$ is equal to some real number L (or $\pm\infty$).

The same result holds if $\frac{f}{g}$ is of the indeterminate form $\frac{\infty}{\infty}$. This means it holds if $\lim_{x \rightarrow c} f(x) = \pm\infty$ and $\lim_{x \rightarrow c} g(x) = \pm\infty$ (instead of both 0). Other indeterminate forms such as $0 \cdot \infty$, $\infty - \infty$, 1^∞ , 0^0 and ∞^0 can often be transformed to the indeterminate forms $\frac{0}{0}$ or $\frac{\infty}{\infty}$.

As an example, to compute

$$\lim_{x \rightarrow a} \frac{\sqrt{x^2 + b} - \sqrt{a^2 + b}}{x - a},$$

since it is of the indeterminate form $\frac{0}{0}$, we have

$$\begin{aligned} \lim_{x \rightarrow a} \frac{\sqrt{x^2 + b} - \sqrt{a^2 + b}}{x - a} &= \lim_{x \rightarrow a} \frac{(\sqrt{x^2 + b} - \sqrt{a^2 + b})'}{(x - a)'} = \lim_{x \rightarrow a} \frac{\frac{1}{2}(x^2 + b)^{-\frac{1}{2}}(2x)}{1} \\ &= \lim_{x \rightarrow a} \frac{x}{\sqrt{x^2 + b}} = \frac{a}{\sqrt{a^2 + b}}. \end{aligned}$$

This agrees with the result we found before, but the steps are much simpler.

In the next problem, we need to use another result stated below.

Theorem 5.3. (Extreme value theorem(極値定理)). Let f be a continuous function defined on a closed and bounded interval $[a, b]$. Then f attains its maximum and minimum values.

This theorem can often be used to show the existence of extremum values. If we know that a differentiable function f has extrema on $[a, b]$, then by proposition 5.3(c), we are sure that the extremum points can only be the solutions to $f'(x) = 0$ or the endpoints a, b . Usually there are only finitely many possibilities to test, so we can easily tell which one is the maximum and which one is the minimum. On the other hand, if we do not know whether f has extrema, then we cannot get any information by solving $f'(x) = 0$, and usually no point is awarded to such an incomplete solution. This suggests the importance of the extreme value theorem.

Problem 5.3. Let x be a real number satisfying $0 < x < \pi$, and let n be a positive integer. Prove that

$$\frac{\sin x}{1} + \frac{\sin 2x}{2} + \cdots + \frac{\sin nx}{n} > 0.$$

Analysis. This is a classical problem which is not easy to solve. One natural idea is to use induction because there is only one extra term in the inductive step. However, it is not always true that $\frac{\sin(n+1)x}{n+1} \geq 0$. Another possible idea is to use differentiation to find the minimum of the function. But first we need to show that the function does have a minimum value (on some domain). To apply the extreme value theorem, we need a closed and bounded interval. Thus, we may extend the function to the domain $[0, \pi]$. Then we can find some more information on the minimum point to derive the inequality.

Solution. We are going to prove

$$\frac{\sin x}{1} + \frac{\sin 2x}{2} + \cdots + \frac{\sin nx}{n} > 0 \quad (1)$$

for any $x \in (0, \pi)$ by induction on n (no matter the inductive hypothesis is useful or not, we can always make such an assumption). The base case $n = 1$ is trivial since $\sin x > 0$ for $0 < x < \pi$. Assume (1) holds for some n , and we now consider

$$f(x) = \frac{\sin x}{1} + \frac{\sin 2x}{2} + \cdots + \frac{\sin nx}{n} + \frac{\sin(n+1)x}{n+1}.$$

Suppose on the contrary that $f(x) \leq 0$ for some $x \in (0, \pi)$. Note that f attains the minimum value on $[0, \pi]$ by the extreme value theorem. As $f(0) = f(\pi) = 0$ and we have assumed $f(x) \leq 0$ for some $x \in (0, \pi)$, we may choose $t \in (0, \pi)$ such that $f(t)$ is the minimum value (hence $f(t) \leq 0$). By proposition 5.3(c), we must have $f'(t) = 0$. Now,

$$f'(t) = \cos t + \cos 2t + \cdots + \cos(n+1)t.$$

There is a standard telescoping method that can compute this sum. Indeed, by some trigonometric identities, we have

$$\begin{aligned} f'(t) &= \frac{1}{2 \sin \frac{t}{2}} \sum_{k=1}^{n+1} 2 \sin \frac{t}{2} \cos kt \\ &= \frac{1}{2 \sin \frac{t}{2}} \sum_{k=1}^{n+1} \left[\sin \left(kt + \frac{t}{2} \right) - \sin \left(kt - \frac{t}{2} \right) \right] \\ &= \frac{1}{2 \sin \frac{t}{2}} \left[\sin \left(nt + \frac{3t}{2} \right) - \sin \frac{t}{2} \right]. \end{aligned}$$

Since $f'(t) = 0$, we have $\sin \left(nt + \frac{3t}{2} \right) = \sin \frac{t}{2}$. There are two possible cases.

- If $nt + \frac{3t}{2} = \frac{t}{2} + 2m\pi$ for some $m \in \mathbb{Z}$, then $(n+1)t = 2m\pi$ so that $\sin(n+1)t = 0$. This yields

$$f(t) = \frac{\sin t}{1} + \frac{\sin 2t}{2} + \dots + \frac{\sin nt}{n} > 0$$

by the inductive hypothesis.

- If $nt + \frac{3t}{2} = (2m+1)\pi - \frac{t}{2}$ for some $m \in \mathbb{Z}$, then $(n+1)t = (2m+1)\pi - t$ so that $\sin(n+1)t > 0$ from $t \in (0, \pi)$. This yields

$$f(t) > \frac{\sin t}{1} + \frac{\sin 2t}{2} + \dots + \frac{\sin nt}{n} > 0$$

by the inductive hypothesis.

In either case, we obtain a contradiction to $f(t) \leq 0$. So $f(x) > 0$ for any $x \in (0, \pi)$, and we are done by induction. $\square$

Next, we introduce one more technique in proving inequalities called the *tangent line trick*. Suppose we are going to prove

$$f(x_1) + f(x_2) + \dots + f(x_n) \geq nf(c)$$

under the constraint $x_1 + x_2 + \dots + x_n = nc$, where c is a constant. Note that an equality case is $x_1 = x_2 = \dots = x_n = c$. One possible way is to prove that

$$f(x) \geq f(c) + f'(c)(x - c) \tag{1}$$

for any x . If this holds, then we plainly obtain

$$f(x_1) + f(x_2) + \dots + f(x_n) \geq nf(c) + f'(c)(x_1 + x_2 + \dots + x_n - nc) = nf(c)$$

as desired. The method also works if the inequality sign is changed. The name of this method is due to the fact that $y = f(c) + f'(c)(x - c)$ is the tangent line to the graph of f at c . The logic is that this tangent line is a good approximation to the function f , and so there is some hope that the inequality (1) holds.

Problem 5.4. Let a, b, c be positive real numbers. Prove that

$$\frac{(4a + b + c)^2}{5a^2 + (b + c)^2} + \frac{(4b + c + a)^2}{5b^2 + (c + a)^2} + \frac{(4c + a + b)^2}{5c^2 + (a + b)^2} \leq 12.$$

Analysis. To use the tangent line trick, we first need to rewrite the expression in the form $f(a) + f(b) + f(c)$ for some function f . This can be done by using a normalization condition $a + b + c = 1$.

Solution. The inequality is homogeneous. Using normalization, we may assume $a + b + c = 1$. Note that

$$\frac{(4a + b + c)^2}{5a^2 + (b + c)^2} = \frac{(3a + 1)^2}{5a^2 + (1 - a)^2} = \frac{(3a + 1)^2}{6a^2 - 2a + 1}.$$

Let $f(x) = \frac{(3x + 1)^2}{6x^2 - 2x + 1}$. Then we have

$$f'(x) = \frac{(6x^2 - 2x + 1)(2)(3x + 1)(3) - (3x + 1)^2(12x - 2)}{(6x^2 - 2x + 1)^2} = \frac{(3x + 1)(-18x + 8)}{(6x^2 - 2x + 1)^2}.$$

The equality case of the desired inequality should be $a = b = c = \frac{1}{3}$. Since $f\left(\frac{1}{3}\right) = 4$ and $f'\left(\frac{1}{3}\right) = 4$, we want to prove

$$f(x) \leq 4 + 4\left(x - \frac{1}{3}\right).$$

Indeed, we have

$$\begin{aligned} f(x) &\leq 4 + 4\left(x - \frac{1}{3}\right) \\ \Leftrightarrow \frac{(3x + 1)^2}{6x^2 - 2x + 1} &\leq \frac{12x + 8}{3} \\ \Leftrightarrow 3(3x + 1)^2 &\leq (12x + 8)(6x^2 - 2x + 1) \\ \Leftrightarrow 0 &\leq 72x^3 - 3x^2 - 22x + 5 \\ \Leftrightarrow 0 &\leq (3x - 1)^2(8x + 5). \end{aligned}$$

This holds for any $x > 0$. Therefore, we have

$$f(a) + f(b) + f(c) \leq 12 + 4\left(a - \frac{1}{3} + b - \frac{1}{3} + c - \frac{1}{3}\right) = 12.$$

Equality holds when $a = b = c$. □

Problem 5.5. Let a, b, c be positive real numbers satisfying $a^2 + b^2 + c^2 = 3$. Prove that

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{4}{3}(a + b + c) \geq 7.$$

Analysis. This time the condition is $a^2 + b^2 + c^2 = 3$, and so it is not too useful even if we obtain an inequality of the form

$$f(x) \geq f(t) + f'(t)(x - t).$$

We may modify the method and aim at proving an inequality of the form

$$f(x) \geq f(t) + C(x^2 - t) \quad (1)$$

for some constants C and t . Firstly, as the equality case should be $a = b = c = 1$, we should choose $t = 1$. Secondly, we not only require both sides of (1) to agree at $t = 1$, but we also want the derivatives of both sides to agree at $t = 1$ (this guarantees $t = 1$ is a double root, and so we can take out the factor $(x - 1)^2$ when proving the inequality). Thus, we would like to choose C such that $f'(x) = 2Cx$ at $x = 1$.

Equivalently, we choose $C = \frac{f'(1)}{2}$. This explains the mysterious solution below.

Solution. Let $f(x) = \frac{1}{x} + \frac{4}{3}x$. Then $f'(x) = -\frac{1}{x^2} + \frac{4}{3}$. As $f(1) = \frac{7}{3}$ and $f'(1) = \frac{1}{3}$, we first prove that

$$f(x) \geq \frac{7}{3} + \frac{1}{6}(x^2 - 1).$$

Indeed,

$$\begin{aligned} f(x) &\geq \frac{7}{3} + \frac{1}{6}(x^2 - 1) \\ \Leftrightarrow \frac{4x^2 + 3}{3x} &\geq \frac{x^2 + 13}{6} \\ \Leftrightarrow 0 &\geq x^3 - 8x^2 + 13x - 6 \\ \Leftrightarrow 0 &\geq (x - 1)^2(x - 6). \end{aligned}$$

Notice that $0 < a, b, c < \sqrt{3} < 6$. Therefore, the above inequality holds when $x = a, b, c$. It follows that

$$f(a) + f(b) + f(c) \geq 7 + \frac{1}{6}(a^2 - 1 + b^2 - 1 + c^2 - 1) = 7.$$

Equality holds when $a = b = c = 1$. □

5.2 Convexity and Jensen's Inequality

In addition to the monotonicity of a function, we may also consider its convexity to study the behaviour of the function to prove some inequalities. The definitions are given below.

Definition 5.6. Let f be a function defined on an interval I . We say that f is a *convex function*(凸函數) if

$$tf(x) + (1-t)f(y) \geq f(tx + (1-t)y)$$

for any $x, y \in I$ and any $0 < t < 1$.

Definition 5.7. Let f be a function defined on an interval I . We say that f is a *concave function*(凹函數) if

$$tf(x) + (1-t)f(y) \leq f(tx + (1-t)y)$$

for any $x, y \in I$ and any $0 < t < 1$.

Geometrically, if we choose any two points on the graph $y = f(x)$ of a convex function f , the segment joining the two points lies above the graph. One example is the graph $y = x^2$ (it is known that $f(x) = x^2$ is convex). Conversely, if we choose any two points on the graph $y = f(x)$ of a concave function f , the segment joining the two points lies below the graph.



In general, a function may not be entirely convex or concave. But in many cases, it can be broken up into intervals each of which is either convex or concave. A point is called an *inflection point*(反曲點/拐點) of a function if it is a point where the function changes its convexity (from convex to concave, or from concave to convex).

In general, it may not be an easy task to check when a function is convex or concave. But there are some simple ways to check if the function f is nice enough.

Proposition 5.4. Let f be a continuous function defined on an interval I . Then f is convex iff

$$\frac{f(x) + f(y)}{2} \geq f\left(\frac{x+y}{2}\right) \quad (1)$$

for any $x, y \in I$. On the other hand, f is concave iff the inequality sign is reversed.

Proof. If f is convex, then

$$tf(x) + (1-t)f(y) \geq f(tx + (1-t)y) \quad (2)$$

for any $x, y \in I$ and any $0 < t < 1$. Putting $t = \frac{1}{2}$ in (2), we obtain (1).

Conversely, if (1) holds for any $x, y \in I$, we first prove that (2) holds for any $t \in (0, 1)$ which is of the form $\frac{k}{2^n}$ for some positive integers k and n . We prove this by induction on n . The inequality (1) is exactly the base case $n = 1$ (since we must have $k = 1$ and so $t = \frac{1}{2}$). Assume the claim holds for some n . We consider the case $t = \frac{k}{2^{n+1}}$ where k is an odd integer.

WLOG assume $k < 2^n$ (otherwise interchange the roles of x and y). We have

$$tx + (1 - t)y = \frac{\frac{k}{2^n}x + \left(2 - \frac{k}{2^n}\right)y}{2} = \frac{\frac{k}{2^n}x + \left(1 - \frac{k}{2^n}\right)y}{2} + \frac{y}{2} = \frac{z + y}{2}$$

where $z = \frac{k}{2^n}x + \left(1 - \frac{k}{2^n}\right)y \in (0, 1)$. Using (1), we deduce

$$f\left(\frac{z + y}{2}\right) \leq \frac{f(z) + f(y)}{2}.$$

By the inductive hypothesis, we have

$$f(z) = f\left(\frac{k}{2^n}x + \left(1 - \frac{k}{2^n}\right)y\right) \leq \frac{k}{2^n}f(x) + \left(1 - \frac{k}{2^n}\right)f(y).$$

Combining all these, we see that

$$f(tx + (1 - t)y) = f\left(\frac{z + y}{2}\right) \leq \frac{k}{2^{n+1}}f(x) + \left(1 - \frac{k}{2^{n+1}}\right)f(y) = tf(x) + (1 - t)f(y).$$

This proves our claim by induction.

Lastly, it suffices to notice that every real number in $(0, 1)$ can be approximated by numbers of the form $\frac{k}{2^n}$ with an arbitrary precision (for example, every r lies in some interval $\left[\frac{k}{2^n}, \frac{k+1}{2^n}\right]$, so that $\left|r - \frac{k}{2^n}\right| \leq \frac{1}{2^n}$, which tends to 0 when $n \rightarrow \infty$). So we can use the continuity to prove (2) for any $t \in (0, 1)$. $\square$

The importance of this result is that we only need to check a weaker condition than the definition to show that f is convex (given that f is continuous). But we seldom use this result since the following result is even more useful for our purpose.

Proposition 5.5. Let f be a twice differentiable function (i.e. f'' exists) defined on an interval I . If $f''(x) \geq 0$ for any $x \in I$, then f is convex. On the other hand, if the inequality sign is reversed, then f is concave.

Proof. Consider

$$g(x) = tf(x) + (1-t)f(y) - f(tx + (1-t)y)$$

with fixed t, y where $0 < t < 1$. Then

$$g'(x) = tf'(x) - tf'(tx + (1-t)y).$$

Since $f''(x) \geq 0$, the function $f'(x)$ is increasing by [proposition 5.3\(a\)](#). Note that

$$\begin{aligned} x &\geq tx + (1-t)y \\ \Leftrightarrow x &\geq y \end{aligned}$$

as $t < 1$. Hence, $g'(x) \geq 0$ for $x \geq y$ and $g'(x) \leq 0$ for $x \leq y$. It follows that g is increasing when $x > y$ and is decreasing when $x < y$. This implies $g(x) \geq g(y) = 0$, which means f is convex by definition. $\square$

How can we use the convexity of a function to find its optimum values? For example, if f is convex on a closed and bounded interval $[a, b]$, then the shape of the graph of f is similar to that of $y = x^2$. In particular, the maximum value of f (which must exist due to the extreme value theorem) can only be attained at an endpoint $x = a$ or $x = b$. Let's see several examples to demonstrate the ideas.

Problem 5.6. (USAMO 1980 Problem 5) Prove that for numbers a, b, c in the interval $[0, 1]$,

$$\frac{a}{b+c+1} + \frac{b}{c+a+1} + \frac{c}{a+b+1} + (1-a)(1-b)(1-c) \leq 1.$$

Analysis. To use the convexity, we regard the left-hand side as a function in one of the variables, say $g(a)$. If we can show that g is a convex function, then it is maximized when $a = 0$ or $a = 1$. Similarly, it suffices to compare the values at the endpoints to find the maximum value.

Solution. Let

$$f(a, b, c) = \frac{a}{b+c+1} + \frac{b}{c+a+1} + \frac{c}{a+b+1} + (1-a)(1-b)(1-c).$$

For fixed $b, c \in [0, 1]$, define $g(a) = f(a, b, c)$. We see that

$$\begin{aligned} g'(a) &= \frac{1}{b+c+1} - \frac{b}{(c+a+1)^2} - \frac{c}{(a+b+1)^2} - (1-b)(1-c), \\ g''(a) &= \frac{2b}{(c+a+1)^3} + \frac{2c}{(a+b+1)^3} \geq 0. \end{aligned}$$

This shows g is convex by [proposition 5.5](#). Therefore, the maximum value of g (which exists by the extreme value theorem) is attained at the endpoint $a = 0$ or $a = 1$. In each case $a = 0$ or $a = 1$, we consider $h(b) = f(a, b, c)$. The same arguments show

that h attains its maximum value at $b = 0$ or $b = 1$. Similarly, in each subcase, the maximum value of $f(a, b, c)$ is attained at $c = 0$ or $c = 1$. Thus, it suffices to consider the 8 cases where each of a, b, c is 0 or 1. We easily find that

$$f(0, 0, 0) = f(0, 0, 1) = f(0, 1, 1) = f(1, 1, 1) = 1.$$

Therefore, the maximum value of $f(a, b, c)$ is 1, and the inequality is proved. Equality holds when $a, b, c \in \{0, 1\}$. $\square$

Remarks. The result can be easily generalized. If we have a function f in n variables such that each variable belongs to the domain $[a, b]$, and f is convex in each variable, then each variable must be a or b when f attains its maximum value.

Without using convexity or calculus, it is rather hard to prove the inequality in this problem. Therefore, it is important to learn this method.

Problem 5.7. (CMO 2016 Problem 6) Fix an integer $n \geq 2$ and positive reals $a < b$. Let $x_1, x_2, \dots, x_n$ be real numbers in the closed interval $[a, b]$. Find the maximum of the following expression:

$$\frac{\frac{x_1^2}{x_2} + \frac{x_2^2}{x_3} + \dots + \frac{x_{n-1}^2}{x_n} + \frac{x_n^2}{x_1}}{x_1 + x_2 + \dots + x_{n-1} + x_n}.$$

Analysis. Directly computing the second derivative of the function is tedious. We may first rearrange the terms to remove the denominator and study the convexity of the function.

Solution. Let

$$S = \frac{S_1}{S_2} = \frac{\frac{x_1^2}{x_2} + \frac{x_2^2}{x_3} + \dots + \frac{x_{n-1}^2}{x_n} + \frac{x_n^2}{x_1}}{x_1 + x_2 + \dots + x_{n-1} + x_n},$$

and let M be the maximum value of S (which exists by the extreme value theorem). We are going to prove $S \leq M$. Equivalently, we need to prove $S_1 - MS_2 \geq 0$. Let

$$f(x_1) = \frac{x_1^2}{x_2} + \frac{x_2^2}{x_3} + \dots + \frac{x_{n-1}^2}{x_n} + \frac{x_n^2}{x_1} - M(x_1 + x_2 + \dots + x_{n-1} + x_n).$$

Then we have

$$\begin{aligned} f'(x_1) &= \frac{2x_1}{x_2} - \frac{x_n^2}{x_1^2} - M, \\ f''(x_1) &= \frac{2}{x_2} + \frac{2x_n^2}{x_1^3} > 0. \end{aligned}$$

By [proposition 5.5](#), the function f is convex. Thus, we must have $x_1 = a$ or $x_1 = b$ when f attains the maximum value. Similarly, it suffices to consider the case when each x_j is a or b .

To proceed, we use the local refinement method. For example, we first see what will happen if we permute the variables. Note that S_2 is unchanged when we do so. Therefore, we just need to maximize S_1 . Suppose there are k copies of a and $n - k$ copies of b among the variables. Note that $a^2 < b^2$ and $\frac{1}{b} < \frac{1}{a}$. So we can use the rearrangement inequality to find an upper bound for S_1 , by regarding it as a random sum of the sequences $\underbrace{a^2, \dots, a^2}_{k \text{ copies}}, \underbrace{b^2, \dots, b^2}_{(n-k) \text{ copies}}$ and $\underbrace{\frac{1}{a}, \dots, \frac{1}{a}}_{k \text{ copies}}, \underbrace{\frac{1}{b}, \dots, \frac{1}{b}}_{(n-k) \text{ copies}}$.

- If $k \leq \frac{n}{2}$, then we have

$$S_1 \leq k \cdot \frac{a^2}{b} + (n - 2k) \cdot \frac{b^2}{b} + k \cdot \frac{b^2}{a} = \frac{k(a^3 - 2ab^2 + b^3) + nab^2}{ab}$$

by the rearrangement inequality. Together with $S_2 = ka + (n - k)b$, we have

$$S \leq \frac{k(a - b)(a^2 + ab - b^2) + nab^2}{ab(k(a - b) + nb)} = \frac{a^2 + ab - b^2}{ab} + \frac{n(b^2 - a^2)}{a(k(a - b) + nb)}.$$

Noting that $b^2 - a^2 > 0$ and $a - b < 0$, this is maximized when k is maximized, i.e. $k = \left\lfloor \frac{n}{2} \right\rfloor$. After simplification, we see that

$$S \leq \begin{cases} \frac{a^2 - ab + b^2}{ab} & \text{if } n \text{ is even,} \\ \frac{n(a^3 + b^3) - (a^3 - 2ab^2 + b^3)}{ab(n(a + b) - (a - b))} & \text{if } n \text{ is odd.} \end{cases}$$

- Similarly, if $k \geq \frac{n}{2}$, then we have

$$S_1 \leq (n - k) \cdot \frac{a^2}{b} + (2k - n) \cdot \frac{a^2}{a} + (n - k) \cdot \frac{b^2}{a} = \frac{n(a^3 - a^2b + b^3) - k(a^3 - 2a^2b + b^3)}{ab},$$

and so

$$S \leq \frac{n(a^3 - a^2b + b^3) - k(a - b)(a^2 - ab - b^2)}{ab(k(a - b) + nb)} = -\frac{a^2 - ab - b^2}{ab} + \frac{n(a^2 - b^2)}{b(k(a - b) + nb)}.$$

Noting that $a^2 - b^2 < 0$ and $a - b < 0$, this is maximized when k is minimized, i.e. $k = \left\lceil \frac{n}{2} \right\rceil$. After simplification, we see that

$$S \leq \begin{cases} \frac{a^2 - ab + b^2}{ab} & \text{if } n \text{ is even,} \\ \frac{n(a^3 + b^3) - (a^3 - 2a^2b + b^3)}{ab(n(a + b) + (a - b))} & \text{if } n \text{ is odd.} \end{cases}$$

Combining the two cases, if n is even, then the maximum value of S is $\frac{a^2 - ab + b^2}{ab}$, which is attained when the sequence is $a, b, a, b, \dots, a, b$, for example.

If n is odd, note that $\frac{n(a^3 + b^3) - (a^3 - 2ab^2 + b^3)}{ab(n(a+b) - (a-b))} \leq \frac{n(a^3 + b^3) - (a^3 - 2a^2b + b^3)}{ab(n(a+b) + (a-b))}$. Indeed, by direct expansion, this is equivalent to

$$2(n-1)(a-b)^3(a+b) \leq 0.$$

Thus, the maximum value of S is $\frac{n(a^3 + b^3) - (a^3 - 2a^2b + b^3)}{ab(n(a+b) + (a-b))}$, which is attained when the sequence is $a, b, a, b, \dots, a, b, a$, for example. $\square$

Problem 5.8. (IMO Shortlist 2020 A4/Problem 2) The real numbers a, b, c, d are such that $a \geq b \geq c \geq d > 0$ and $a + b + c + d = 1$. Prove that

$$(a + 2b + 3c + 4d)a^a b^b c^c d^d < 1. \quad (1)$$

Analysis. It can be shown that $x^x \rightarrow 1$ when $x \rightarrow 0$ (which will be proved in the following solution). Therefore, the left-hand side of the inequality tends to 1 when $a \rightarrow 1$ and $b, c, d \rightarrow 0$. In view of this, we may bound the expression crudely as long as the ‘equality’ holds when $b, c, d = 0$. This gives us an upper bound for the expression which is a function in a . Then we can study this one variable function to see if it is bounded above by 1 or not.

Solution. Obviously, we have $b, c, d < 1 - a$. Also, we have

$$a + 2b + 3c + 4d \leq a + 3b + 3c + 3d = 3 - 2a.$$

Thus, we see that

$$(a + 2b + 3c + 4d)a^a b^b c^c d^d < (3 - 2a)a^a (1 - a)^b (1 - a)^c (1 - a)^d = (3 - 2a)a^a (1 - a)^{1-a}.$$

Let $f(a) = (3 - 2a)a^a (1 - a)^{1-a}$. It may be tedious to work on this function directly, so we consider

$$g(a) = \ln f(a) = \ln(3 - 2a) + a \ln a + (1 - a) \ln(1 - a)$$

instead. Now, we have

$$\begin{aligned} g'(a) &= -\frac{2}{3 - 2a} + \ln a - \ln(1 - a), \\ g''(a) &= -\frac{4}{(3 - 2a)^2} + \frac{1}{a} + \frac{1}{1 - a} = \frac{8(a - 1)^2 + 1}{a(1 - a)(3 - 2a)^2} > 0. \end{aligned}$$

It follows that g is convex on $(0, 1)$ by [proposition 5.5](#). Thus, we compute

$$\lim_{a \rightarrow 1^-} g(a) = 0 + 0 + \lim_{a \rightarrow 1^-} (1 - a) \ln(1 - a) = \lim_{a \rightarrow 1^-} (1 - a) \ln(1 - a).$$

This is of the indeterminate form $0 \cdot \infty$, so we can use L'Hôpital's rule as follows to compute this limit. Indeed,

$$\lim_{a \rightarrow 1^-} (1-a) \ln(1-a) = \lim_{a \rightarrow 1^-} \frac{\ln(1-a)}{(1-a)^{-1}} = \lim_{a \rightarrow 1^-} \frac{-(1-a)^{-1}}{(1-a)^{-2}} = \lim_{a \rightarrow 1^-} -(1-a) = 0.$$

This implies $\lim_{a \rightarrow 1^-} g(a) = 0$. By the convexity, for any $a \in [k, 1)$ (with fixed k), we can deduce $g(a) \leq \max\{g(k), 0\}$. Note that the best bound for a is $a \geq \frac{1}{4}$. However, it turns out that $g\left(\frac{1}{4}\right) > 0$, so we cannot prove the inequality $f(a) = e^{g(a)} < 1$ immediately. So we need to find an alternative method when a is small.

When a is small (e.g. close to $\frac{1}{4}$), the above argument does not work because the bounds $b, c, d < 1-a$ are too weak. In that case, the given bounds $b, c, d \leq a$ should work much better. Indeed, we also have

$$(a + 2b + 3c + 4d)a^a b^b c^c d^d < (3-2a)a^a a^b a^c a^d = (3-2a)a^{a+b+c+d} = (3-2a)a.$$

It is less than 1 iff $(a-1)(2a-1) > 0$, i.e. $a < \frac{1}{2}$.

Therefore, we can now work out a complete proof. If $a < \frac{1}{2}$, we can use the latter argument to deduce (1). If $a \geq \frac{1}{2}$, we can use the former argument to deduce

$$g(a) \leq \max\left\{g\left(\frac{1}{2}\right), 0\right\}.$$

It is not hard to check that

$$g\left(\frac{1}{2}\right) = \ln 2 + \frac{1}{2} \ln \frac{1}{2} + \frac{1}{2} \ln \frac{1}{2} = 0.$$

Thus, we obtain $g(a) \leq 0$, and so $f(a) = e^{g(a)} \leq 1$. This also implies (1). So we are done in both cases. $\square$

Remarks. For some harder problems, we may not be able to solve it completely by using the convexity. However, we may still solve part of the problem (e.g. the case $a \geq \frac{1}{2}$ in the above solution) and have some partial progress.

Apart from the geometric meaning of a convex/concave function, we can also use its algebraic definition directly to deduce some useful inequalities. Among these, the most important and useful result is Jensen's inequality, as given below.

Theorem 5.4. (Jensen's inequality (琴生不等式)) Let f be a convex function defined on an interval I , and let $n \geq 2$. For any real numbers $a_1, a_2, \dots, a_n \in I$ and $w_1, w_2, \dots, w_n > 0$, we have

$$\frac{w_1 f(a_1) + w_2 f(a_2) + \dots + w_n f(a_n)}{w_1 + w_2 + \dots + w_n} \geq f\left(\frac{w_1 a_1 + w_2 a_2 + \dots + w_n a_n}{w_1 + w_2 + \dots + w_n}\right).$$

If f'' is nonzero except for a finite number of points, equality holds iff $a_1 = a_2 = \dots = a_n$. On the other hand, the inequality sign is reversed if f is concave.

Proof. We shall prove by induction on n . For $n = 2$, the inequality

$$\frac{w_1 f(a_1) + w_2 f(a_2)}{w_1 + w_2} \geq f\left(\frac{w_1 a_1 + w_2 a_2}{w_1 + w_2}\right)$$

is just the same as the definition of a convex function, by taking $t = \frac{w_1}{w_1 + w_2}$. Now, suppose the claim is true for all $n \leq k$ where $k \geq 2$.

When $n = k + 1$, by the inductive hypothesis, we have

$$\begin{aligned} & \frac{w_1 f(a_1) + w_2 f(a_2) + \dots + w_k f(a_k) + w_{k+1} f(a_{k+1})}{w_1 + w_2 + \dots + w_k + w_{k+1}} \\ & \geq \frac{(w_1 + w_2 + \dots + w_k) f\left(\frac{w_1 a_1 + w_2 a_2 + \dots + w_k a_k}{w_1 + w_2 + \dots + w_k}\right) + w_{k+1} f(a_{k+1})}{w_1 + w_2 + \dots + w_k + w_{k+1}} \quad (\text{the case } n = k) \\ & \geq f\left(\frac{(w_1 + w_2 + \dots + w_k) \cdot \frac{w_1 a_1 + w_2 a_2 + \dots + w_k a_k}{w_1 + w_2 + \dots + w_k} + w_{k+1} a_{k+1}}{(w_1 + w_2 + \dots + w_k) + w_{k+1}}\right) \quad (\text{the case } n = 2) \\ & = f\left(\frac{w_1 a_1 + w_2 a_2 + \dots + w_{k+1} a_{k+1}}{w_1 + w_2 + \dots + w_{k+1}}\right). \end{aligned}$$

This proves the theorem by induction. □

We usually apply the following form of Jensen's inequality:

$$w_1 f(a_1) + w_2 f(a_2) + \dots + w_n f(a_n) \geq (w_1 + w_2 + \dots + w_n) f\left(\frac{w_1 a_1 + w_2 a_2 + \dots + w_n a_n}{w_1 + w_2 + \dots + w_n}\right).$$

In particular, when $w_1 = w_2 = \dots = w_n$, we have

$$f(a_1) + f(a_2) + \dots + f(a_n) \geq n f\left(\frac{a_1 + a_2 + \dots + a_n}{n}\right).$$

In other words, the average of the images of any n points is at least as the image of their average.

Jensen's inequality can be used to prove various inequalities, including many well-known inequalities.

Problem 5.9. (Weighted AM-GM inequality) Let $n \geq 2$. Let $a_1, a_2, \dots, a_n, w_1, w_2, \dots, w_n$ be positive real numbers. Prove that

$$\frac{w_1 a_1 + w_2 a_2 + \dots + w_n a_n}{w_1 + w_2 + \dots + w_n} \geq (a_1^{w_1} a_2^{w_2} \dots a_n^{w_n})^{\frac{1}{w_1 + w_2 + \dots + w_n}}.$$

Analysis. To use Jensen's inequality, the key step is to choose a suitable function.

The left-hand side of the weighted AM-GM inequality looks like some expressions in Jensen's inequality. But we probably should not use the identity function to apply Jensen's inequality, so it does not give us any clue in choosing the function.

For the right-hand side, we have a product of separated terms. In view of Jensen's inequality, which mainly consists of a sum of separated terms, we need to convert the product to a sum. One standard way is to use the logarithm function.

Solution. Consider $f(x) = \ln x$. Since $f'(x) = \frac{1}{x}$ and $f''(x) = -\frac{1}{x^2} < 0$, f is a concave function by [proposition 5.5](#). By Jensen's inequality, we have

$$\frac{w_1 \ln a_1 + w_2 \ln a_2 + \dots + w_n \ln a_n}{w_1 + w_2 + \dots + w_n} \leq \ln \left(\frac{w_1 a_1 + w_2 a_2 + \dots + w_n a_n}{w_1 + w_2 + \dots + w_n} \right).$$

This implies

$$\ln (a_1^{w_1} a_2^{w_2} \dots a_n^{w_n})^{\frac{1}{w_1 + w_2 + \dots + w_n}} \leq \ln \left(\frac{w_1 a_1 + w_2 a_2 + \dots + w_n a_n}{w_1 + w_2 + \dots + w_n} \right),$$

and hence

$$(a_1^{w_1} a_2^{w_2} \dots a_n^{w_n})^{\frac{1}{w_1 + w_2 + \dots + w_n}} \leq \frac{w_1 a_1 + w_2 a_2 + \dots + w_n a_n}{w_1 + w_2 + \dots + w_n}.$$

This proves the weighted AM-GM inequality. □

Problem 5.10. (Power mean inequality) Let $n \geq 2$. Let $a_1, a_2, \dots, a_n$ be positive real numbers, and let $r < s$ be nonzero real numbers. Prove that

$$\left(\frac{a_1^r + a_2^r + \dots + a_n^r}{n} \right)^{\frac{1}{r}} \leq \left(\frac{a_1^s + a_2^s + \dots + a_n^s}{n} \right)^{\frac{1}{s}}.$$

Analysis. We already get a sum of separated terms in the expressions. One may tend to use the function $f(x) = x^r$ to generate the expression on the left. But this is independent of s , so it could hardly lead to a proof to the desired inequality. Rather than that, we may regard $a_1^r, a_2^r, \dots, a_n^r$ as variables, and find a function that can generate the terms a_i^s . For example, we may find f such that $f(a_i^r) = a_i^s$ for each i , which means $f(x) = x^{\frac{s}{r}}$.

Solution. Consider $f(x) = x^{\frac{s}{r}}$. Then $f'(x) = \frac{s}{r} x^{\frac{s}{r}-1}$ and $f''(x) = \frac{s}{r} \left(\frac{s}{r} - 1 \right) x^{\frac{s}{r}-2}$.

Note that $x^{\frac{s}{r}-2}$ is always positive for $x > 0$. Also, $\frac{s}{r} \left(\frac{s}{r} - 1 \right) = \frac{s(s-r)}{r^2}$ is positive if $s > 0$, and is negative if $s < 0$ (since $s > r$). So there are two separated cases to consider.

Case 1. $s > 0$

From above, $f''(x) > 0$ so that f is convex on $(0, \infty)$ by [proposition 5.5](#). By Jensen's inequality, we get

$$\frac{f(a_1^r) + f(a_2^r) + \cdots + f(a_n^r)}{n} \geq f\left(\frac{a_1^r + a_2^r + \cdots + a_n^r}{n}\right).$$

This gives

$$\frac{a_1^s + a_2^s + \cdots + a_n^s}{n} \geq \left(\frac{a_1^r + a_2^r + \cdots + a_n^r}{n}\right)^{\frac{s}{r}}.$$

Since $s > 0$, this implies

$$\left(\frac{a_1^s + a_2^s + \cdots + a_n^s}{n}\right)^{\frac{1}{s}} \geq \left(\frac{a_1^r + a_2^r + \cdots + a_n^r}{n}\right)^{\frac{1}{r}}.$$

Case 2. $s < 0$

Similarly, $f''(x) < 0$ so that f is concave on $(0, \infty)$ by [proposition 5.5](#). By Jensen's inequality, we get

$$\frac{f(a_1^r) + f(a_2^r) + \cdots + f(a_n^r)}{n} \leq f\left(\frac{a_1^r + a_2^r + \cdots + a_n^r}{n}\right).$$

This gives

$$\frac{a_1^s + a_2^s + \cdots + a_n^s}{n} \leq \left(\frac{a_1^r + a_2^r + \cdots + a_n^r}{n}\right)^{\frac{s}{r}}.$$

Since $s < 0$, this implies

$$\left(\frac{a_1^s + a_2^s + \cdots + a_n^s}{n}\right)^{\frac{1}{s}} \geq \left(\frac{a_1^r + a_2^r + \cdots + a_n^r}{n}\right)^{\frac{1}{r}}.$$

This proves the power mean inequality. □

Remarks. The weighted power mean inequality can be proved via this method using the weighted form of Jensen's inequality. Also, the case where one of r, s equals 0 can be done by continuity in view of its construction. Alternatively, one may also consider some functions involving logarithm to prove the same result using Jensen's inequality.

Recall that [proposition 3.1](#) is proved by the power mean inequality. So it can also be proved by Jensen's inequality directly. For example, to prove

$$a_1 \sqrt[k]{b_1} + a_2 \sqrt[k]{b_2} + \cdots + a_n \sqrt[k]{b_n} \leq \sqrt[k]{(a_1 + a_2 + \cdots + a_n)^{k-1} (a_1 b_1 + a_2 b_2 + \cdots + a_n b_n)},$$

it suffices to apply Jensen's inequality to the concave function $\sqrt[k]{x}$.

Problem 5.11. (Hölder's inequality) Let $m, n \geq 2$. Let $a_{1,1}, a_{1,2}, \dots, a_{1,n}, a_{2,1}, a_{2,2}, \dots, a_{2,n}, \dots, a_{m,1}, a_{m,2}, \dots, a_{m,n}$ and $w_1, w_2, \dots, w_m$ be positive real numbers satisfying $w_1 + w_2 + \dots + w_m = 1$. Prove that

$$(a_{1,1} + a_{1,2} + \dots + a_{1,n})^{w_1} (a_{2,1} + a_{2,2} + \dots + a_{2,n})^{w_2} \dots (a_{m,1} + a_{m,2} + \dots + a_{m,n})^{w_m} \\ \geq a_{1,1}^{w_1} a_{2,1}^{w_2} \dots a_{m,1}^{w_m} + a_{1,2}^{w_1} a_{2,2}^{w_2} \dots a_{m,2}^{w_m} + \dots + a_{1,n}^{w_1} a_{2,n}^{w_2} \dots a_{m,n}^{w_m}.$$

Analysis. As there are a lot of variables in the expression, it is tedious to handle everything at one time. A useful method is to use induction. One may find that everything essentially comes from the case $m = 2$, which is also the most common application of Hölder's inequality.

Solution. We shall prove the inequality by induction on m . When $m = 2$, we need to prove

$$(a_1 + a_2 + \dots + a_n)^s (b_1 + b_2 + \dots + b_n)^{1-s} \geq a_1^s b_1^{1-s} + a_2^s b_2^{1-s} + \dots + a_n^s b_n^{1-s}$$

for $0 < s < 1$. Indeed, consider $f(x) = x^{\frac{1}{1-s}}$. Then we get $f'(x) = \frac{1}{1-s} x^{\frac{s}{1-s}}$ and $f''(x) = \frac{s}{(1-s)^2} x^{\frac{2s-1}{1-s}} > 0$ for $x > 0$. By [proposition 5.5](#), f is convex. Applying Jensen's inequality, we have

$$\sum_{i=1}^n a_i f\left(\left(\frac{b_i}{a_i}\right)^{1-s}\right) \geq \left(\sum_{i=1}^n a_i\right) f\left(\frac{\sum_{i=1}^n a_i^s b_i^{1-s}}{\sum_{i=1}^n a_i}\right).$$

This implies

$$\sum_{i=1}^n b_i \geq \left(\sum_{i=1}^n a_i\right)^{-\frac{s}{1-s}} \left(\sum_{i=1}^n a_i^s b_i^{1-s}\right)^{\frac{1}{1-s}},$$

or simply

$$\left(\sum_{i=1}^n a_i\right)^s \left(\sum_{i=1}^n b_i\right)^{1-s} \geq \sum_{i=1}^n a_i^s b_i^{1-s}.$$

This establishes the base case. Now, suppose

$$(b_{1,1} + b_{1,2} + \dots + b_{1,n})^{s_1} (b_{2,1} + b_{2,2} + \dots + b_{2,n})^{s_2} \dots (b_{m-1,1} + b_{m-1,2} + \dots + b_{m-1,n})^{s_{m-1}} \\ \geq b_{1,1}^{s_1} b_{2,1}^{s_2} \dots b_{m-1,1}^{s_{m-1}} + b_{1,2}^{s_1} b_{2,2}^{s_2} \dots b_{m-1,2}^{s_{m-1}} + \dots + b_{1,n}^{s_1} b_{2,n}^{s_2} \dots b_{m-1,n}^{s_{m-1}}$$

for positive variables where $s_1 + s_2 + \dots + s_{m-1} = 1$. To prove that

$$(a_{1,1} + a_{1,2} + \dots + a_{1,n})^{w_1} (a_{2,1} + a_{2,2} + \dots + a_{2,n})^{w_2} \dots (a_{m,1} + a_{m,2} + \dots + a_{m,n})^{w_m} \\ \geq a_{1,1}^{w_1} a_{2,1}^{w_2} \dots a_{m,1}^{w_m} + a_{1,2}^{w_1} a_{2,2}^{w_2} \dots a_{m,2}^{w_m} + \dots + a_{1,n}^{w_1} a_{2,n}^{w_2} \dots a_{m,n}^{w_m}$$

for $w_1 + w_2 + \dots + w_m = 1$, we first choose $t = \frac{1}{1-w_m}$ so that $tw_1 + tw_2 + \dots + tw_{m-1} = 1$.

By the inductive hypothesis, we have

$$(a_{1,1} + a_{1,2} + \dots + a_{1,n})^{tw_1} \dots (a_{m-1,1} + a_{m-1,2} + \dots + a_{m-1,n})^{tw_{m-1}} \\ \geq a_{1,1}^{tw_1} a_{2,1}^{tw_2} \dots a_{m-1,1}^{tw_{m-1}} + \dots + a_{1,n}^{tw_1} a_{2,n}^{tw_2} \dots a_{m-1,n}^{tw_{m-1}}.$$

Replacing $a_{1,i}^{w_1} a_{2,i}^{w_2} \cdots a_{m-1,i}^{w_{m-1}}$ by c_i , it suffices to prove

$$(c_1^t + c_2^t + \cdots + c_n^t)^{\frac{1}{t}} (a_{m,1} + a_{m,2} + \cdots + a_{m,n})^{w_m} \geq c_1 a_{m,1}^{w_m} + c_2 a_{m,2}^{w_m} + \cdots + c_n a_{m,n}^{w_m}.$$

This is just Hölder's inequality when $n = 2$ by noting $\frac{1}{t} + w_m = 1$. The proof is thus complete by induction. $\square$

Problem 5.12. Let x, y, z be positive real numbers such that $x^2 + y^2 + z^2 = 1$. Prove that

$$\begin{aligned} \frac{1}{x} + \frac{1}{y} + \frac{1}{z} - (x + y + z) &\geq 2\sqrt{3}, \\ \frac{1}{x} + \frac{1}{y} + \frac{1}{z} + (x + y + z) &\geq 4\sqrt{3}. \end{aligned}$$

Analysis. The two results are quite similar. To apply Jensen's inequality, note that it is better to have the sum of variables being a constant as we can obtain a constant after applying the inequality. This suggests we should consider x^2, y^2, z^2 as variables instead of x, y, z .

Solution. Consider

$$f(a) = \frac{1}{\sqrt{a}} - \sqrt{a}.$$

Then

$$\begin{aligned} f'(a) &= -\frac{1}{2}a^{-\frac{3}{2}} - \frac{1}{2}a^{-\frac{1}{2}}, \\ f''(a) &= \frac{3}{4}a^{-\frac{5}{2}} + \frac{1}{4}a^{-\frac{3}{2}} > 0. \end{aligned}$$

Therefore, f is convex by [proposition 5.5](#). By Jensen's inequality, we have

$$f(x^2) + f(y^2) + f(z^2) \geq 3f\left(\frac{x^2 + y^2 + z^2}{3}\right) = 3f\left(\frac{1}{3}\right) = 2\sqrt{3}.$$

Similarly, we deduce the other inequality by considering $g(a) = \frac{1}{\sqrt{a}} + \sqrt{a}$. Note that

$$g''(a) = \frac{3}{4}a^{-\frac{5}{2}} - \frac{1}{4}a^{-\frac{3}{2}} = \frac{3-a}{4a^{\frac{5}{2}}} > 0$$

in the range $a \in (0, 1)$, which is the range of x, y, z . $\square$

Remarks. Jensen's inequality is useful if the inequality is of the form

$$f(a) + f(b) + f(c) \geq k$$

where k is a constant (and probably under the constraint $a + b + c = 1$).

Problem 5.13. Let a, b, c be positive real numbers. Prove that

$$\frac{a}{(b+c)^k} + \frac{b}{(c+a)^k} + \frac{c}{(a+b)^k} \geq \frac{3^k}{2^k(a+b+c)^{k-1}}$$

for any real number $k \geq 1$.

Analysis. The inequality is homogeneous as the degree of each term is $-(k-1)$. One may normalize it to obtain a separable inequality so as to apply Jensen's inequality.

Solution. WLOG assume $a+b+c=3$. The inequality is simply

$$\frac{a}{(3-a)^k} + \frac{b}{(3-b)^k} + \frac{c}{(3-c)^k} \geq \frac{3}{2^k}.$$

Consider

$$f(x) = \frac{x}{(3-x)^k}.$$

Then

$$\begin{aligned} f'(x) &= \frac{3 + (k-1)x}{(3-x)^{k+1}}, \\ f''(x) &= \frac{6k + k(k-1)x}{(3-x)^{k+2}} > 0 \end{aligned}$$

as $k \geq 1$ and $0 < x < 3$. This shows f is convex. By Jensen's inequality, we have

$$f(a) + f(b) + f(c) \geq 3f\left(\frac{a+b+c}{3}\right) = 3f(1) = \frac{3}{2^k}$$

as desired. □

Note that Jensen's inequality only suggests the optimal case occurs in the average case. But sometimes it is more desirable to claim that the expression becomes larger (or smaller) when the variables are getting closer. This can be done by the following stronger result.

Theorem 5.5. ([Majorization inequality/Karamata's inequality](#) (盖不等式)) Let f be a convex function defined on an interval I . For any real numbers $x_1, x_2, \dots, x_n, y_1, y_2, \dots, y_n \in I$ with $(x_1, x_2, \dots, x_n) > (y_1, y_2, \dots, y_n)$, we have

$$f(x_1) + f(x_2) + \dots + f(x_n) \geq f(y_1) + f(y_2) + \dots + f(y_n).$$

If f'' is nonzero except for a finite number of points, equality holds iff $(x_1, x_2, \dots, x_n) = (y_1, y_2, \dots, y_n)$. On the other hand, the inequality sign is reversed if f is concave.

Proof. We first prove the following property of a convex function.

Claim. Let f be a convex function defined on an interval I . For any $a, b, c, d \in I$ such that $a \leq c$ and $b \leq d$, we have

$$\frac{f(d) - f(c)}{d - c} \geq \frac{f(b) - f(a)}{b - a}.$$

Geometrically, this means the slope of the line joining two points on the graph of f is monotonic increasing when the x -coordinates of the points increase. This is quite obvious in view of the geometric meaning of a convex function. To prove this **claim** algebraically, it suffices to prove

$$\frac{f(d) - f(c)}{d - c} \geq \frac{f(b) - f(c)}{b - c} \geq \frac{f(b) - f(a)}{b - a}.$$

We only prove the second one since the other is essentially the same. There are several cases.

- If $b \leq a \leq c$, the inequality can be rewritten as

$$\frac{c - a}{c - b}f(b) + \frac{a - b}{c - b}f(c) \geq f(a).$$

This holds by the convexity of f since $\frac{c - a}{c - b} + \frac{a - b}{c - b} = 1$ and $\frac{c - a}{c - b}b + \frac{a - b}{c - b}c = a$.

- If $a \leq b \leq c$, the inequality can be rewritten as

$$\frac{c - b}{c - a}f(a) + \frac{b - a}{c - a}f(c) \geq f(b).$$

This holds by the convexity of f since $\frac{c - b}{c - a} + \frac{b - a}{c - a} = 1$ and $\frac{c - b}{c - a}a + \frac{b - a}{c - a}c = b$.

- If $a \leq c \leq b$, the inequality can be rewritten as

$$\frac{b - c}{b - a}f(a) + \frac{c - a}{b - a}f(b) \geq f(c).$$

This holds by the convexity of f since $\frac{b - c}{b - a} + \frac{c - a}{b - a} = 1$ and $\frac{b - c}{b - a}a + \frac{c - a}{b - a}b = c$.

So the **claim** is proved.

We return to prove the main theorem. WLOG assume $x_i \neq y_i$ for every i (otherwise we can remove the two equal terms). We define $z_i = \frac{f(x_i) - f(y_i)}{x_i - y_i}$ for each i . By the **claim**, we have $z_i \geq z_{i+1}$ for each $i = 1, 2, \dots, n - 1$. Let

$$A_j = \sum_{i=1}^j x_i \quad \text{and} \quad B_j = \sum_{i=1}^j y_i$$

for each $j = 0, 1, \dots, n$ (so $A_0 = B_0 = 0$). Then we have

$$\begin{aligned}
\sum_{i=1}^n (f(x_i) - f(y_i)) &= \sum_{i=1}^n z_i (x_i - y_i) \\
&= \sum_{i=1}^n z_i ((A_i - A_{i-1}) - (B_i - B_{i-1})) \\
&= \sum_{i=1}^n z_i (A_i - B_i) - \sum_{i=1}^n z_i (A_{i-1} - B_{i-1}) \\
&= \sum_{i=1}^{n-1} z_i (A_i - B_i) - \sum_{i=1}^{n-1} z_{i+1} (A_i - B_i) \quad (\text{since } A_n = B_n \text{ and } A_0 = B_0) \\
&= \sum_{i=1}^{n-1} (z_i - z_{i+1}) (A_i - B_i).
\end{aligned}$$

The majorization condition guarantees $A_i \geq B_i$. Also, $z_i \geq z_{i+1}$ from above. This proves

$$\sum_{i=1}^n (f(x_i) - f(y_i)) \geq 0$$

as desired. □

By taking $y_1 = y_2 = \dots = y_n$ in the majorization inequality, it is reduced to the usual form of Jensen's inequality when $w_1 = w_2 = \dots = w_n = 1$. The majorization inequality is useful when we are dealing with discrete things. For example, suppose we need to obtain a lower bound for $2^{a_1} + 2^{a_2} + \dots + 2^{a_n}$ given the sum $a_1 + a_2 + \dots + a_n$, where all variables are nonnegative integers. We may consider the convex function $f(x) = 2^x$ and claim that the minimum occurs when every pair of variables differ by at most 1, since this is the case where the sequence is majorized by all other sequences of nonnegative integers with the same sum. For instance, given $a + b + c = 10$, we get $2^a + 2^b + 2^c \geq 2^4 + 2^3 + 2^3$ by the majorization inequality as $(a, b, c) \succ (4, 3, 3)$. This is also useful in combinatorics where the variables are usually positive integers. The fact that the binomial coefficient $\binom{x}{2}$ is a convex function is quite useful.

5.3 Integration

Integration is not as useful as differentiation for the purpose of proving some inequalities. But still there are some kinds of applications which are worth to introduce.

There are different definitions of the integral of a function. The most widely used one is the Riemann integral (also commonly known as the *definite integral*(定積分)).

Definition 5.8. Let f be a function defined on a closed and bounded interval $[a, b]$. Consider any partition $a = x_0 < x_1 < \cdots < x_n = b$ of $[a, b]$, and consider any point $t_i \in [x_{i-1}, x_i]$ for $i = 1, 2, \dots, n$. We define the *Riemann integral* (黎曼積分) of f by

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(t_i)(x_i - x_{i-1}),$$

where the limit is taken over all partitions of $[a, b]$ such that $\max_{1 \leq i \leq n} (x_i - x_{i-1})$ approaches 0 (if the limit exists). In that case, we say that the function f is *integrable* (可積).

Note that $f(t_i)(x_i - x_{i-1})$ is the area of a rectangle with height $f(t_i)$ and width $x_i - x_{i-1}$, which gives a good approximation to the area under the graph of $f(x)$ over the interval $[x_{i-1}, x_i]$ when $x_i - x_{i-1}$ is small. Therefore, $\sum_{i=1}^n f(t_i)(x_i - x_{i-1})$ is an approximation to the total area under the graph. As a result, the integral of f represents the actual area under the graph.

The following gives some basic results about integration.

Proposition 5.6. Let f, g be integrable functions defined on an interval $I = [a, b]$. Then the following hold.

- (a) $f + g$ is integrable and $\int_a^b (f + g)(x) dx = \int_a^b f(x) dx + \int_a^b g(x) dx$
- (b) cf is integrable and $\int_a^b (cf)(x) dx = c \int_a^b f(x) dx$
- (c) for any $a < c < b$, we have $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$
- (d) if $f(x) \leq g(x)$ for any $x \in [a, b]$, then $\int_a^b f(x) dx \leq \int_a^b g(x) dx$
- (e) $|f|$ is integrable and $\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$

To compute the integral of a function f , we normally do not use the definition. Indeed, the fundamental theorem of calculus allows us to compute it using the concept of differentiation.

Definition 5.9. Let f be a function defined on an interval I . We say that a function F is an *antiderivative* (原函數) of f if $F'(x) = f(x)$ for any $x \in I$.

Theorem 5.6. (Fundamental theorem of calculus(微積分基本定理)) Let f be an integrable function defined on a closed and bounded interval $[a, b]$. If F is an antiderivative of f , then

$$\int_a^b f(x) dx = F(b) - F(a).$$

There is another part of the fundamental theorem of calculus. We only introduce this part here because it is more useful for the purpose of computing the integral. Indeed, the main task is just to find a function F such that $F'(x) = f(x)$. For example, to compute

$$\int_1^2 x^k dx$$

where k is a real number different from -1 , we just need to notice $\frac{d}{dx} \left(\frac{1}{k+1} x^{k+1} \right) = x^k$. By the fundamental theorem of calculus, we find that

$$\int_1^2 x^k dx = \frac{1}{k+1} (2)^{k+1} - \frac{1}{k+1} (1)^{k+1} = \frac{2^{k+1} - 1}{k+1}.$$

On the other hand, for $k = -1$, since $\frac{d}{dx}(\ln x) = \frac{1}{x}$, we have

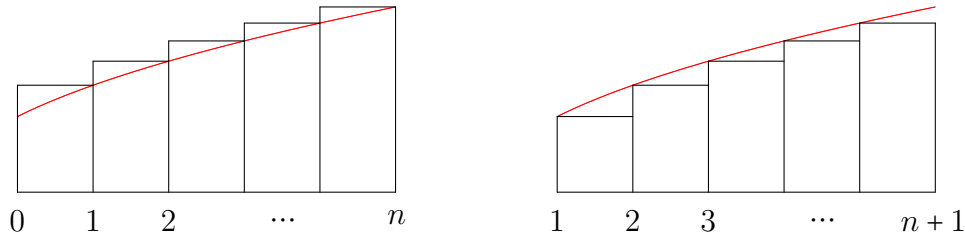
$$\int_1^2 \frac{1}{x} dx = \ln 2 - \ln 1 = \ln 2.$$

The main result about inequalities that we can use is the following.

Proposition 5.7. Let f be a strictly increasing integrable function defined on $\mathbb{R}^+$. For any positive integer n , we have

$$\int_0^n f(x) dx < \sum_{i=1}^n f(i) < \int_1^{n+1} f(x) dx.$$

Proof. For the inequality on the left, consider the partition $0 < 1 < \dots < n$ of the interval $[0, n]$. The summand $f(i)$ can be regarded as the area of a rectangle with height $f(i)$ and length 1 over the interval $[i-1, i]$. As shown in the figure below, the sum gives the total area of all such rectangles, which contain the region under the graph of $f(x)$ (the red curve) over the interval $[0, n]$ since f is increasing. This proves the inequality by the geometric meaning of integrals.



Similarly, for the inequality on the right, consider the partition $1 < 2 < \dots < n + 1$ of the interval $[1, n + 1]$. The summand $f(i)$ can be regarded as the area of a rectangle with height $f(i)$ and length 1 over the interval $[i, i + 1]$. This time the rectangles are contained inside the region under the graph of $f(x)$ over the interval $[1, n + 1]$. This proves the inequality. $\square$

The above result allows us to estimate the sum $\sum_{i=1}^n f(i)$ using integration. This is convenient if an antiderivative of $f(x)$ is easy to find. There are many variations of this result (e.g. f is decreasing, the sum is $f(2) + f(4) + \dots + f(2n)$, etc.). One just needs to use the ideas in the proof and the geometric meaning of integrals to derive the corresponding results.

Problem 5.14. (IMO HK TST 2017 Test 1 Problem 5) Find the first digit after the decimal point of the number $\frac{1}{1009} + \frac{1}{1010} + \dots + \frac{1}{2016}$.

Analysis. There is no closed form of the harmonic sum $\sum_{k=a}^b \frac{1}{k}$. But it is easy to estimate this sum using integration since an antiderivative of $\frac{1}{x}$ is $\ln x$. It turns out that the approximation is very good. In the following solution, we need to use the (well-known) results $e < 3$ and $\ln 2 \approx 0.69$ (correct to 2 decimal places).

Solution. Let $S = \sum_{k=1009}^{2016} \frac{1}{k}$. Note that the function $\frac{1}{x}$ is decreasing on $(0, \infty)$. Using a variation of **proposition 5.7**, we have

$$\int_{1009}^{2017} \frac{1}{x} dx < S < \int_{1008}^{2016} \frac{1}{x} dx.$$

As $\ln x$ is an antiderivative of $\frac{1}{x}$, we have

$$\int_{1009}^{2017} \frac{1}{x} dx = \ln 2017 - \ln 1009 = \ln \frac{2017}{1009}$$

and

$$\int_{1008}^{2016} \frac{1}{x} dx = \ln 2016 - \ln 1008 = \ln \frac{2016}{1008} = \ln 2$$

by the fundamental theorem of calculus. This gives

$$\ln \frac{2017}{1009} < S < \ln 2 < 0.7.$$

Note that $\ln \frac{2017}{1009} \approx \ln 2 \approx 0.69$, and so it is a good guess that the answer should be 6. Thus, it remains to prove that $\ln \frac{2017}{1009} > 0.6$, i.e. $\frac{2017}{1009} > e^{0.6}$. By the binomial

theorem, we have

$$\left(\frac{2017}{1009}\right)^5 = \left(2 - \frac{1}{1009}\right)^5 > 2^5 - \binom{5}{1}2^4 \cdot \frac{1}{1009} > 27 > e^3.$$

This establishes the bound $0.6 < S < 0.7$, and so the answer is 6. $\square$

Problem 5.15. Considering all possible choices of the positive integer n and the positive real numbers $a_1, a_2, \dots, a_n$ with sum 1, find the smallest real number C such that

$$\frac{a_1}{(1+a_1)^2} + \frac{a_2}{(1+a_1+a_2)^2} + \cdots + \frac{a_n}{(1+a_1+a_2+\cdots+a_n)^2} < C.$$

Analysis. This time we still use the same idea as the proof of [proposition 5.7](#) to estimate the sum on the left. Instead of using rectangles of the same width, we may also use rectangles of different widths over different intervals.

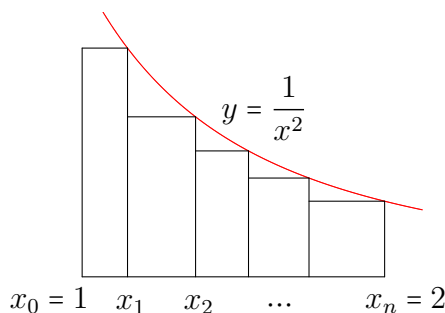
Solution. Define $f(x) = \frac{1}{x^2}$ and let $x_j = 1 + a_1 + \cdots + a_j$ for $0 \leq j \leq n$ (so $x_0 = 1$ and $x_n = 2$). Consider the partition $x_0 < x_1 < \cdots < x_n$ of $[x_0, x_n]$. For each interval $[x_{i-1}, x_i]$, we construct a rectangle of height $f(x_i)$ and width $x_i - x_{i-1} = a_i$. The total areas of all these rectangles is exactly

$$S = \sum_{i=1}^n a_i f(x_i) = \frac{a_1}{(1+a_1)^2} + \frac{a_2}{(1+a_1+a_2)^2} + \cdots + \frac{a_n}{(1+a_1+a_2+\cdots+a_n)^2}.$$

Since f is decreasing, its graph lies above all the rectangles. Thus, we have

$$S < \int_1^2 \frac{1}{x^2} dx = -\frac{1}{2} + \frac{1}{1} = \frac{1}{2}$$

(since an antiderivative of $\frac{1}{x^2}$ is $-\frac{1}{x}$). Also, by the definition of integration, this bound is best possible since S approaches $\frac{1}{2}$ when $n \rightarrow \infty$ and $\max_{1 \leq j \leq n} \{a_j\} \rightarrow 0$. It follows that the minimum C is $\frac{1}{2}$.



$\square$

Another way to use integration is as follows. Suppose we have established an inequality $f(t) \geq g(t)$ in a variable t (possibly together with other variables which are regarded as constants) where f and g are integrable functions. Then we can use [proposition 5.6\(d\)](#) to deduce $\int_a^b f(t) dt \geq \int_a^b g(t) dt$ for some suitable constants a and b . This sometimes gives us an entirely different inequality.

Problem 5.16. Let a, b, c be positive real numbers. Prove that

$$\frac{1}{2a} + \frac{1}{2b} + \frac{1}{2c} \geq \frac{1}{a+b} + \frac{1}{b+c} + \frac{1}{c+a}.$$

Analysis. We first demonstrate the method with this simple inequality (which can be proved by noting $\frac{1}{4a} + \frac{1}{4b} \geq \frac{1}{a+b}$, for example).

Solution. It is well-known that

$$x^2 + y^2 + z^2 \geq xy + yz + zx$$

for any positive real numbers x, y, z . Putting $x = t^{a-\frac{1}{2}}$, $y = t^{b-\frac{1}{2}}$ and $z = t^{c-\frac{1}{2}}$, we obtain

$$t^{2a-1} + t^{2b-1} + t^{2c-1} \geq t^{a+b-1} + t^{b+c-1} + t^{c+a-1}.$$

By [proposition 5.6\(d\)](#), we get

$$\int_0^1 (t^{2a-1} + t^{2b-1} + t^{2c-1}) dt \geq \int_0^1 (t^{a+b-1} + t^{b+c-1} + t^{c+a-1}) dt.$$

An antiderivative of $t^{2a-1} + t^{2b-1} + t^{2c-1}$ is $\frac{1}{2a}t^{2a} + \frac{1}{2b}t^{2b} + \frac{1}{2c}t^{2c}$, while an antiderivative of $t^{a+b-1} + t^{b+c-1} + t^{c+a-1}$ is $\frac{1}{a+b}t^{a+b} + \frac{1}{b+c}t^{b+c} + \frac{1}{c+a}t^{c+a}$. Thus, we obtain the desired inequality

$$\frac{1}{2a} + \frac{1}{2b} + \frac{1}{2c} \geq \frac{1}{a+b} + \frac{1}{b+c} + \frac{1}{c+a}.$$

□

Problem 5.17. Let a, b, c be positive real numbers. Prove that

$$\frac{1}{5a+b} + \frac{1}{5b+c} + \frac{1}{5c+a} \geq \frac{1}{3a+2b+c} + \frac{1}{3b+2c+a} + \frac{1}{3c+2a+b}.$$

Analysis. The way of proving this inequality is similar to the above problem. This time we explain more on the choice of the functions. Firstly, in view of the solution of [problem 5.16](#), the term $\frac{1}{5a+b}$ may come from a function of the form $\frac{1}{5a+b}t^k$ for some exponent k . If we would like its derivative to be a simple function, one way is

to set $k = 5a + b$. After differentiating the function, we obtain t^{5a+b-1} . Similarly, it suffices to prove

$$t^{5a+b-1} + t^{5b+c-1} + t^{5c+a-1} \geq t^{3a+2b+c-1} + t^{3b+2c+a-1} + t^{3c+2a+b-1}.$$

The next step is to replace some powers of t by x, y, z to obtain an inequality in the new variables. For example, we may set $(x, y, z) = (t^{5a+b-1}, t^{5b+c-1}, t^{5c+a-1})$. To express $t^{3a+2b+c-1}$ in terms of x, y, z , we consider $t^{3a+2b+c-1} = x^\alpha y^\beta z^\gamma$. This gives the system

$$\begin{cases} 5\alpha + \gamma = 3, \\ \alpha + 5\beta = 2, \\ \beta + 5\gamma = 1, \\ \alpha + \beta + \gamma = 1. \end{cases}$$

It is not hard to get the solution $(\alpha, \beta, \gamma) = \left(\frac{4}{7}, \frac{2}{7}, \frac{1}{7}\right)$. Thus, it remains to prove

$$x + y + z \geq x^{\frac{4}{7}} y^{\frac{2}{7}} z^{\frac{1}{7}} + y^{\frac{4}{7}} z^{\frac{2}{7}} x^{\frac{1}{7}} + z^{\frac{4}{7}} x^{\frac{2}{7}} y^{\frac{1}{7}}.$$

Replacing x, y, z by x^7, y^7, z^7 , this can be easily transformed to a polynomial inequality

$$x^7 + y^7 + z^7 \geq x^4 y^2 z + y^4 z^2 x + z^4 x^2 y.$$

This explains the mysterious solution below.

Solution. We first prove that

$$x^7 + y^7 + z^7 \geq x^4 y^2 z + y^4 z^2 x + z^4 x^2 y \quad (1)$$

for any positive real numbers x, y, z . Indeed, by the AM-GM inequality, we have

$$\frac{4x^7 + 2y^7 + z^7}{7} \geq x^4 y^2 z.$$

Summing up similar inequalities, we obtain (1). Putting $x = t^{\frac{5a+b-1}{7}}$, $y = t^{\frac{5b+c-1}{7}}$ and $z = t^{\frac{5c+a-1}{7}}$, this inequality becomes

$$t^{5a+b-1} + t^{5b+c-1} + t^{5c+a-1} \geq t^{3a+2b+c-1} + t^{3b+2c+a-1} + t^{3c+2a+b-1}.$$

Integrating both sides from 0 to 1, we obtain the desired inequality

$$\frac{1}{5a+b} + \frac{1}{5b+c} + \frac{1}{5c+a} \geq \frac{1}{3a+2b+c} + \frac{1}{3b+2c+a} + \frac{1}{3c+2a+b}.$$

□

Apart from calculus, some analytic ideas may also be useful. For example, sometimes we may see what will happen if some variables approach $\pm\infty$ or 0 (for positive variables). We have already seen some examples using such consideration as listed in the summary below.

- using differentiation to determine the monotonicity ([problems 5.1, 5.2](#))
- using differentiation to determine all possible extremum points ([problem 5.3](#))
- tangent line trick ([problems 5.4, 5.5](#))
- using convexity ([problems 5.6, 5.7, 5.8](#))
- Jensen's inequality ([problems 5.9, 5.10, 5.11, 5.12, 5.13](#))
- majorization inequality
- using integration to approximate a discrete sum ([problems 5.14, 5.15](#))
- integrating an inequality ([problems 5.16, 5.17](#))
- finding limits when some variables approach some points ([problems 5.2, 5.8](#))

6 Modern Inequalities

In view of the current trend of inequality problems in the Olympiad, more problems can be solved without a heavy use of the classical inequalities and theorems. Only very little knowledge is needed to prove such modern inequalities. In particular, there are two kinds of basic ideas with growing importance, namely, mathematical induction and combinatorial ideas. We shall go through some more examples of this kinds in this section.

6.1 Mathematical Induction

Induction is one of the most important proving techniques in mathematics. It can also be used to prove many inequalities involving some positive integer parameters (e.g. the number of variables, the exponents of some variables).

Problem 6.1. Let $n \geq 2$, and let $a_1, a_2, \dots, a_n$ be real numbers such that $a_j > 1$ for $j = 1, 2, \dots, n$, and $|a_j - a_{j+1}| < 1$ for $j = 1, 2, \dots, n-1$. Prove that

$$\frac{a_1}{a_2} + \frac{a_2}{a_3} + \dots + \frac{a_{n-1}}{a_n} + \frac{a_n}{a_1} < 2n - 1.$$

Analysis. For inequality problems involving a sequence, induction is a very useful method. For this problem, all conditions still hold when we move from the case $n = k$ to the case $n = k + 1$. Also, the inequalities for two consecutive cases only differ by a few terms, so there is a good chance that induction can solve the problem.

Solution. We prove the inequality by induction on n .

When $n = 2$, we have $a_1, a_2 > 1$ and $|a_1 - a_2| < 1$, and we need to prove

$$\frac{a_1}{a_2} + \frac{a_2}{a_1} < 3.$$

This inequality is equivalent to $a_1^2 + a_2^2 < 3a_1a_2$, or $(a_1 - a_2)^2 < a_1a_2$. This holds since $(a_1 - a_2)^2 < 1 < a_1a_2$.

Assume the case $n = k$ holds, i.e.

$$\frac{a_1}{a_2} + \frac{a_2}{a_3} + \dots + \frac{a_{k-1}}{a_k} + \frac{a_k}{a_1} < 2k - 1.$$

Consider the case $n = k + 1$. We need to prove

$$\frac{a_1}{a_2} + \frac{a_2}{a_3} + \dots + \frac{a_{k-1}}{a_k} + \frac{a_k}{a_{k+1}} + \frac{a_{k+1}}{a_1} < 2k + 1. \quad (1)$$

Firstly, by the inductive hypothesis, we have

$$\begin{aligned} & \frac{a_1}{a_2} + \frac{a_2}{a_3} + \cdots + \frac{a_{k-1}}{a_k} + \frac{a_k}{a_{k+1}} + \frac{a_{k+1}}{a_1} \\ &= \left(\frac{a_1}{a_2} + \frac{a_2}{a_3} + \cdots + \frac{a_{k-1}}{a_k} + \frac{a_k}{a_1} \right) + \frac{a_k}{a_{k+1}} + \frac{a_{k+1} - a_k}{a_1} \\ &< 2k - 1 + \frac{a_k}{a_{k+1}} + \frac{a_{k+1} - a_k}{a_1}. \end{aligned}$$

Comparing with our goal (1), it suffices to prove

$$\frac{a_k}{a_{k+1}} + \frac{a_{k+1} - a_k}{a_1} \leq 2. \quad (2)$$

This is a rather simple inequality, and we just need some basic manipulations on inequalities.

- If $a_k \leq a_{k+1}$, then $\frac{a_k}{a_{k+1}} \leq 1$. Also, we have $\frac{a_{k+1} - a_k}{a_1} < \frac{1}{1} = 1$. Adding these, we obtain (2).
- If $a_k > a_{k+1}$, then $\frac{a_{k+1} - a_k}{a_1} < 0$. Also, we have $\frac{a_k}{a_{k+1}} < \frac{a_{k+1} + 1}{a_{k+1}} < 2$ since $a_{k+1} > 1$. Adding these, we obtain (2).

Thus, (2) holds in any case, and we are done by induction. $\square$

Remarks. As can be seen from the solution, we do not need any advanced theorems in solving problems like this (and such theorems do not help much). This is one feature of the modern inequalities that we are studying in this section.

Problem 6.2. (India MO 2020 Problem 4) Let $n \geq 2$ be an integer and let $1 < a_1 \leq a_2 \leq \cdots \leq a_n$ be n real numbers such that $a_1 + a_2 + \cdots + a_n = 2n$. Prove that

$$a_1 a_2 \cdots a_{n-1} + a_1 a_2 \cdots a_{n-2} + \cdots + a_1 a_2 + a_1 + 2 \leq a_1 a_2 \cdots a_n.$$

Analysis. Suppose we intend to use induction to solve this problem. In the inductive step (case $n + 1$), we cannot directly use the result of case n because the condition $a_1 + a_2 + \cdots + a_n = 2n$ needs not hold (we only have $a_1 + a_2 + \cdots + a_{n+1} = 2(n + 1)$).

To get rid of this issue, we may first apply the inductive hypothesis to the variables $a_1, a_2, \dots, a_{n-1}, b$ where b is chosen such that $a_1 + a_2 + \cdots + a_{n-1} + b = 2n$. Here we choose to replace a_n by b because a_n only occurs once in the inequality while other variables occur more number of times. Then when we proceed to prove the inequality in the case $n + 1$, we only need to handle fewer terms.

Solution. We prove this by induction on n . For the base case $n = 2$, we have $a_1 + a_2 = 4$, and so

$$\begin{aligned} & a_1 + 2 \leq a_1 a_2 \\ \Leftrightarrow & a_1 + 2 \leq a_1(4 - a_1) \\ \Leftrightarrow & a_1^2 - 3a_1 + 2 \leq 0 \\ \Leftrightarrow & (a_1 - 1)(a_1 - 2) \leq 0. \end{aligned}$$

This is true since $1 < a_1 \leq a_2$ and $a_1 + a_2 = 4$ (which yields $1 < a_1 \leq 2$).

Assume the case n holds. Consider the case $n + 1$. Define $b = a_n + a_{n+1} - 2$, so that

$$a_1 + a_2 + \cdots + a_{n-1} + b = 2(n + 1) - 2 = 2n.$$

Also, we need to prove the inequality $a_{n-1} \leq b$ in order to use the inductive hypothesis.

Indeed, we have $a_{n+1} \geq \frac{2(n+1)}{n+1} = 2$ since a_{n+1} is the largest among $a_1, a_2, \dots, a_{n+1}$ (with sum $2(n+1)$). This yields

$$b = a_n + a_{n+1} - 2 \geq a_{n-1} + 2 - 2 = a_{n-1}.$$

Therefore, we can apply the inductive hypothesis to the terms $a_1, a_2, \dots, a_{n-1}, b$. This gives

$$a_1 a_2 \cdots a_{n-1} + a_1 a_2 \cdots a_{n-2} + \cdots + a_1 a_2 + a_1 + 2 \leq a_1 a_2 \cdots a_{n-1} b.$$

Comparing with our goal, which is

$$a_1 a_2 \cdots a_n + a_1 a_2 \cdots a_{n-1} + a_1 a_2 \cdots a_{n-2} + \cdots + a_1 a_2 + a_1 + 2 \leq a_1 a_2 \cdots a_n a_{n+1},$$

it suffices to prove

$$a_1 a_2 \cdots a_n + a_1 a_2 \cdots a_{n-1} b \leq a_1 a_2 \cdots a_{n+1}.$$

Now,

$$\begin{aligned} & a_1 a_2 \cdots a_n + a_1 a_2 \cdots a_{n-1} b \leq a_1 a_2 \cdots a_{n+1} \\ \Leftrightarrow & a_n + (a_n + a_{n+1} - 2) \leq a_n a_{n+1} \\ \Leftrightarrow & 0 \leq (a_n - 1)(a_{n+1} - 2). \end{aligned}$$

This holds because $a_n > 1$ and $a_{n+1} \geq 2$ (proved above). Therefore, the inductive step is established, and the inequality is proved by induction. Equality holds when $a_1 = a_2 = \cdots = a_n = 2$. $\square$

Problem 6.3. Let $x_1, x_2, \dots, x_n$ be nonnegative real numbers satisfying

$$x_1 + x_2 + \cdots + x_n \leq \frac{1}{2}.$$

Prove that

$$(1 - x_1)(1 - x_2) \cdots (1 - x_n) \geq \frac{1}{2}.$$

Analysis. At first glance, it does not seem to be possible to prove this inequality by induction. This is because if we apply the inductive hypothesis

$$(1 - x_1)(1 - x_2) \cdots (1 - x_n) \geq \frac{1}{2}$$

to estimate $(1 - x_1)(1 - x_2) \cdots (1 - x_n)(1 - x_{n+1})$, we only obtain

$$(1 - x_1)(1 - x_2) \cdots (1 - x_n)(1 - x_{n+1}) \geq \frac{1}{2}(1 - x_{n+1}).$$

Obviously, we cannot show that this is at least $\frac{1}{2}$.

One reason to explain this failure is that, the condition $x_1 + x_2 + \cdots + x_{n+1} \leq \frac{1}{2}$ in the inductive step is stronger than the one needed in the inductive hypothesis (which is $x_1 + x_2 + \cdots + x_n \leq \frac{1}{2}$). So the result we derive from it is weaker. In such situations, we may consider to strengthen the inequality to prove. This sounds like we need to prove something even harder. But in fact it also gives us a stronger inductive hypothesis to use in the inductive step, so it is possible that the inequality can be proved under this idea.

In other words, we try to prove an inequality of the form

$$(1 - x_1)(1 - x_2) \cdots (1 - x_n) \geq \frac{1}{2} + f(x_1, x_2, \dots, x_n)$$

for some nonnegative function f . It is not hard to see that the equality case of the desired inequality is $(x_1, x_2, \dots, x_n) = \left(\frac{1}{2}, 0, 0, \dots, 0\right)$ or its permutation. So we need $f\left(\frac{1}{2}, 0, 0, \dots, 0\right) = 0$, and of course f should be symmetric. For example, noticing that the sum of the variables is a given condition, a simple choice would be

$$f(x_1, x_2, \dots, x_n) = \frac{1}{2} - (x_1 + x_2 + \cdots + x_n).$$

Solution. We shall prove by induction that $x_1 + x_2 + \cdots + x_n \leq \frac{1}{2}$ implies

$$(1 - x_1)(1 - x_2) \cdots (1 - x_n) \geq 1 - (x_1 + x_2 + \cdots + x_n). \quad (1)$$

When $n = 1$, the inequality is indeed an equality. Assume (1) holds for some integer $n = k \geq 1$. When $n = k + 1$, we need to prove $x_1 + x_2 + \cdots + x_{k+1} \leq \frac{1}{2}$ implies

$$(1 - x_1)(1 - x_2) \cdots (1 - x_k)(1 - x_{k+1}) \geq 1 - (x_1 + x_2 + \cdots + x_{k+1}). \quad (2)$$

Since the variables are nonnegative, the condition implies $x_1 + x_2 + \cdots + x_k \leq \frac{1}{2}$ and $x_{k+1} < 1$. Applying the inductive hypothesis to the first k terms of (2), we have

$$(1 - x_1)(1 - x_2) \cdots (1 - x_k)(1 - x_{k+1}) \geq [1 - (x_1 + x_2 + \cdots + x_k)](1 - x_{k+1}).$$

Letting $y = x_1 + x_2 + \cdots + x_k$, it remains to prove

$$(1 - y)(1 - x_{k+1}) \geq 1 - (y + x_{k+1}).$$

After simplification, this is simply $yx_{k+1} \geq 0$, which is obviously true. This proves (2), hence (1) holds by induction.

Lastly, it follows easily that

$$(1 - x_1)(1 - x_2) \cdots (1 - x_n) \geq 1 - (x_1 + x_2 + \cdots + x_n) \geq \frac{1}{2}.$$

Equality holds when one of $x_1, x_2, \dots, x_n$ is $\frac{1}{2}$ and the others are 0. $\square$

There is an induction method called the *Cauchy induction*. The method says if we can prove that the case n implies the case $2n$ as well as the case $n - 1$, and the base case $n = 1$ is established, then the result holds for every positive integer n . This is because from $n = 1$ and the implication $n \Rightarrow 2n$, we can obtain every case of the form $n = 2^k$ by the usual induction (on k). Afterwards, since every positive integer is smaller than some power of 2, we can go backwards via the implication $n \Rightarrow n - 1$ to obtain the desired case. Thus, all cases are covered.

For example, the Cauchy induction can be used to prove the AM-GM inequality. In that well-known proof, if we replace one of the n variables by the arithmetic mean of the other variables, we are able to show that the case n implies the case $n - 1$. One may try to fill in the details of this proof. In general, this method suggests sometimes we may think of some special ways to use induction, rather than sticking to the most standard idea or the standard order.

Problem 6.4. (IMO Shortlist 2016 A1) Let a, b and c be positive real numbers such that $\min\{ab, bc, ca\} \geq 1$. Prove that

$$\sqrt[3]{(a^2 + 1)(b^2 + 1)(c^2 + 1)} \leq \left(\frac{a + b + c}{3}\right)^2 + 1.$$

Analysis. We can consider an analogous problem with n variables instead of 3 variables. The case $n = 2$ is easy to solve. It does not seem to be easy to proceed from $n = 2$ to $n = 3$, but it is rather easy to go from $n = 2$ to $n = 4$ directly. So we may think of the Cauchy induction (or at least its idea).

Solution. We first prove the following results.

Claim 1. For any positive real numbers x, y with $xy \geq 1$, we have

$$(x^2 + 1)(y^2 + 1) \leq \left(\left(\frac{x + y}{2}\right)^2 + 1\right)^2.$$

Proof. Firstly, we rewrite the left-hand side as a function of $x + y$ and xy , i.e.

$$(x^2 + 1)(y^2 + 1) = (xy - 1)^2 + (x + y)^2.$$

Since the right-hand side of the inequality is fixed when $x + y$ is fixed, it is natural to bound the above expression by using

$$xy \leq \frac{(x + y)^2}{4}.$$

Since $xy \geq 1$, this implies

$$(xy - 1)^2 \leq \left(\frac{(x + y)^2}{4} - 1 \right)^2.$$

It follows that

$$(x^2 + 1)(y^2 + 1) \leq \left(\frac{(x + y)^2}{4} - 1 \right)^2 + (x + y)^2 = \left(\left(\frac{x + y}{2} \right)^2 + 1 \right)^2.$$

□

Claim 2. For any positive real numbers a, b, c, d with $ab, cd \geq 1$, we have

$$(a^2 + 1)(b^2 + 1)(c^2 + 1)(d^2 + 1) \leq \left(\left(\frac{a + b + c + d}{4} \right)^2 + 1 \right)^4.$$

Proof. This can be proved using [claim 1](#). (When one formulates this claim, one may impose the constraints $ab, ac, ad, bc, bd, cd \geq 1$. But it turns out we only need a weaker condition.) Indeed, we have

$$(a^2 + 1)(b^2 + 1)(c^2 + 1)(d^2 + 1) \leq \left(\left(\frac{a + b}{2} \right)^2 + 1 \right)^2 \left(\left(\frac{c + d}{2} \right)^2 + 1 \right)^2$$

by [claim 1](#). Also, by setting $x = \frac{a + b}{2}$ and $y = \frac{c + d}{2}$, we can use [claim 1](#) again to deduce

$$\left(\left(\frac{a + b}{2} \right)^2 + 1 \right) \left(\left(\frac{c + d}{2} \right)^2 + 1 \right) \leq \left(\left(\frac{a + b + c + d}{4} \right)^2 + 1 \right)^2$$

(note that $xy \geq \sqrt{ab}\sqrt{cd} \geq 1$ by the AM-GM inequality). This implies

$$(a^2 + 1)(b^2 + 1)(c^2 + 1)(d^2 + 1) \leq \left(\left(\frac{a + b + c + d}{4} \right)^2 + 1 \right)^4$$

as desired. □

Now, we return to the original question. Let a, b, c be positive real numbers such that $ab, bc, ca \geq 1$. WLOG assume $c \geq b \geq a$, so that $c \geq 1$. Based on the idea of the

proof of the AM-GM inequality mentioned above, we would like to set $d = \frac{a+b+c}{3}$ in [claim 2](#). First we need to check the necessary conditions. In particular, we have to check

$$cd = \frac{c(a+b+c)}{3} = \frac{ca+bc+c^2}{3} \geq \frac{1+1+1}{3} = 1.$$

Thus, using [claim 2](#) with $d = \frac{a+b+c}{3}$, we find that

$$(a^2+1)(b^2+1)(c^2+1)(d^2+1) \leq \left(\left(\frac{a+b+c+d}{4} \right)^2 + 1 \right)^4 = (d^2+1)^4.$$

This implies $(a^2+1)(b^2+1)(c^2+1) \leq (d^2+1)^3$, and hence

$$\sqrt[3]{(a^2+1)(b^2+1)(c^2+1)} \leq \left(\frac{a+b+c}{3} \right)^2 + 1.$$

□

Problem 6.5. (IMO Shortlist 2020 A7) Let n and k be positive integers. Prove that for $a_1, \dots, a_n \in [1, 2^k]$ one has

$$\sum_{i=1}^n \frac{a_i}{\sqrt{a_1^2 + \dots + a_i^2}} \leq 4\sqrt{kn}.$$

Analysis. There are two parameters n and k in this problem. So we need to decide which one is the parameter for the induction. (Of course we may use induction on both n and k at the same time, which means we assume all cases $(n', k') \neq (n, k)$ with $n' \leq n$ and $k' \leq k$ hold and proceed to the case (n, k) . But it turns out we need not use this stronger assumption.)

If we proceed from the case n to the case $n+1$, there is one extra term on the left, which can be large. So we cannot directly use the inductive hypothesis to prove it. If we proceed from the case k to the case $k+1$, the sizes of some terms can be larger. It is not hard to see (and to prove using a local refinement idea) we may assume the larger terms have larger indices, and so these terms only affect the tail part of the sum. So it is possible to use the inductive hypothesis. This time we have a much better bound for the additional terms because each large term has a stronger lower bound than the trivial bound 1. Therefore, it should be the correct way to use the induction.

Solution. Let

$$f(a_1, a_2, \dots, a_n) = \sum_{i=1}^n \frac{a_i}{\sqrt{a_1^2 + \dots + a_i^2}}.$$

We first prove that we may permute $a_1, a_2, \dots, a_n$ in increasing order as this does not decrease the value of f . Indeed, consider a permutation which maximizes f . It exists

since there are finitely many permutations. If there exists $j = 1, 2, \dots, n-1$ such that $a_j > a_{j+1}$, we interchange the two terms a_j and a_{j+1} . Let $x = a_1^2 + a_2^2 + \dots + a_{j-1}^2$. Note that

$$\begin{aligned}
& f(a_1, \dots, a_{j+1}, a_j, \dots, a_n) - f(a_1, \dots, a_j, a_{j+1}, \dots, a_n) \\
&= \frac{a_{j+1}}{\sqrt{x + a_{j+1}^2}} + \frac{a_j}{\sqrt{x + a_{j+1}^2 + a_j^2}} - \frac{a_j}{\sqrt{x + a_j^2}} - \frac{a_{j+1}}{\sqrt{x + a_j^2 + a_{j+1}^2}} \\
&= \left(\frac{a_{j+1}}{\sqrt{x + a_{j+1}^2}} - \frac{a_{j+1}}{\sqrt{x + a_j^2 + a_{j+1}^2}} \right) - \left(\frac{a_j}{\sqrt{x + a_j^2}} - \frac{a_j}{\sqrt{x + a_{j+1}^2 + a_j^2}} \right) \\
&= \frac{a_{j+1} \left(\sqrt{x + a_j^2 + a_{j+1}^2} - \sqrt{x + a_{j+1}^2} \right)}{\sqrt{(x + a_{j+1}^2)(x + a_j^2 + a_{j+1}^2)}} - \frac{a_j \left(\sqrt{x + a_j^2 + a_{j+1}^2} - \sqrt{x + a_j^2} \right)}{\sqrt{(x + a_j^2)(x + a_{j+1}^2 + a_j^2)}} \\
&= \frac{a_j^2 a_{j+1}}{\sqrt{(x + a_{j+1}^2)(x + a_j^2 + a_{j+1}^2)} \left(\sqrt{x + a_j^2 + a_{j+1}^2} + \sqrt{x + a_{j+1}^2} \right)} \\
&\quad - \frac{a_j a_{j+1}^2}{\sqrt{(x + a_j^2)(x + a_{j+1}^2 + a_j^2)} \left(\sqrt{x + a_j^2 + a_{j+1}^2} + \sqrt{x + a_j^2} \right)}.
\end{aligned}$$

This is positive because $a_j^2 a_{j+1} > a_j a_{j+1}^2$ and $x + a_{j+1}^2 < x + a_j^2$. Therefore, we derive a contradiction that f is not maximized. Thus, we may now assume $a_1 \leq a_2 \leq \dots \leq a_n$.

We now prove the desired inequality by induction on k . When $k = 1$, we need to prove

$$\sum_{i=1}^n \frac{a_i}{\sqrt{a_1^2 + \dots + a_i^2}} \leq 4\sqrt{n}$$

for any $a_1, \dots, a_n$ satisfying $1 \leq a_i \leq 2$. This can be done by using the trivial bounds. Indeed,

$$\sum_{i=1}^n \frac{a_i}{\sqrt{a_1^2 + \dots + a_i^2}} \leq \sum_{i=1}^n \frac{2}{\sqrt{1^2 + \dots + 1^2}} = \sum_{i=1}^n \frac{2}{\sqrt{i}}.$$

To estimate this sum, we can use integration. Since the function $\frac{2}{\sqrt{x}}$ is decreasing, we have

$$\sum_{i=1}^n \frac{2}{\sqrt{i}} < \int_0^n \frac{2}{\sqrt{x}} dx = 4\sqrt{n} - 4\sqrt{0} = 4\sqrt{n}.$$

This establishes the base case.

Now, assume the result holds for all smaller cases, and we consider the case $k \geq 2$. We may assume $a_n > 2^{k-1}$, or we can replace k by $k-1$ (and the resultant inequality is even stronger). Then we choose the index m such that $a_m \leq 2^{k-1} < a_{m+1}$ (possibly $m = 0$, with $a_0 = 0$). Note that

$$\sum_{i=1}^m \frac{a_i}{\sqrt{a_1^2 + \dots + a_i^2}} \leq 4\sqrt{(k-1)m}$$

by the inductive hypothesis applied to $a_1, a_2, \dots, a_m \in [1, 2^{k-1}]$ (which still holds if $m = 0$). Also, as in the base case, we have

$$\begin{aligned} \sum_{i=m+1}^n \frac{a_i}{\sqrt{a_1^2 + \dots + a_i^2}} &\leq \sum_{i=m+1}^n \frac{a_i}{\sqrt{a_{m+1}^2 + \dots + a_i^2}} \leq \sum_{i=m+1}^n \frac{2^k}{\sqrt{(2^{k-1})^2 + \dots + (2^{k-1})^2}} \\ &= \sum_{i=m+1}^n \frac{2}{\sqrt{i-m}} < \int_m^n \frac{2}{\sqrt{x-m}} dx = 4\sqrt{n-m}. \end{aligned}$$

Combining these, we obtain

$$\sum_{i=1}^n \frac{a_i}{\sqrt{a_1^2 + \dots + a_i^2}} \leq 4\sqrt{(k-1)m} + 4\sqrt{n-m}.$$

Therefore, it remains to prove

$$\sqrt{(k-1)m} + \sqrt{n-m} \leq \sqrt{kn}.$$

This is elementary. For instance, it follows from the Cauchy-Schwarz inequality since

$$\sqrt{kn} = \sqrt{((k-1) + (1))((m) + (n-m))} \geq \sqrt{(k-1)m} + \sqrt{n-m}.$$

This completes the inductive step, and we are done. $\square$

6.2 Combinatorial Inequalities

Combinatorial ideas are often required when solving problems in algebra and number theory. If one cannot solve some problems using the standard tricks and classical results, one may think of such ideas and test if they are useful or not. In this subsection, we shall go through several inequality problems which have some combinatorial flavour, and recall some basic techniques in combinatorics that may help.

Problem 6.6. (Tuymaada Olympiad 2020 Senior Problem 2) Given positive real numbers $a_1, a_2, \dots, a_n$, let

$$m = \min \left\{ a_1 + \frac{1}{a_2}, a_2 + \frac{1}{a_3}, \dots, a_{n-1} + \frac{1}{a_n}, a_n + \frac{1}{a_1} \right\}.$$

Prove the inequality

$$\sqrt[n]{a_1 a_2 \dots a_n} + \frac{1}{\sqrt[n]{a_1 a_2 \dots a_n}} \geq m.$$

Analysis. To prove the inequality, a simple (and equivalent) way is to find a term $a_{j-1} + \frac{1}{a_j}$ which is bounded above by $\sqrt[n]{a_1 a_2 \dots a_n} + \frac{1}{\sqrt[n]{a_1 a_2 \dots a_n}}$. In view of the two expressions, it is even better if we can find an index j such that $a_{j-1} \leq \sqrt[n]{a_1 a_2 \dots a_n}$

and $a_j \geq \sqrt[n]{a_1 a_2 \cdots a_n}$. This transforms the problem to an existence problem in combinatorics.

Now, the key observation is that the geometric mean $\sqrt[n]{a_1 a_2 \cdots a_n}$ is a kind of average, and so there exist ‘small numbers’ as well as ‘large numbers’ (relative to this average) among $a_1, a_2, \dots, a_n$. What we need is that there exist two consecutive numbers, which are small and large respectively. This is obvious because we may suppose all terms are put in a circle in view of the cyclic symmetry. When we move from a small number to a large number around the circle, there must be some point for which the size just changes from small to large.

Solution. In the following, all indices are taken modulo n . Let $g = \sqrt[n]{a_1 a_2 \cdots a_n}$ be the geometric mean of the n real numbers. Then there exist distinct indices i and j such that $a_i \leq g \leq a_j$. In particular, we may choose j to be the smallest such index for which $i < j$. By this minimality, we must have $a_{j-1} \leq g$, or otherwise we can replace j by $j-1$ (note that if $a_{j-1} > g$, we must have $j-1 \neq i$). Thus, we obtain

$$g + \frac{1}{g} \geq a_{j-1} + \frac{1}{a_j} \geq m$$

as desired. □

Remarks. Considering some extremal cases (such as the smallest index j satisfying some properties) is a useful technique in solving some combinatorial problems. Such a consideration may also be useful in other branches.

Problem 6.7. (International Zhautykov Olympiad 2019 Problem 2) Find the biggest real number C , such that for any different positive real numbers $a_1, a_2, \dots, a_{2019}$, the inequality

$$\frac{a_1}{|a_2 - a_3|} + \frac{a_2}{|a_3 - a_4|} + \cdots + \frac{a_{2019}}{|a_1 - a_2|} > C$$

is satisfied.

Analysis. It more or less means we need to find the minimum value of the expression on the left (it is more correct to call this the *infimum* (下确界) in analysis, because there is no equality case). First we need to guess the answer and provide an ‘equality’ case. Second we need to prove the inequality for that particular choice of C .

Strictly speaking, there is no equality case but there exist cases which approach the equality. This is mainly because of the positivity of the numbers. Thus, it is very likely that the extremal case is attained when some variables tend to 0. As no two consecutive numbers should be close to each other unless the preceding term is close to 0 (otherwise one of the fractions is very large), a natural example that may come up is the sequence $0, x, 0, x, \dots, 0, x, y$. With some trial and error, we can actually take $y \approx x$, and x can be arbitrary. However, since we need $a_i > 0$ and $a_i \neq a_j$ for any $i \neq j$, we have to change each value by a little bit. This is the reason why we need

to add or subtract some ε terms in the solution below. In any case, we now have a good guess that $C = 1010$.

Next, to prove the inequality when $C = 1010$, the first idea is based on the above example. When a term a_i is close to 0, we do not lose too much by ignoring the term $\frac{a_i}{|a_{i+1} - a_{i+2}|}$ and replacing the term $\frac{a_{i-1}}{|a_i - a_{i+1}|}$ by $\frac{a_{i-1}}{a_{i+1}}$. In other words, we can remove the absolute values from $|a_i - a_{i+1}|$ by ignoring the smaller term a_i , which may simplify the expression. To have a better understanding on the terms remained after this simplification, we associate each term a_{i-1} to the larger term among a_i and a_{i+1} . Such an association can be better described by constructing a (directed) graph. This is the place where some combinatorial languages are used.

Solution. The largest C is 1010. In the following, all indices are taken modulo 2019.

Firstly, consider

$$a_n = \begin{cases} n\varepsilon^2 & \text{if } n \text{ is odd and } n \neq 2019, \\ 1 - n\varepsilon & \text{if } n \text{ is even or } n = 2019 \end{cases}$$

where $\varepsilon > 0$ is a sufficiently small constant (this is essentially the example $0, 1, 0, 1, \dots, 0, 1, 1$ mentioned above). Clearly, all numbers are distinct. When $\varepsilon \rightarrow 0^+$, we have

$$\begin{aligned} \frac{a_n}{|a_{n+1} - a_{n+2}|} &\rightarrow \frac{0}{1-0} = 0 & \text{for } n = 1, 3, 5, \dots, 2015, \\ \frac{a_n}{|a_{n+1} - a_{n+2}|} &\rightarrow \frac{1}{1-0} = 1 & \text{for } n = 2, 4, 6, \dots, 2018, 2019, \\ \frac{a_{2017}}{|a_{2018} - a_{2019}|} &= 2017\varepsilon \rightarrow 0. \end{aligned}$$

(In the construction, we use ε for some cases and use ε^2 for other cases. The main purpose is to allow $\frac{a_{2017}}{|a_{2018} - a_{2019}|} \rightarrow 0$.) For this sequence, we have

$$\frac{a_1}{|a_2 - a_3|} + \frac{a_2}{|a_3 - a_4|} + \dots + \frac{a_{2019}}{|a_1 - a_2|} \rightarrow 1010,$$

so we must have $C \leq 1010$.

Secondly, we prove the inequality

$$\frac{a_1}{|a_2 - a_3|} + \frac{a_2}{|a_3 - a_4|} + \dots + \frac{a_{2019}}{|a_1 - a_2|} > 1010$$

for any such sequence. Consider a directed graph G of 2019 vertices as follows. The vertices are the indices $1, 2, \dots, 2019$. For each $i = 1, 2, \dots, 2019$, we draw an edge from i to $i+1$ if $a_{i+1} > a_{i+2}$, and draw an edge from i to $i+2$ if $a_{i+2} > a_{i+1}$. Thus, exactly one edge is drawn from each i to some $f(i)$. For simplicity, we define the *step* of the edge $i, f(i)$ to be $f(i) - i$ (which is 1 or 2).

From our discussion, we should first remove the absolute values as follows:

$$\frac{a_1}{|a_2 - a_3|} + \frac{a_2}{|a_3 - a_4|} + \dots + \frac{a_{2019}}{|a_1 - a_2|} > \frac{a_1}{a_{f(1)}} + \frac{a_2}{a_{f(2)}} + \dots + \frac{a_{2019}}{a_{f(2019)}}. \quad (1)$$

It is still not easy to give a lower bound for this expression because the denominators are quite arbitrary. We should focus on those fractions whose denominators (and numerators) have more relationships. So we try to look for more structure of the graph G .

As there are finitely many vertices in G and the out-degree of each vertex is 1 (which is positive), G must consist of a cycle $v_1, v_2, \dots, v_k, v_1$ (imagine that when one travels along the edges from any starting point, one must eventually repeat some vertices and stay in the same cycle forever). When we go from v_1 back to itself, we have travelled at least 2019 steps. Thus, the number of edges in this cycle is at least $\left\lceil \frac{2019}{2} \right\rceil = 1010$, i.e. $k \geq 1010$.

It turns out that we can simply focus on the indices appearing in this cycle to estimate (1). Indeed, since $f(v_i) = v_{i+1}$ by definition, we have

$$\frac{a_1}{a_{f(1)}} + \frac{a_2}{a_{f(2)}} + \dots + \frac{a_{2019}}{a_{f(2019)}} \geq \frac{a_{v_1}}{a_{v_2}} + \frac{a_{v_2}}{a_{v_3}} + \dots + \frac{a_{v_k}}{a_{v_1}} \geq k \geq 1010$$

by the AM-GM inequality. This establishes the lower bound, and the proof is complete. $\square$

Remarks. One merit of using graph theory is that there are many standard terminologies like edges, paths, cycles, etc., so that we do not need to define new things. Another merit is that there are some well-known results that we can use directly without proof.

Problem 6.8. (IMO Shortlist 2015 C5/Problem 6) The sequence $a_1, a_2, \dots$ of integers satisfies the following conditions:

- (i) $1 \leq a_j \leq 2015$ for all $j \geq 1$;
- (ii) $k + a_k \neq \ell + a_\ell$ for all $1 \leq k < \ell$.

Prove that there exist two positive integers b and N such that

$$\left| \sum_{j=m+1}^n (a_j - b) \right| \leq 1007^2$$

for all integers m and n satisfying $n > m \geq N$.

Analysis. Although the problem statement is about sequences and inequalities, this problem is classified as a combinatorics problem in the shortlist. Though one should not know the classification in the real IMO, it is not hard to figure out that some combinatorial ideas should be useful in solving this problem after some trial.

Similar to the previous problem, we first construct a (directed) graph G that relates some pairs of the indices. This is a common way of using graph theory in problems involving a sequence. For this problem, the main condition is $k + a_k \neq \ell + a_\ell$ for any

$k \neq \ell$. In view of this, it may be a good idea to draw an edge from each vertex k to the vertex $k + a_k$. Then the condition means that each vertex is the terminal point of at most one edge. So G is simply the union of some disjoint infinite paths.

Again we define the *step* of the edge $k, k + a_k$ to be a_k . Then $\sum_{j=m+1}^n a_j$ is the sum of steps of the edges drawing from $m + 1, \dots, n$. Using double counting, the average of this sum is approximately the number of paths in G . Indeed, suppose there are t paths. The sum of steps of each path (for the part that more or less covers the vertices $m + 1$ and n) is about $n - m$. Since there are $n - m$ terms in the sum, the average is about $\frac{t(n - m)}{n - m} = t$. Therefore, it is natural to take $b = t$ so that $\left| \sum_{j=m+1}^n (a_j - b) \right|$ is minimized. The remaining parts are just some more careful estimations.

Solution. Consider an infinite directed graph G as follows. The vertices are the positive integers. For each positive integer k , we draw an edge from k to $k + a_k$, and define its *step* to be a_k . Since exactly one edge is drawn from each k , if we start from an arbitrary vertex, we can move along the edges and travel in some infinite paths in a unique way. Also, the given condition $k + a_k \neq \ell + a_\ell$ for $k \neq \ell$ guarantees that each vertex is the terminal point of at most one edge. Therefore, the paths are disjoint.

Moreover, we can show that there are finitely many (actually, at most 2015) paths. Indeed, if there are 2016 paths, then for some sufficiently large M , each path should contain some vertex among $M + 1, M + 2, \dots, M + 2015$, since otherwise a path must contain an edge of step ≥ 2016 , contradicting $a_k \leq 2015$ for all k . But this is also impossible as the paths cannot be disjoint by the pigeonhole principle (as there are only 2015 vertices among $M + 1, M + 2, \dots, M + 2015$). So the number of paths is at most 2015. This gives us a clear picture on the graph G (it is the union of at most 2015 disjoint infinite paths).

Now, we choose b to be the number of paths in G , and choose a sufficiently large N which is larger than the initial vertex of every path (this is quite a natural choice because we have to use the structure of G). It remains to estimate the sum

$$\left| \sum_{j=m+1}^n (a_j - b) \right|$$

for any $n > m \geq N$. We fix any such m and n . Recall that

$$S = \sum_{j=m+1}^n a_j$$

is the sum of steps of the edges drawing from $m + 1, \dots, n$, and we just need a more careful double counting to estimate this sum. In particular, we can estimate the contribution from each path one by one.

For each path, consider the vertices $v_1, v_2, \dots, v_r$ in this order such that v_1 is the smallest index for which $v_1 > m$, and v_r is the smallest index for which $v_r > n$. Then

the contribution of this path to S is

$$a_{v_1} + a_{v_2} + \cdots + a_{v_{r-1}} = (v_2 - v_1) + (v_3 - v_2) + \cdots + (v_r - v_{r-1}) = v_r - v_1.$$

(This is quite obvious if one understands the meaning of the steps of edges.)

Therefore, if we let $x_1, x_2, \dots, x_b$ be the smallest indices of the paths for which $x_i > m$, and let $y_1, y_2, \dots, y_b$ be the smallest indices of the paths for which $y_i > n$ (so x_i, y_i are the v_1, v_r above), then we have

$$S = \sum_{i=1}^b (y_i - x_i).$$

Note that each x_i is close to m , and each y_i is close to n . Thus, we rewrite this as

$$\sum_{j=m+1}^n (a_j - b) = S - (n - m)b = \sum_{i=1}^b ((y_i - x_i) - (n - m)) = \sum_{i=1}^b (y_i - n) - \sum_{i=1}^b (x_i - m). \quad (1)$$

This is exactly the sum to estimate.

Now, the numbers $y_1 - n, y_2 - n, \dots, y_b - n$ must be distinct and they can only be $1, 2, \dots, 2015$ since each $y_i - n$ is at most the step of an edge. More importantly, one of them is equal to 1 since the vertex $n + 1$ belongs to some path. It follows that

$$1 + \sum_{i=1}^{b-1} (i + 1) \leq \sum_{i=1}^b (y_i - n) \leq 1 + \sum_{i=1}^{b-1} (2016 - b + i). \quad (2)$$

Similarly, we have

$$1 + \sum_{i=1}^{b-1} (i + 1) \leq \sum_{i=1}^b (x_i - m) \leq 1 + \sum_{i=1}^{b-1} (2016 - b + i). \quad (3)$$

Combining (1), (2) and (3), we obtain

$$\left| \sum_{j=m+1}^n (a_j - b) \right| \leq \sum_{i=1}^{b-1} ((2016 - b + i) - (i + 1)) = (2015 - b)(b - 1).$$

Lastly, by the AM-GM inequality, this is at most

$$\left(\frac{(2015 - b) + (b - 1)}{2} \right)^2 = 1007^2$$

as desired. □

Problem 6.9. (Iran MO 2nd Round 2021 Problem 5) 1400 real numbers are given. Prove that one can choose three of them like x, y, z such that

$$\left| \frac{(x - y)(y - z)(z - x)}{x^4 + y^4 + z^4 + 1} \right| < 0.009.$$

Analysis. Note that the fraction is small if x, y, z are close to each other (since $|(x-y)(y-z)(z-x)|$ is small), or one of x, y, z is large (since $x^4 + y^4 + z^4$ is large). So a possible plan is as follows. If there exists a large number x , then we are done by choosing such x and any (smaller) y, z . If there is no large number, then all numbers are bounded in some intervals, and we can use the pigeonhole principle to show that 3 of them must be sufficiently close and we are done. It remains to set some explicit bounds to define ‘large’ and ‘close’.

Solution. WLOG assume at least half of the numbers are nonnegative. In the following, we only deal with these 700 nonnegative real numbers. Let C be a large constant to be determined. Suppose there exists a number x larger than C and two smaller numbers $y > z$. Then we have

$$\left| \frac{(x-y)(y-z)(z-x)}{x^4 + y^4 + z^4 + 1} \right| = \frac{(x-y)(y-z)(x-z)}{x^4 + y^4 + z^4 + 1} < \frac{(x)(x)(x)}{x^4} = \frac{1}{x} < \frac{1}{C}.$$

So we are done if $C = \frac{1}{0.009}$. In other words, we are done if the largest number is larger than C . Thus, we now assume all numbers are at most C .

Now, there are 700 numbers in the interval $[0, C]$. As we need to find 3 close numbers, we should partition this interval into fewer than $\frac{700}{3-1} = 350$ intervals to use the pigeonhole principle. Thus, we partition $[0, C]$ into 349 small intervals

$$\left[0, \frac{C}{349}\right], \left[\frac{C}{349}, \frac{2C}{349}\right], \dots, \left[\frac{348C}{349}, C\right].$$

By the pigeonhole principle, $\left\lceil \frac{700}{349} \right\rceil = 3$ of the numbers $x > y > z$ must belong to the same interval. So the difference of any two of them is at most $\frac{C}{349}$. We have

$$\left| \frac{(x-y)(y-z)(z-x)}{x^4 + y^4 + z^4 + 1} \right| = \frac{(x-y)(y-z)(x-z)}{x^4 + y^4 + z^4 + 1} \leq (x-y)(y-z)(x-z).$$

This can be estimated using the AM-GM inequality. Indeed, as $x-y$ and $y-z$ are positive, we have

$$(x-y)(y-z) \leq \left(\frac{(x-y) + (y-z)}{2} \right)^2 = \frac{(x-z)^2}{4}.$$

Combining these, we obtain

$$\left| \frac{(x-y)(y-z)(z-x)}{x^4 + y^4 + z^4 + 1} \right| \leq \frac{(x-z)^3}{4} \leq \frac{1}{4} \left(\frac{C}{349} \right)^3.$$

It remains to check this is less than 0.009, i.e. $4(0.009)(349 \times 0.009)^3 > 1$. This holds because

$$\begin{aligned} 4(0.009)(349 \times 0.009)^3 &= 0.036(3.141)^3 > 0.036(3.1)^3 = 0.036 \times 29.791 \\ &> 0.036 \times 29 = 1.044 > 1. \end{aligned}$$

This proves the desired result. □

Remarks. The pigeonhole principle is another combinatorial result which is commonly used in many other areas. This is often used when we have many objects, and we need to find some of them which satisfy certain properties.

Problem 6.10. (IMO Shortlist 2016 A2) Find the smallest real constant C such that for any positive real numbers a_1, a_2, a_3, a_4 and a_5 (not necessarily distinct), one can always choose distinct subscripts i, j, k and ℓ such that

$$\left| \frac{a_i}{a_j} - \frac{a_k}{a_\ell} \right| \leq C.$$

Analysis. We first need to guess the correct answer, or otherwise it is hard to know whether the bound obtained is the best or not. As multiplying all numbers by the same constant does not affect the result, we may assume the smallest number is 1. And then a simple idea is just to test various examples with different features, such as

$$(1, 1, 1, 1, n), (1, 1, 1, n, \infty), (1, 1, n, n, \infty), (1, n, n, n, \infty), (1, x, y, \infty, \infty)$$

(where ∞ denotes a number which tends to infinity). In particular, the example $(1, 2, 2, 2, \infty)$ seems to give us the best bound $\frac{1}{2}$. So we are going to prove the assertion when $C = \frac{1}{2}$.

After C is fixed, we want to find two fractions which are close to each other. For distinct indices i, j with $a_i \leq a_j$, we probably want to focus on the fraction $\frac{a_i}{a_j}$ rather than $\frac{a_j}{a_i}$ since all such smaller fractions lie in the interval $[0, 1]$, and there is a higher chance for them to be close to each other. (More rigorously, if $r, s \geq 1$, we have $\left| \frac{1}{r} - \frac{1}{s} \right| = \left| \frac{r-s}{rs} \right| \leq |r-s|$. So we should focus on the reciprocals of r and s .) This leaves us 10 such fractions, and again we can use the pigeonhole principle together with some exhaustive arguments to prove the assertion.

Solution. The smallest C is $\frac{1}{2}$.

We first show that $C \geq \frac{1}{2}$. Consider the numbers $1, 2, 2, 2, n$ where n is a large real number. The fractions formed by two of these numbers in ascending order are $\frac{1}{n}, \frac{2}{n}, \frac{1}{2}, \frac{2}{2}, \frac{2}{1}, \frac{n}{2}, \frac{n}{1}$ respectively. Since the indices i, j, k, ℓ are distinct, $\frac{1}{n}$ and $\frac{2}{n}$ cannot be chosen simultaneously. Therefore the minimum value of $\left| \frac{a_i}{a_j} - \frac{a_k}{a_\ell} \right|$ is $\frac{1}{2} - \frac{2}{n}$. When n tends to infinity, this value approaches $\frac{1}{2}$, and so $C \geq \frac{1}{2}$.

Next, we show that $C = \frac{1}{2}$ is a possible choice. For any positive real numbers $a_1 \leq a_2 \leq a_3 \leq a_4 \leq a_5$, there are $\binom{5}{2} = 10$ fractions of the form $\frac{a_i}{a_j}$ where $1 \leq i < j \leq 5$. All these fractions lie in the interval $(0, 1]$. By the pigeonhole principle, we can find 5 of them which belong to $\left(0, \frac{1}{2}\right]$ or $\left(\frac{1}{2}, 1\right]$. This means there are 5 fractions such that every pair of the fractions has a difference less than $\frac{1}{2}$. It remains to show that there must be 2 of them whose indices are pairwise distinct.

This question can now be transformed to a question in graph theory (mainly to demonstrate how we can use combinatorial ideas). Let G be a graph whose vertices are $1, 2, 3, 4, 5$. For $i < j$, we draw an edge $\{i, j\}$ if $\frac{a_i}{a_j}$ is one of the 5 fractions. We want to show that if G has exactly 5 edges, then we can always find 2 edges without common vertices.

WLOG assume one of the edges is $\{1, 2\}$. All other edges must have 1 or 2 as one of the endpoints, or we are done by choosing that edge together with the edge $\{1, 2\}$. As there are 4 more edges, one of them is not incident on the vertex 1, say $\{2, 3\}$. Similarly, one edge is not incident on the vertex 2, and it can only be $\{1, 3\}$ (if it is $\{1, 4\}$, we can take $\{2, 3\}$ and $\{1, 4\}$). But then any other edge cannot have a common endpoint with all of $\{1, 2\}, \{1, 3\}, \{2, 3\}$ simultaneously. So we can always find such a pair of edges.

To conclude, $C = \frac{1}{2}$ is the smallest possible constant. □

Remarks. The official solution provides a more elegant way to use the pigeonhole principle to conclude there are two fractions whose difference is at most $\frac{1}{2}$. Indeed, 3 of the fractions

$$\frac{a_1}{a_2}, \frac{a_3}{a_4}, \frac{a_1}{a_5}, \frac{a_2}{a_3}, \frac{a_4}{a_5}$$

must belong to the same interval $\left(0, \frac{1}{2}\right]$ or $\left(\frac{1}{2}, 1\right]$. And among these 3 fractions, 2 of them must be consecutive in the above list (where $\frac{a_1}{a_2}$ and $\frac{a_4}{a_5}$ are regarded as consecutive). Clearly, they have distinct indices, and we are done.

Problem 6.11. (USA TST 2011 Problem 9) Determine whether or not there exist two different sets A, B , each consisting of at most 2011^2 positive integers, such that every x with $0 < x < 1$ satisfies the following inequality:

$$\left| \sum_{a \in A} x^a - \sum_{b \in B} x^b \right| < (1 - x)^{2011}.$$

Analysis. The first simple observation is that, if we add the same large constant C to every element in A, B , then the quantity on the left is multiplied by x^C . For a fixed x , we have $x^C \rightarrow 0$ when $C \rightarrow \infty$. This suggests we mainly need to take care of those x which are close to 1. In other words, if we can find some sets A, B such that the inequality holds for any $x \in (1 - \varepsilon, 1)$ where $\varepsilon > 0$ is fixed (and perhaps small), then the inequality holds for any $x \in (0, 1)$ after increasing all elements by a large constant C . It looks like we are very close to the goal (of course it is not!), so the answer should be ‘yes’.

As we should focus on those x which are close to 1, looking at the factor $1 - x$ should be helpful. Suppose the quantity on the left can be expressed as $(1 - x)^k g(x)$ where $1 - x$ does not divide $g(x)$. Then the inequality does not hold if $k < 2011$ (since $g(1) \neq 0$, we cannot have $g(x) < (1 - x)^{2011-k}$ when $x \rightarrow 1$). On the other hand, if $k > 2011$, then it remains to prove $(1 - x)^{k-2011} g(x) < 1$, which should hold when x is close to 1. Therefore, now the major task is to find A and B such that $(1 - x)^{2012}$ divides

$$f(x) = \sum_{a \in A} x^a - \sum_{b \in B} x^b.$$

Using some basic calculus, we know that this holds iff $1 - x$ divides $f(x), f'(x), f''(x), \dots, f^{(2011)}(x)$ (where $f^{(j)}(x)$ means taking the derivative of $f(x)$ for j times). As

$$f^{(j)}(x) = \sum_{a \in A} a(a-1)\cdots(a-j+1)x^{a-j} - \sum_{b \in B} b(b-1)\cdots(b-j+1)x^{b-j},$$

what we need is $|A| = |B|$ (since $f(1) = 0$) and

$$\sum_{a \in A} a(a-1)\cdots(a-j+1) = \sum_{b \in B} b(b-1)\cdots(b-j+1)$$

for $j = 1, 2, \dots, 2011$. This transforms the problem to an existence problem, which can again be solved using the pigeonhole principle (and in fact it is similar to some well-known applications of the pigeonhole principle).

Solution. We first prove the following results.

Claim 1. There exist two sets $A \neq B$ of positive integers such that $|A| = |B| = 2011^2$ and

$$\sum_{a \in A} a(a-1)\cdots(a-j+1) = \sum_{b \in B} b(b-1)\cdots(b-j+1)$$

for $j = 1, 2, \dots, 2011$.

Proof. Let M be a sufficiently large positive integer. We consider all subsets of $\{1, 2, \dots, M\}$ consisting of exactly 2011^2 elements. For every such subset X , define

$$s_j = \sum_{x \in X} x(x-1)\cdots(x-j+1)$$

for $j = 1, 2, \dots, 2011$. We want to show that there are two subsets corresponding to the same tuple $(s_1, s_2, \dots, s_{2011})$. A standard idea is to show that there are more subsets than the number of possible tuples. Indeed, since

$$0 \leq s_j < \sum_{x \in X} M^j = 2011^2 M^j,$$

there are at most $2011^2 M^j$ choices for s_j . Therefore, there are at most

$$\prod_{j=1}^{2011} (2011^2 M^j) = (2011^2)^{2011} M^{1+2+\dots+2011} = 2011^{2022} M^{1006 \times 2011}$$

different tuples. On the other hand, the number of subsets is

$$\binom{M}{2011^2} = \frac{M(M-1)\dots(M-2011^2+1)}{2011^2!} > \frac{1}{2011^2!} \left(\frac{M}{2}\right)^{2011^2}$$

if $\frac{M}{2} > 2011^2 - 1$. Note that

$$\begin{aligned} & \frac{1}{2011^2!} \left(\frac{M}{2}\right)^{2011^2} > 2011^{2022} M^{1006 \times 2011} \\ \Leftrightarrow & M^{2011^2 - 1006 \times 2011} > 2^{2011^2} \cdot 2011^{2022} \cdot 2011^2! \\ \Leftrightarrow & M^{1005 \times 2011} > 2^{2011^2} \cdot 2011^{2022} \cdot 2011^2!. \end{aligned}$$

This holds for sufficiently large M . Therefore, by the pigeonhole principle, there are two subsets A and B corresponding to the same tuple. This proves [claim 1](#). $\square$

Claim 2. Let $f(x)$ be a polynomial such that $f(1) = f'(1) = \dots = f^{(2011)}(1) = 0$. Then $(1-x)^{2012}$ divides $f(x)$.

Proof. This is a well-known result. Indeed, it is obvious that we can rewrite any polynomial as

$$f(x) = a_0 + a_1(x-1) + a_2(x-1)^2 + \dots + a_d(x-1)^d$$

for some constants $a_0, a_1, \dots, a_d$ (for example, by rewriting $f(x+1)$ as the usual polynomial form). Putting $x = 1$, we obtain $a_0 = 0$. Also, note that

$$f^{(j)}(x) = \sum_{i=j}^d a_i i(i-1)\dots(i-j+1)(x-1)^{i-j}$$

for $j = 1, 2, \dots, 2011$. Putting $x = 1$, we obtain $a_j = 0$. This implies

$$f(x) = \sum_{i=2012}^d a_i (x-1)^i,$$

which is clearly divisible by $(1-x)^{2012}$. $\square$

Now, we choose a pair of sets A, B satisfying the conditions in [claim 1](#). Let

$$f(x) = \sum_{a \in A} x^a - \sum_{b \in B} x^b.$$

Then the conditions in [claim 1](#) simply mean that $f(1) = f'(1) = \dots = f^{(2011)}(1) = 0$. By [claim 2](#), we can write

$$f(x) = (1-x)^{2012} g(x)$$

for some polynomial $g(x)$. The continuous function $|g(x)|$ is bounded by some constant $K > 0$ on the interval $[0, 1]$ by the extreme value theorem. Let $\varepsilon = \min \left\{ \frac{1}{K}, 1 \right\}$, and let C be a sufficiently large positive integer depending on ε . As suggested above, we add C to every element in A and B . Then we have to estimate

$$\left| \sum_{a \in A} x^{a+C} - \sum_{b \in B} x^{b+C} \right| = x^C |f(x)| = x^C (1-x)^{2012} |g(x)|.$$

- For $x \in (1 - \varepsilon, 1)$, we have

$$x^C (1-x)^{2012} |g(x)| < (1)(1)^{2011}(\varepsilon)(K) \leq 1$$

since $1 - x < \varepsilon$ and $\varepsilon \leq \frac{1}{K}$.

- For $x \in (0, 1 - \varepsilon]$, we have

$$x^C (1-x)^{2012} |g(x)| < (1 - \varepsilon)^C (1)^{2012}(K) < 1$$

because $(1 - \varepsilon)^C \rightarrow 0$ when $C \rightarrow \infty$.

Therefore, the inequality is established if we choose the sets $\{a + C : a \in A\}$ and $\{b + C : b \in B\}$. $\square$

Lastly, we summarize some useful techniques in solving some modern inequality problems.

- mathematical induction
 - standard induction ([problems 6.1, 6.2, 6.5](#))
 - modifying the inductive hypothesis ([problem 6.3](#))
 - Cauchy induction ([problem 6.4](#))
- combinatorial ideas
 - considering extremal cases ([problem 6.6](#))
 - using graph theory ([problems 6.7, 6.8, 6.10](#))
 - double counting ([problem 6.8](#))
 - pigeonhole principle ([problems 6.9, 6.10, 6.11](#))
- analytic ideas, $\varepsilon \rightarrow 0$ or $n \rightarrow \infty$ ([problems 6.7, 6.10, 6.11](#))

Links

Theorems

- [AM-GM inequality](#)
- [Cauchy-Schwarz inequality](#)
- [Chain rule](#)
- [Chebyshev's inequality](#)
- [Chebyshev's sum inequality](#)
- [Extreme value theorem](#)
- [Fundamental theorem of calculus](#)
- [Generalized Chebyshev's inequality](#)
- [Generalized Turkevich's inequality](#)
- [Hölder's inequality](#)
- [Jensen's inequality](#)
- [Karamata's inequality](#)
- [L'Hôpital's rule](#)
- [Majorization inequality](#)
- [Mean value theorem](#)
- [Muirhead's theorem](#)
- [Power mean inequality](#)
- [Rearrangement inequality](#)
- [Schur's inequality](#)
- [Suranyi's inequality](#)
- [Turkevich's inequality](#)
- [Vornicu-Schur's inequality](#)
- [Weighted AM-GM inequality](#)
- [Weighted power mean inequality](#)

Propositions

- [1.1](#): Basic facts about inequalities
- [1.2](#): Nonnegativity of the square of real numbers
- [1.3](#): Criterion for a real polynomial to be nonnegative

- 1.4: Quadratic inequalities
- 2.1: Corollary of the Cauchy-Schwarz inequality
- 3.1: Corollary of the power mean inequality
- 5.1: Derivatives of elementary functions
- 5.2: Basic differentiable rules
- 5.3: Derivatives of monotonic functions
- 5.4: Criterion for a continuous function to be convex/concave
- 5.5: Criterion for a twice differentiable function to be convex/concave
- 5.6: Basic facts about integration
- 5.7: Estimation of a discrete sum using integration

Problems

- 1.1: Completing square 1
- 1.2: Completing square 2
- 1.3: Factorization
- 1.4: Single variable inequality
- 1.5: Discriminant
- 1.6: CMO 1988 Problem 4(1)
- 1.7: Quadratic inequality
- 2.1: AM-GM inequality 1
- 2.2: AM-GM inequality 2
- 2.3: IMO HK TST 2007 Test 1 Problem 3
- 2.4: AM-GM inequality 3
- 2.5: AM-GM inequality 4
- 2.6: AM-GM inequality 5
- 2.7: IMO Shortlist 1998 A3
- 2.8: Power mean inequality 1
- 2.9: Power mean inequality 2
- 2.10: Cauchy-Schwarz inequality 1
- 2.11: Cauchy-Schwarz inequality 2
- 2.12: CGMO 2013 Problem 5
- 2.13: Triangle inequality

- 2.14: Cauchy-Schwarz inequality 3
- 2.15: Hölder's inequality 1
- 2.16: Hölder's inequality 2
- 2.17: Chebyshev's inequality 1
- 2.18: Chebyshev's inequality 2
- 3.1: Iran TST 1996/CHKMO 2010 Problem 4
- 3.2: IMO 2000 Problem 2
- 3.3: CSMO 2006 Problem 6
- 3.4: Homogenization
- 3.5: IMO 2001 Problem 2
- 3.6: IMO Shortlist 2004 A5
- 3.7: APMO 2022 Problem 5
- 3.8: Normalization 1
- 3.9: Normalization 2
- 4.1: Local refinement method
- 4.2: IMO Shortlist 2005 A3
- 4.3: Proof of the AM-GM inequality (mixing variables)
- 4.4: Mixing variables 1
- 4.5: Mixing variables 2
- 4.6: CMO 1988 Problem 4(2)
- 4.7: CMO 2011 Problem 5
- 5.1: Basic differentiation
- 5.2: Differentiation and limits
- 5.3: Differentiation and the extreme value theorem
- 5.4: Tangent line trick 1
- 5.5: Tangent line trick 2
- 5.6: USAMO 1980 Problem 5
- 5.7: CMO 2016 Problem 6
- 5.8: IMO 2020 Problem 2
- 5.9: Proof of the weighted AM-GM inequality
- 5.10: Proof of the power mean inequality
- 5.11: Proof of Hölder's inequality

- 5.12: Jensen's inequality 1
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- 5.14: IMO HK TST 2017 Test 1 Problem 5
- 5.15: Integration
- 5.16: Integrating an inequality 1
- 5.17: Integrating an inequality 2
- 6.1: Induction 1
- 6.2: India MO 2020 Problem 4
- 6.3: Induction 2
- 6.4: IMO Shortlist 2016 A1
- 6.5: IMO Shortlist 2020 A7
- 6.6: Tuymaada Olympiad 2020 Senior Problem 2
- 6.7: International Zhaitykov Olympiad 2019 Problem 2
- 6.8: IMO 2015 Problem 6
- 6.9: Iran MO 2nd Round 2021 Problem 5
- 6.10: IMO Shortlist 2016 A2
- 6.11: USA TST 2011 Problem 9

Terminologies and Notations

- $\mathbb{R}$
- $\mathbb{R}^+$
- $\mathbb{R}_0^+$
- antiderivative
- arithmetic mean
- Cauchy induction
- concave function
- continuous
- continuous function
- convex function
- critical point
- cyclic summation sign $\sum_{cyc}$

- definite integral $\int_a^b f(x) dx$
- derivative $\frac{df}{dx}, f'$
- differentiable
- differentiable function
- direct sum
- discontinuous
- discriminant Δ
- geometric mean
- harmonic mean
- homogeneous
- homogenization
- infimum
- inflexion point
- integrable
- left-hand limit $\lim_{x \rightarrow c^-} f(x)$
- limit $\lim_{x \rightarrow c} f(x)$
- local refinement method
- majorization
- majorize $>, <$
- method of completing square
- mixing variables
- nonnegative polynomial
- normalization
- one-sided limit
- positive polynomial
- random sum
- reverse sum
- Riemann integral $\int_a^b f(x) dx$
- right-hand limit $\lim_{x \rightarrow c^+} f(x)$
- second derivative f''

- smoothing method
- symmetric summation sign $\sum_{\text{sym}}$
- tangent line trick

References

- Inequalities — Methods and Olympiad Problems (by ???)
(Available from: <https://epdf.tips/inequalities-methods-and-olympiad-problems.html>)
- Olympiad Inequalities — Ideas and Tricks (by Pham Văn Thuận, Lê Vĩ)
- Secrets in Inequalities (volume 1) (by Pham Kim Hung)

「天之道，損有餘而補不足，是故虛勝實，不足勝有餘。」

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